CONTRACT FOR CONSULTANT'S SERVICES
(Time-Based)

Project Name: Dasu Hydropower Transmission Line Project

Loan No. IDA: 5497-5498

Contract No. DHTLP-2014

between

National Transmission & Despatch Company Limited (NTDCL), Pakistan

and

Korea Electric Power Corporation (KEPCO), Korea

January, 2015

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I. Form of Contract

(TIME-BASED)

This CONTRACT (hereinafter called the "Contract") is made the ------- day of the month of *January 2015*, between, on the one hand, **National Transmission & Despatch Company Limited (NTDCL)** (hereinafter called the "Client") and, on the other hand, **Korea Electric Power Corporation (KEPCO)** (hereinafter called the "Consultant").

WHEREAS

(a) The Client has requested the Consultant to provide certain consulting services as defined in this Contract (hereinafter called the "Services");

(b) the Consultant, having represented to the Client that it has the required professional skills, expertise and technical resources, has agreed to provide the Services on the terms and conditions set forth in this Contract;

(c) the Client has received a loan from the International Development Association (IDA) toward the cost of the Services and intends to apply a portion of the proceeds of this loan to eligible payments under this Contract, it being understood that (i) payments by the Bank will be made only at the request of the Client and upon approval by the Bank; (ii) such payments will be subject, in all respects, to the terms and conditions of the loan agreement, including prohibitions of withdrawal from the loan account for the purpose of any payment to persons or entities, or for any import of goods, if such payment or import, to the knowledge of the Bank, is prohibited by the decision of the United Nations Security council taken under Chapter VII of the Charter of the United Nations; and (iii) no party other than the Client shall derive any rights from the loan agreement or have any claim to the loan proceeds;

NOW THEREFORE the parties hereto hereby agree as follows:

1. The following documents attached hereto shall be deemed to form an integral part of this Contract:

   (a) The General Conditions of Contract (including Attachment 1 "Bank Policy – Corrupt and Fraudulent Practices);
   (b) The Special Conditions of Contract;
   (c) Appendices:

   Appendix A: Terms of Reference (including Form Tech-5)
   Appendix B: Key Experts (including form Tech-6)
   Appendix C: Remuneration Cost Estimates

Appendix D: Reimbursable Cost Estimates
Appendix E: Form of Advance Payments Guarantee
Appendix F: Contract Negotiation minutes

In the event of any inconsistency between the documents, the following order of precedence shall prevail: the Special Conditions of Contract; the General Conditions of Contract, including Attachment 1; Appendix A; Appendix B; Appendix C and Appendix D; Appendix E; Appendix F. Any reference to this Contract shall include, where the context permits, a reference to its Appendices.

2. The mutual rights and obligations of the Client and the Consultant shall be as set forth in the Contract, in particular:

(a) the Consultant shall carry out the Services in accordance with the provisions of the Contract; and
(b) the Client shall make payments to the Consultant in accordance with the provisions of the Contract.

IN WITNESS WHEREOF, the Parties hereto have caused this Contract to be signed in their respective names as of the day and year first above written.

For and on behalf of National Transmission & Despatch Company Limited (NTDCL)

_____

(Engr. Mahmood Ahmad)
Chief Engineer/ Project Manager Dasu

For and on behalf of Korea Electric Power Corporation (KEPCO)

_____

(JaeHeun Park)
Senior Manager/Overseas Business Division

General Conditions of Contract

A. GENERAL PROVISIONS

1. Definitions

1.1. Unless the context otherwise requires, the following terms whenever used in this Contract have the following meanings:

(a) "Applicable Guidelines" means the Guidelines for Selection and Employment of Consultants under IBRD Loans and IDA Credits & Grants by World Bank Borrowers, dated January 2011 ("Consultants' Guidelines").

(b) "Applicable Law" means the laws and any other instruments having the force of law in the Client's country, or in such other country as may be specified in the **Special Conditions of Contract (SCC)**, as they may be issued and in force from time to time.

(c) "Bank" means the International Bank for Reconstruction and Development (IBRD) or the International Development Association (IDA).

(d) "Borrower" means the Government, Government agency or other entity that signs the financing agreement with the Bank.

(e) "Client" means the implementing agency that signs the Contract for the Services with the Selected Consultant.

(f) "Consultant" means a legally-established professional consulting firm or entity selected by the Client to provide the Services under the signed Contract.

(g) "Contract" means the legally binding written agreement signed between the Client and the Consultant and which includes all the attached documents listed in its paragraph 1 of the Form of Contract (the General Conditions (GCC), the Special Conditions (SCC), and the Appendices).

(h) "Day" means a working day unless indicated otherwise.

(i) "Effective Date" means the date on which this Contract comes into force and effect pursuant to Clause GCC 11.

(j) "Experts" means, collectively, Key Experts, Non-Key Experts, or any other personnel of the Consultant, Sub-consultant or JV member(s) assigned by the Consultant to perform the Services or any part thereof under the Contract.

(k) "Foreign Currency" means any currency other than the currency

of the Client's country.

(l) "GCC" means these General Conditions of Contract.

(m) "Government" means the government of the Client's country.

(n) "Joint Venture (JV)" means an association with or without a legal personality distinct from that of its members, of more than one entity where one member has the authority to conduct all businesses for and on behalf of any and all the members of the JV, and where the members of the JV are jointly and severally liable to the Client for the performance of the Contract.

(o) "Key Expert(s)" means an individual professional whose skills, qualifications, knowledge and experience are critical to the performance of the Services under the Contract and whose Curricula Vitae (CV) was taken into account in the technical evaluation of the Consultant's proposal.

(p) "Local Currency" means the currency of the Client's country.

(q) "Non-Key Expert(s)" means an individual professional provided by the Consultant or its Sub-consultant to perform the Services or any part thereof under the Contract.

(r) "Party" means the Client or the Consultant, as the case may be, and "Parties" means both of them.

(s) "SCC" means the Special Conditions of Contract by which the GCC may be amended or supplemented but not over-written.

(t) "Services" means the work to be performed by the Consultant pursuant to this Contract, as described in Appendix A hereto.

(u) "Sub-consultants" means an entity to whom/which the Consultant subcontracts any part of the Services while remaining solely liable for the execution of the Contract.

(v) "Third Party" means any person or entity other than the Government, the Client, the Consultant or a Sub-consultant.

2. Relationship between the Parties

2.1. Nothing contained herein shall be construed as establishing a relationship of master and servant or of principal and agent as between the Client and the Consultant. The Consultant, subject to this Contract, has complete charge of the Experts and Sub-consultants, if any, performing the Services and shall be fully responsible for the Services performed by them or on their behalf hereunder.

3. Law Governing

3.1. This Contract, its meaning and interpretation, and the relation

	Contract	between the Parties shall be governed by the Applicable Law.
4.	**Language**	4.1. This Contract has been executed in the language specified in the **SCC**, which shall be the binding and controlling language for all matters relating to the meaning or interpretation of this Contract.
5.	**Headings**	5.1. The headings shall not limit, alter or affect the meaning of this Contract.
6.	**Communications**	6.1. Any communication required or permitted to be given or made pursuant to this Contract shall be in writing in the language specified in Clause GCC 4. Any such notice, request or consent shall be deemed to have been given or made when delivered in person to an authorized representative of the Party to whom the communication is addressed, or when sent to such Party at the address specified in the **SCC**. 6.2. A Party may change its address for notice hereunder by giving the other Party any communication of such change to the address specified in the **SCC**.
7.	**Location**	7.1. The Services shall be performed at such locations as are specified in **Appendix A** hereto and, where the location of a particular task is not so specified, at such locations, whether in the Government's country or elsewhere, as the Client may approve.
8.	**Authority of Member in Charge**	8.1. In case the Consultant is a Joint Venture, the members hereby authorize the member specified in the **SCC** to act on their behalf in exercising all the Consultant's rights and obligations towards the Client under this Contract, including without limitation the receiving of instructions and payments from the Client.
9.	**Authorized Representatives**	9.1. Any action required or permitted to be taken, and any document required or permitted to be executed under this Contract by the Client or the Consultant may be taken or executed by the officials specified in the **SCC.**
10.	**Corrupt and Fraudulent Practices**	10.1. The Bank requires compliance with its policy in regard to corrupt and fraudulent practices as set forth in **Attachment 1** to the GCC.
	a. Commissions and Fees	10.2. The Client requires the Consultant to disclose any commissions or fees that may have been paid or are to be paid to agents or any other party with respect to the selection process or execution of the Contract. The information disclosed must include at least the name and address of the agent or other party, the amount and currency, and the purpose of the commission, gratuity or fee. Failure to disclose such commissions, gratuities or fees may result in

termination of the Contract and/or sanctions by the Bank.

B. COMMENCEMENT, COMPLETION, MODIFICATION AND TERMINATION OF CONTRACT

11. Effectiveness of Contract

11.1. This Contract shall come into force and effect on the date (the "Effective Date") of the Client's notice to the Consultant instructing the Consultant to begin carrying out the Services. This notice shall confirm that the effectiveness conditions, if any, listed in the **SCC** have been met.

12. Termination of Contract for Failure to Become Effective

12.1. If this Contract has not become effective within such time period after the date of Contract signature as specified in the **SCC**, either Party may, by not less than twenty two (22) days written notice to the other Party, declare this Contract to be null and void, and in the event of such a declaration by either Party, neither Party shall have any claim against the other Party with respect hereto.

13. Commencement of Services

13.1. The Consultant shall confirm availability of Key Experts and begin carrying out the Services not later than the number of days after the Effective Date specified in the **SCC**.

14. Expiration of Contract

14.1. Unless terminated earlier pursuant to Clause GCC 19 hereof, this Contract shall expire at the end of such time period after the Effective Date as specified in the **SCC**.

15. Entire Agreement

15.1. This Contract contains all covenants, stipulations and provisions agreed by the Parties. No agent or representative of either Party has authority to make, and the Parties shall not be bound by or be liable for, any statement, representation, promise or agreement not set forth herein.

16. Modifications or Variations

16.1. Any modification or variation of the terms and conditions of this Contract, including any modification or variation of the scope of the Services, may only be made by written agreement between the Parties. However, each Party shall give due consideration to any proposals for modification or variation made by the other Party.

16.2. In cases of substantial modifications or variations, the prior written consent of the Bank is required.

17. Force Majeure

a. Définition

17.1. For the purposes of this Contract, "Force Majeure" means an event which is beyond the reasonable control of a Party, is not foreseeable, is unavoidable, and makes a Party's performance of its obligations hereunder impossible or so impractical as reasonably to be considered impossible under the circumstances, and subject to

those requirements, includes, but is not limited to, war, riots, civil disorder, earthquake, fire, explosion, storm, flood or other adverse weather conditions, strikes, lockouts or other industrial action confiscation or any other action by Government agencies.

17.2. Force Majeure shall not include (i) any event which is caused by the negligence or intentional action of a Party or such Party's Experts, Sub-consultants or agents or employees, nor (ii) any event which a diligent Party could reasonably have been expected to both take into account at the time of the conclusion of this Contract, and avoid or overcome in the carrying out of its obligations hereunder.

17.3. Force Majeure shall not include insufficiency of funds or failure to make any payment required hereunder.

b. No Breach of Contract

17.4. The failure of a Party to fulfill any of its obligations hereunder shall not be considered to be a breach of, or default under, this Contract insofar as such inability arises from an event of Force Majeure, provided that the Party affected by such an event has taken all reasonable precautions, due care and reasonable alternative measures, all with the objective of carrying out the terms and conditions of this Contract.

c. Measures to be Taken

17.5. A Party affected by an event of Force Majeure shall continue to perform its obligations under the Contract as far as is reasonably practical, and shall take all reasonable measures to minimize the consequences of any event of Force Majeure.

17.6. A Party affected by an event of Force Majeure shall notify the other Party of such event as soon as possible, and in any case not later than fourteen (14) calendar days following the occurrence of such event, providing evidence of the nature and cause of such event, and shall similarly give written notice of the restoration of normal conditions as soon as possible.

17.7. Any period within which a Party shall, pursuant to this Contract, complete any action or task, shall be extended for a period equal to the time during which such Party was unable to perform such action as a result of Force Majeure.

17.8. During the period of their inability to perform the Services as a result of an event of Force Majeure, the Consultant, upon instructions by the Client, shall either:

(a) demobilize, in which case the Consultant shall be reimbursed for additional costs they reasonably and necessarily incurred, and, if required by the Client, in

reactivating the Services; or

(b) continue with the Services to the extent reasonably possible, in which case the Consultant shall continue to be paid under the terms of this Contract and be reimbursed for additional costs reasonably and necessarily incurred.

17.9. In the case of disagreement between the Parties as to the existence or extent of Force Majeure, the matter shall be settled according to Clauses GCC 48 & 49.

18. Suspension

18.1. The Client may, by written notice of suspension to the Consultant, suspend all payments to the Consultant hereunder if the Consultant fails to perform any of its obligations under this Contract, including the carrying out of the Services, provided that such notice of suspension (i) shall specify the nature of the failure, and (ii) shall request the Consultant to remedy such failure within a period not exceeding thirty (30) calendar days after receipt by the Consultant of such notice of suspension.

19. Termination

19. 1 This Contract may be terminated by either Party as per provisions set up below:

a. By the Client

19.1.1 The Client may terminate this Contract in case of the occurrence of any of the events specified in paragraphs (a) through (f) of this Clause. In such an occurrence the Client shall give at least thirty (30) calendar days' written notice of termination to the Consultant in case of the events referred to in (a) through (d); at least sixty (60) calendar days' written notice in case of the event referred to in (e); and at least five (5) calendar days' written notice in case of the event referred to in (f):

(a) If the Consultant fails to remedy a failure in the performance of its obligations hereunder, as specified in a notice of suspension pursuant to Clause GCC 18;

(b) If the Consultant becomes (or, if the Consultant consists of more than one entity, if any of its members becomes) insolvent or bankrupt or enter into any agreements with their creditors for relief of debt or take advantage of any law for the benefit of debtors or go into liquidation or receivership whether compulsory or voluntary;

(c) If the Consultant fails to comply with any final decision reached as a result of arbitration proceedings pursuant to Clause GCC 49.1;

(d) If, as the result of Force Majeure, the Consultant is

unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;

(e) If the Client, in its sole discretion and for any reason whatsoever, decides to terminate this Contract;

(f) If the Consultant fails to confirm availability of Key Experts as required in Clause GCC 13.

19.1.2 Furthermore, if the Client determines that the Consultant has engaged in corrupt, fraudulent, collusive, coercive or obstructive practices, in competing for or in executing the Contract, then the Client may, after giving fourteen (14) calendar days written notice to the Consultant, terminate the Consultant's employment under the Contract.

b. By the Consultant

19.1.3 The Consultant may terminate this Contract, by not less than thirty (30) calendar days' written notice to the Client, in case of the occurrence of any of the events specified in paragraphs (a) through (d) of this Clause.

(a) If the Client fails to pay any money due to the Consultant pursuant to this Contract and not subject to dispute pursuant to Clauses GCC 49.1 within forty-five (45) calendar days after receiving written notice from the Consultant that such payment is overdue.

(b) If, as the result of Force Majeure, the Consultant is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days.

(c) If the Client fails to comply with any final decision reached as a result of arbitration pursuant to Clause GCC 49.1.

(d) If the Client is in material breach of its obligations pursuant to this Contract and has not remedied the same within forty-five (45) days (or such longer period as the Consultant may have subsequently approved in writing) following the receipt by the Client of the Consultant's notice specifying such breach.

c. Cessation of Rights and Obligations

19.1.4 Upon termination of this Contract pursuant to Clauses GCC 12 or GCC 19 hereof, or upon expiration of this Contract pursuant to Clause GCC 14, all rights and obligations of the Parties hereunder shall cease, except (i) such rights and obligations as may have accrued on the date of termination or expiration, (ii) the obligation of confidentiality set forth in Clause GCC 22, (iii) the Consultant's obligation to permit

inspection, copying and auditing of their accounts and records set forth in Clause GCC 25, and (iv) any right which a Party may have under the Applicable Law.

d. Cessation of Services

19.1.5 Upon termination of this Contract by notice of either Party to the other pursuant to Clauses GCC 19a or GCC 19b, the Consultant shall, immediately upon dispatch or receipt of such notice, take all necessary steps to bring the Services to a close in a prompt and orderly manner and shall make every reasonable effort to keep expenditures for this purpose to a minimum. With respect to documents prepared by the Consultant and equipment and materials furnished by the Client, the Consultant shall proceed as provided, respectively, by Clauses GCC 27 or GCC 28.

e. Payment upon Termination

19.1.6 Upon termination of this Contract, the Client shall make the following payments to the Consultant:

(a) remuneration for Services satisfactorily performed prior to the effective date of termination, and reimbursable expenditures for expenditures actually incurred prior to the effective date of termination; and pursuant to Clause 42;

(b) in the case of termination pursuant to paragraphs (d) and (e) of Clause GCC 19.1.1, reimbursement of any reasonable cost incidental to the prompt and orderly termination of this Contract, including the cost of the return travel of the Experts.

C. OBLIGATIONS OF THE CONSULTANT

20. General

a. Standard of Performance

20.1 The Consultant shall perform the Services and carry out the Services with all due diligence, efficiency and economy, in accordance with generally accepted professional standards and practices, and shall observe sound management practices, and employ appropriate technology and safe and effective equipment, machinery, materials and methods. The Consultant shall always act, in respect of any matter relating to this Contract or to the Services, as a faithful adviser to the Client, and shall at all times support and safeguard the Client's legitimate interests in any dealings with the third parties.

20.2 The Consultant shall employ and provide such qualified and experienced Experts and Sub-consultants as are required to carry out the Services.

20.3 The Consultant may subcontract part of the Services to an

extent and with such Key Experts and Sub-consultants as may be approved in advance by the Client. Notwithstanding such approval, the Consultant shall retain full responsibility for the Services.

b. Law Applicable to Services

20.4 The Consultant shall perform the Services in accordance with the Contract and the Applicable Law and shall take all practicable steps to ensure that any of its Experts and Sub-consultants, comply with the Applicable Law.

20.5 Throughout the execution of the Contract, the Consultant shall comply with the import of goods and services prohibitions in the Client's country when

(a) as a matter of law or official regulations, the Borrower's country prohibits commercial relations with that country; or

(b) by an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, the Borrower's Country prohibits any import of goods from that country or any payments to any country, person, or entity in that country.

20.6 The Client shall notify the Consultant in writing of relevant local customs, and the Consultant shall, after such notification, respect such customs.

21. Conflict of Interests

21.1 The Consultant shall hold the Client's interests paramount, without any consideration for future work, and strictly avoid conflict with other assignments or their own corporate interests.

a. Consultant Not to Benefit from Commissions, Discounts, etc.

21.1.1 The payment of the Consultant pursuant to GCC F (Clauses GCC 41 through 46) shall constitute the Consultant's only payment in connection with this Contract and, subject to Clause GCC 21.1.3, the Consultant shall not accept for its own benefit any trade commission, discount or similar payment in connection with activities pursuant to this Contract or in the discharge of its obligations hereunder, and the Consultant shall use its best efforts to ensure that any Sub-consultants, as well as the Experts and agents of either of them, similarly shall not receive any such additional payment.

21.1.2 Furthermore, if the Consultant, as part of the Services, has the responsibility of advising the Client on the procurement of goods, works or services, the Consultant shall comply with the Bank's Applicable Guidelines, and shall at all times exercise such responsibility in the best interest of the Client. Any discounts or commissions obtained by the Consultant in the exercise of such

procurement responsibility shall be for the account of the Client.

b. Consultant and Affiliates Not to Engage in Certain Activities

21.1.3 The Consultant agrees that, during the term of this Contract and after its termination, the Consultant and any entity affiliated with the Consultant, as well as any Sub-consultants and any entity affiliated with such Sub-consultants, shall be disqualified from providing goods, works or non-consulting services resulting from or directly related to the Consultant's Services for the preparation or implementation of the project, unless otherwise indicated in the **SCC**.

c. Prohibition of Conflicting Activities

21.1.4 The Consultant shall not engage, and shall cause its Experts as well as its Sub-consultants not to engage, either directly or indirectly, in any business or professional activities that would conflict with the activities assigned to them under this Contract.

d. Strict Duty to Disclose Conflicting Activities

21.1.5 The Consultant has an obligation and shall ensure that its Experts and Sub-consultants shall have an obligation to disclose any situation of actual or potential conflict that impacts their capacity to serve the best interest of their Client, or that may reasonably be perceived as having this effect. Failure to disclose said situations may lead to the disqualification of the Consultant or the termination of its Contract.

22. Confidentiality

22.1 Except with the prior written consent of the Client, the Consultant and the Experts shall not at any time communicate to any person or entity any confidential information acquired in the course of the Services, nor shall the Consultant and the Experts make public the recommendations formulated in the course of, or as a result of, the Services.

23. Liability of the Consultant

23.1 Subject to additional provisions, if any, set forth in the **SCC**, the Consultant's liability under this Contract shall be as determined under the Applicable Law.

24. Insurance to be Taken out by the Consultant

24.1 The Consultant (i) shall take out and maintain, and shall cause any Sub-consultants to take out and maintain, at its (or the Sub-consultants', as the case may be) own cost but on terms and conditions approved by the Client, insurance against the risks, and for the coverage specified in the **SCC,** and (ii) at the Client's request, shall provide evidence to the Client showing that such insurance has been taken out and maintained and that the current premiums therefore have been paid. The Consultant shall ensure that such insurance is in place prior to commencing the Services as stated in Clause GCC 13.

25. Accounting, Inspection and

25.1 The Consultant shall keep, and shall make all reasonable efforts to cause its Sub-consultants to keep, accurate and systematic accounts and records in respect of the Services in such form and

Auditing

detail as will clearly identify relevant time changes and costs.

25.2. The Consultant shall permit and shall cause its Sub-consultants to permit, the Bank and/or persons appointed by the Bank to inspect the Site and/or all accounts and records relating to the performance of the Contract and the submission of the Proposal to provide the Services, and to have such accounts and records audited by auditors appointed by the Bank if requested by the Bank. The Consultant's attention is drawn to Clause GCC 10 which provides, inter alia, that acts intended to materially impede the exercise of the Bank's inspection and audit rights provided for under this Clause GCC25.2 constitute a prohibited practice subject to contract termination (as well as to a determination of ineligibility under the Bank's prevailing sanctions procedures.)

26. Reporting Obligations

26.1 The Consultant shall submit to the Client the reports and documents specified in **Appendix A**, in the form, in the numbers and within the time periods set forth in the said Appendix.

27. Proprietary Rights of the Client in Reports and Records

27.1 Unless otherwise indicated in the **SCC**, all reports and relevant data and information such as maps, diagrams, plans, databases, other documents and software, supporting records or material compiled or prepared by the Consultant for the Client in the course of the Services shall be confidential and become and remain the absolute property of the Client. The Consultant shall, not later than upon termination or expiration of this Contract, deliver all such documents to the Client, together with a detailed inventory thereof. The Consultant may retain a copy of such documents, data and/or software but shall not use the same for purposes unrelated to this Contract without prior written approval of the Client.

27.2 If license agreements are necessary or appropriate between the Consultant and third parties for purposes of development of the plans, drawings, specifications, designs, databases, other documents and software, the Consultant shall obtain the Client's prior written approval to such agreements, and the Client shall be entitled at its discretion to require recovering the expenses related to the development of the program(s) concerned. Other restrictions about the future use of these documents and software, if any, shall be specified in the **SCC**.

28. Equipment, Vehicles and Materials

28.1 Equipment, vehicles and materials made available to the Consultant by the Client, or purchased by the Consultant wholly or partly with funds provided by the Client, shall be the property of the Client and shall be marked accordingly. Upon termination or expiration of this Contract, the Consultant shall make available to the Client an inventory of such equipment, vehicles and materials and shall dispose of such equipment, vehicles and materials in accordance with the Client's instructions. While in possession of such equipment,

vehicles and materials, the Consultant, unless otherwise instructed by the Client in writing, shall insure them at the expense of the Client in an amount equal to their full replacement value.

28.2 Any equipment or materials brought by the Consultant or its Experts into the Client's country for the use either for the project or personal use shall remain the property of the Consultant or the Experts concerned, as applicable.

D. CONSULTANT'S EXPERTS AND SUB-CONSULTANTS

29. Description of Key Experts

29.1 The title, agreed job description, minimum qualification and time-input estimates to carry out the Services of each of the Consultant's Key Experts are described in **Appendix B.**

29.2 If required to comply with the provisions of Clause GCC 20a, adjustments with respect to the estimated time-input of Key Experts set forth in **Appendix B** may be made by the Consultant by a written notice to the Client, provided (i) that such adjustments shall not alter the original time-input estimates for any individual by more than 10% or one week, whichever is larger; and (ii) that the aggregate of such adjustments shall not cause payments under this Contract to exceed the ceilings set forth in Clause GCC 41.2.

29.3 If additional work is required beyond the scope of the Services specified in **Appendix A**, the estimated time-input for the Key Experts may be increased by agreement in writing between the Client and the Consultant. In case where payments under this Contract exceed the ceilings set forth in Clause GCC 41.1, the Parties shall sign a Contract amendment.

30. Replacement of Key Experts

30.1 Except as the Client may otherwise agree in writing, no changes shall be made in the Key Experts.

30.2 Notwithstanding the above, the substitution of Key Experts during Contract execution may be considered only based on the Consultant's written request and due to circumstances outside the reasonable control of the Consultant, including but not limited to death or medical incapacity. In such case, the Consultant shall forthwith provide as a replacement, a person of equivalent or better qualifications and experience, and at the same rate of remuneration.

31. Approval of Additional Key Experts

31.1 If during execution of the Contract, additional Key Experts are required to carry out the Services, the Consultant shall submit to the Client for review and approval a copy of their Curricula Vitae (CVs). If the Client does not object in writing (stating the reasons for the objection) within twenty two (22) days from the date of

receipt of such CVs, such additional Key Experts shall be deemed to have been approved by the Client.

The rate of remuneration payable to such new additional Key Experts shall be based on the rates for other Key Experts position which require similar qualifications and experience.

32. Removal of Experts or Sub-consultants

32.1 If the Client finds that any of the Experts or Sub-consultant has committed serious misconduct or has been charged with having committed a criminal action, or shall the Client determine that Consultant's Expert of Sub-consultant have engaged in corrupt, fraudulent, collusive, coercive or obstructive practice while performing the Services, the Consultant shall, at the Client's written request, provide a replacement.

32.2 In the event that any of Key Experts, Non-Key Experts or Sub-consultants is found by the Client to be incompetent or incapable in discharging assigned duties, the Client, specifying the grounds therefore, may request the Consultant to provide a replacement.

32.3 Any replacement of the removed Experts or Sub-consultants shall possess better qualifications and experience and shall be acceptable to the Client.

33. Replacement/ Removal of Experts – Impact on Payments

33.1 Except as the Client may otherwise agree, (i) the Consultant shall bear all additional travel and other costs arising out of or incidental to any removal and/or replacement, and (ii) the remuneration to be paid for any of the Experts provided as a replacement shall not exceed the remuneration which would have been payable to the Experts replaced or removed.

34. Working Hours, Overtime, Leave, etc.

34.1 Working hours and holidays for Experts are set forth in **Appendix B**. To account for travel time to/from the Client's country, experts carrying out Services inside the Client's country shall be deemed to have commenced or finished work in respect of the Services such number of days before their arrival in, or after their departure from, the Client's country as is specified in **Appendix B**.

34.2 The Experts shall not be entitled to be paid for overtime nor to take paid sick leave or vacation leave except as specified in **Appendix B**, and the Consultant's remuneration shall be deemed to cover these items.

34.3 Any taking of leave by Key Experts shall be subject to the prior approval by the Consultant who shall ensure that absence for leave purposes will not delay the progress and or impact adequate supervision of the Services.

E. OBLIGATIONS OF THE CLIENT

35. Assistance and Exemptions

35.1 Unless otherwise specified in the **SCC**, the Client shall use its best efforts to:

(a) Assist the Consultant with obtaining work permits and such other documents as shall be necessary to enable the Consultant to perform the Services.

(b) Assist the Consultant with promptly obtaining, for the Experts and, if appropriate, their eligible dependents, all necessary entry and exit visas, residence permits, exchange permits and any other documents required for their stay in the Client's country while carrying out the Services under the Contract.

(c) Facilitate prompt clearance through customs of any property required for the Services and of the personal effects of the Experts and their eligible dependents.

Issue to officials, agents and representatives of the Government all such instructions and information as may be necessary or appropriate for the prompt and effective implementation of the Services.

(d) Assist the Consultant and the Experts and any Sub-consultants employed by the Consultant for the Services with obtaining exemption from any requirement to register or obtain any permit to practice their profession or to establish themselves either individually or as a corporate entity in the Client's country according to the applicable law in the Client's country.

(e) Assist the Consultant, any Sub-consultants and the Experts of either of them with obtaining the privilege, pursuant to the applicable law in the Client's country, of bringing into the Client's country reasonable amounts of foreign currency for the purposes of the Services or for the personal use of the Experts and of withdrawing any such amounts as may be earned therein by the Experts in the execution of the Services.

(f) Provide to the Consultant any such other assistance as may be specified in the **SCC**.

36. Access to Project Site

36.1 The Client warrants that the Consultant shall have, free of charge, unimpeded access to the project site in respect of which access is required for the performance of the Services. The Client will be responsible for any damage to the project site or any property thereon resulting from such access and will indemnify the Consultant and each of the experts in respect of liability for any such damage, unless such damage is caused by the willful default or negligence of

the Consultant or any Sub-consultants or the Experts of either of them.

37. Change in the Applicable Law Related to Taxes and Duties

37.1 If, after the date of this Contract, there is any change in the applicable law in the Client's country with respect to taxes and duties which increases or decreases the cost incurred by the Consultant in performing the Services, then the remuneration and reimbursable expenses otherwise payable to the Consultant under this Contract shall be increased or decreased accordingly by agreement between the Parties hereto, and corresponding adjustments shall be made to the ceiling amounts specified in Clause GCC 41.1

38. Services, Facilities and Property of the Client

38.1 The Client shall make available to the Consultant and the Experts, for the purposes of the Services and free of any charge, the services, facilities and property described in the Terms of Reference (**Appendix A)** at the times and in the manner specified in said **Appendix A.**

38.2 In case that such services, facilities and property shall not be made available to the Consultant as and when specified in **Appendix A**, the Parties shall agree on (i) any time extension that it may be appropriate to grant to the Consultant for the performance of the Services, (ii) the manner in which the Consultant shall procure any such services, facilities and property from other sources, and (iii) the additional payments, if any, to be made to the Consultant as a result thereof pursuant to Clause GCC 41.3.

39. Counterpart Personnel

39.1 The Client shall make available to the Consultant free of charge such professional and support counterpart personnel, to be nominated by the Client with the Consultant's advice, if specified in **Appendix A**.

39.2 If counterpart personnel are not provided by the Client to the Consultant as and when specified in **Appendix A**, the Client and the Consultant shall agree on (i) how the affected part of the Services shall be carried out, and (ii) the additional payments, if any, to be made by the Client to the Consultant as a result thereof pursuant to Clause GCC 41.3.

39.3 Professional and support counterpart personnel, excluding Client's liaison personnel, shall work under the exclusive direction of the Consultant. If any member of the counterpart personnel fails to perform adequately any work assigned to such member by the Consultant that is consistent with the position occupied by such member, the Consultant may request the replacement of such member, and the Client shall not unreasonably refuse to act upon such request.

40. Payment Obligation

40.1 In consideration of the Services performed by the Consultant under this Contract, the Client shall make such payments to the Consultant and in such manner as is provided by GCC F below.

F. PAYMENTS TO THE CONSULTANT

41. Ceiling Amount

41.1 An estimate of the cost of the Services is set forth in **Appendix C**(Remuneration) and **Appendix D**(Reimbursable expenses).

41.2 Payments under this Contract shall not exceed the ceilings in foreign currency and in local currency specified in the **SCC**.

41.3 For any payments in excess of the ceilings specified in GCC41.2, an amendment to the Contract shall be signed by the Parties referring to the provision of this Contract that evokes such amendment.

42. Remuneration and Reimbursable Expenses

42.1 The Client shall pay to the Consultant (i) remuneration that shall be determined on the basis of time actually spent by each Expert in the performance of the Services after the date of commencing of Services or such other date as the Parties shall agree in writing; and (ii) reimbursable expenses that are actually and reasonably incurred by the Consultant in the performance of the Services.

42.2 All payments shall be at the rates set forth in **Appendix C** and **Appendix D**.

42.3 Unless the **SCC** provides for the price adjustment of the remuneration rates, said remuneration shall be fixed for the duration of the Contract.

42.4 The remuneration rates shall cover: (i) such salaries and allowances as the Consultant shall have agreed to pay to the Experts as well as factors for social charges and overheads (bonuses or other means of profit-sharing shall not be allowed as an element of overheads), (ii) the cost of backstopping by home office staff not included in the Experts' list in **Appendix B**, (iii) the Consultant's profit, and (iv) any other items as specified in the **SCC**.

42.5 Any rates specified for Experts not yet appointed shall be provisional and shall be subject to revision, with the written approval of the Client, once the applicable remuneration rates and allowances are known.

43. Taxes and Duties

43.1 The Consultant, Sub-consultants and Experts are responsible for meeting any and all tax liabilities arising out of the Contract

unless it is stated otherwise in the **SCC**.

43.2 As an exception to the above and as stated in the **SCC**, all local identifiable indirect taxes (itemized and finalized at Contract negotiations) are reimbursed to the Consultant or are paid by the Client on behalf of the Consultant.

44. Currency of Payment

44.1 Any payment under this Contract shall be made in the currency(ies) specified in the **SCC.**

45. Mode of Billing and Payment

45.1 Billings and payments in respect of the Services shall be made as follows:

(a) *Advance payment*. Within the number of days after the Effective Date, the Client shall pay to the Consultant an advance payment as specified in the **SCC**. Unless otherwise indicated in the **SCC**, an advance payment shall be made against an advance payment bank guarantee acceptable to the Client in an amount (or amounts) and in a currency (or currencies) specified in the **SCC**. Such guarantee (i) is to remain effective until the advance payment has been fully set off, and (ii) is to be in the form set forth in **Appendix E**, or in such other form as the Client shall have approved in writing. The advance payments will be set off by the Client in equal installments against the statements for the number of months of the Services specified in the **SCC** until said advance payments have been fully set off.

(b) *The Itemized Invoices.* As soon as practicable and not later than fifteen (15) days after the end of each calendar month during the period of the Services, or after the end of each time interval otherwise indicated in the **SCC**, the Consultant shall submit to the Client, in duplicate, itemized invoices, accompanied by the receipts or other appropriate supporting documents, of the amounts payable pursuant to Clauses GCC 44 and GCC 45 for such interval, or any other period indicated in the **SCC**. Separate invoices shall be submitted for expenses incurred in foreign currency and in local currency. Each invoice shall show remuneration and reimbursable expenses separately.

(c) The Client shall pay the Consultant's invoices within sixty (60) days after the receipt by the Client of such itemized invoices with supporting documents. Only such portion of an invoice that is not satisfactorily supported may be withheld from payment. Should any discrepancy be found to exist between actual payment and costs authorized to be incurred by the Consultant, the Client may add or subtract the difference from any subsequent payments.

(d) *The Final Payment* .The final payment under this Clause shall be made only after the final report and a final invoice, identified as such, shall have been submitted by the Consultant and approved as satisfactory by the Client. The Services shall be deemed completed and finally accepted by the Client and the final report and final invoice shall be deemed approved by the Client as satisfactory ninety (90) calendar days after receipt of the final report and final invoice by the Client unless the Client, within such ninety (90) calendar day period, gives written notice to the Consultant specifying in detail deficiencies in the Services, the final report or final invoice. The Consultant shall thereupon promptly make any necessary corrections, and thereafter the foregoing process shall be repeated. Any amount that the Client has paid or has caused to be paid in accordance with this Clause in excess of the amounts payable in accordance with the provisions of this Contract shall be reimbursed by the Consultant to the Client within thirty (30) days after receipt by the Consultant of notice thereof. Any such claim by the Client for reimbursement must be made within twelve (12) calendar months after receipt by the Client of a final report and a final invoice approved by the Client in accordance with the above.

(e) All payments under this Contract shall be made to the accounts of the Consultant specified in the **SCC**.

(f) With the exception of the final payment under (d) above, payments do not constitute acceptance of the Services nor relieve the Consultant of any obligations hereunder.

46. Interest on Delayed Payments

46.1 If the Client had delayed payments beyond fifteen (15) days after the due date stated in Clause GCC 45.1 (c), interest shall be paid to the Consultant on any amount due by, not paid on, such due date for each day of delay at the annual rate stated in the **SCC.**

G. FAIRNESS AND GOOD FAITH

47. Good Faith

47.1 The Parties undertake to act in good faith with respect to each other's rights under this Contract and to adopt all reasonable measures to ensure the realization of the objectives of this Contract.

H. SETTLEMENT OF DISPUTES

48. Amicable Settlement

48.1 The Parties shall seek to resolve any dispute amicably by mutual consultation.

48.2 If either Party objects to any action or inaction of the other

Party, the objecting Party may file a written Notice of Dispute to the other Party providing in detail the basis of the dispute. The Party receiving the Notice of Dispute will consider it and respond in writing within fourteen (14) days after receipt. If that Party fails to respond within fourteen (14) days, or the dispute cannot be amicably settled within fourteen (14) days following the response of that Party, Clause GCC 49.1 shall apply.

49. Dispute Resolution

49.1 Any dispute between the Parties arising under or related to this Contract that cannot be settled amicably may be referred to by either Party to the adjudication/arbitration in accordance with the provisions specified in the **SCC**.

Bank's Policy – Corrupt and Fraudulent Practices

Guidelines for Selection and Employment of Consultants under IBRD Loans and IDA Credits & Grants by World Bank Borrowers, dated January 2011:

"**Fraud and Corruption**

1.23 It is the Bank's policy to require that Borrowers (including beneficiaries of Bank loans), consultants, and their agents (whether declared or not), sub-contractors, sub-consultants, service providers, or suppliers, and any personnel thereof, observe the highest standard of ethics during the selection and execution of Bank-financed contracts [footnote: In this context, any action taken by a consultant or any of its personnel, or its agents, or its sub-consultants, sub-contractors, services providers, suppliers, and/or their employees, to influence the selection process or contract execution for undue advantage is improper.]. In pursuance of this policy, the Bank:

(a) defines, for the purposes of this provision, the terms set forth below as follows:

(i) "corrupt practice" is the offering, giving, receiving, or soliciting, directly or indirectly, of anything of value to influence improperly the actions of another party[1];

(ii) "fraudulent practice" is any act or omission, including misrepresentation, that knowingly or recklessly misleads, or attempts to mislead, a party to obtain financial or other benefit or to avoid an obligation[2];

(iii) "collusive practices" is an arrangement between two or more parties designed to achieve an improper purpose, including to influence improperly the actions of another party[3];

(iv) "coercive practices" is impairing or harming, or threatening to impair or harm, directly or indirectly, any party or the property of the party to influence improperly the actions of a party[4];

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[1] *For the purpose of this sub-paragraph, "another party" refers to a public official acting in relation to the selection process or contract execution. In this context "public official" includes World Bank staff and employees of other organizations taking or reviewing selection decisions.*

[2] *For the purpose of this sub-paragraph, "party" refers to a public official; the terms "benefit" and "obligation" relate to the selection process or contract execution; and the "act or omission" is intended to influence the selection process or contract execution.*

[3] *For the purpose of this sub-paragraph, "parties" refers to participants in the procurement or selection process (including public officials) attempting either themselves, or through another person or entity not participating in the procurement or selection process, to simulate competition or to establish prices at artificial, non-competitive levels, or are privy to each other's bid prices or other conditions.*

[4] For the purpose of this sub-paragraph, "party" refers to a participant in the selection process or contract execution.

(v) "obstructive practice" is

(aa) deliberately destroying, falsifying, altering, or concealing of evidence material to the investigation or making false statements to investigators in order to materially impede a Bank investigation into allegations of a corrupt, fraudulent, coercive, or collusive practice; and/or threatening, harassing, or intimidating any party to prevent it from disclosing its knowledge of matters relevant to the investigation or from pursuing the investigation, or

(bb) acts intended to materially impede the exercise of the Bank's inspection and audit rights;

(b) will reject a proposal for award if it determines that the consultant recommended for award or any of its personnel, or its agents, or its sub-consultants, sub-contractors, services providers, suppliers, and/or their employees, has, directly or indirectly, engaged in corrupt, fraudulent, collusive, coercive, or obstructive practices in competing for the contract in question;

(c) will declare misprocurement and cancel the portion of the Loan allocated to a contract if it determines at any time that representatives of the Borrower or of a recipient of any part of the proceeds of the Loan were engaged in corrupt, fraudulent, collusive, coercive, or obstructive practices during the selection process or the implementation of the contract in question, without the Borrower having taken timely and appropriate action satisfactory to the Bank to address such practices when they occur, including by failing to inform the Bank in a timely manner they knew of the practices;

(d) will sanction a firm or an individual at any time, in accordance with prevailing Bank's sanctions procedures[5], including by publicly declaring such firm or an ineligible, either indefinitely or for a stated period of time: (i) to be awarded a Bank-financed contract, and (ii) to be a nominated[6] sub-consultant, supplier, or service provider of an otherwise eligible firm being awarded a Bank-financed contract.

[5] A firm or an individual may be declared ineligible to be awarded a Bank-financed contract upon (i) completion of the Bank's sanctions proceedings as per its sanctions procedures, including inter alia: cross-debarment as agreed with other International Financial Institutions, including Multilateral Development Banks, and through the application of the World Bank Group corporate administrative procurement sanctions procedures for fraud and corruption; and (ii) as a result of temporary suspension or early temporary suspension in connection with an ongoing sanctions proceedings. See footnote 12 and paragraph 8 of Appendix 1 of these Guidelines.

[6] A nominated sub-consultant, supplier, or service provider is one which has been either (i) included by the consultant in its proposal because it brings specific and critical experience and know-how that are accounted for in the technical evaluation of the consultant's proposal for the particular services; or (ii) appointed by the Borrower.

Special Conditions of Contract

Number of GC Clause	Amendments of, and Supplements to, Clauses in the General Conditions of Contract
1.1(b) and 3.1	The Contract shall be construed in accordance with the laws of**:** Islamic Republic of Pakistan
4.1	**The language is:** English
6.1 and 6.2	**The addresses are:** Client : National Transmission & Despatch Company (NTDC) Attention : Chief Engineer/ Project Director Dasu Transmission Line Project NTDC 4th floor, PIA Tower, Lahore – Pakistan Phone No. 0092-42-99200695 Fax No. 0092-42-99200696 Email. cepdpmudasu@gmail.com Consultant: M/s Korea Electric Power Corporation (KEPCO). 55 Jeollyeok-ro, Naju, Jeonnam, 525-350, South Korea. Attention : Mr. Jaeheun Park (Senior Manager) Naju, South Korea Phone No. +82-61-345 5878 Fax No. +82-61-345 5879 E-mail: jaeheun.park@kepco.co.kr
8.1	The Lead Member on behalf of the JV is**: N/A**
9.1	**The Authorized Representatives are:** **For the Client:** Chief Engineer/ Project Director Dasu Transmission Line Project NTDC 4th floor, PIA Tower, Lahore – Pakistan Phone No. 0092-42-99200695 Fax No. 0092-42-99200696 Email. cepdpmudasu@gmail.com **For the Consultant:** Mr. Jaeheun Park (Project Manager) M/s Korea Electric Power Corporation (KEPCO). 55 Jeollyeok-ro, Naju, Jeonnam, 525-350, South Korea.

	Naju, South Korea Phone No. +82-61-345 5878 Fax No. +82-61-345 5879 E-mail: jaeheun.park@kepco.co.kr
11.1	**The effectiveness conditions are the following**: This Contract shall be effective on the date it is executed by Client and Consultant subsequent to the concurrence from the WB.***provided that*** the Services under this Contract shall commence on the date on which the Client requires such Services to be performed pursuant to the Client's notice to proceed.
12.1	**Termination of Contract for Failure to Become Effective:** **The time period shall be:** 90 days
13.1	**Commencement of Services:** **The number of days shall be:** Fifteen (15) days from the date of effectiveness of contract. Confirmation of Key Experts' availability to start the Assignment shall be submitted to the Client in writing as a written statement signed by each Key Expert.
14.1	**Expiration of Contract:** The time period of expiration/completion of services under this Contract shall be 12 months for Assignment A and 36 months for Assignment B plus 1 year for defect liability period. The scope of work under this Contract shall include (i) Assignment-A and (ii) Assignment-B. However, commencement of Assignment-B subject to (i) completion of Assignment-A, (ii) satisfactory performance of consultants and its approval by NTDC and World Bank, and (iii) the Client's notice (GCC. Clause: 11.1: Effectiveness of Contract) to the Consultant instructing the Consultant to begin carrying out the Services for Assignment-B. "Completion of Services" means completion of all services and obligations as defined in TOR and submissions as required in the same.
21 b.	**The Client reserves the right to determine on a case-by-case basis whether the Consultant should be disqualified from providing goods, works or non-consulting services due to a conflict of a nature described in Clause GCC 21.1.3** Yes

23.1	Limitation of the Consultant's Liability towards the Client: (a) Except in the case of gross negligence or willful misconduct on the part of the Consultant or on the part of any person or a firm acting on behalf of the Consultant in carrying out the Services, the Consultant, with respect to damage caused by the Consultant to the Client's property, shall not be liable to the Client: (i) for any consequential loss or damage; and (ii) for any direct loss or damage within two times the total value of the Contract; (b) This limitation of liability shall not (i) affect the Consultant's liability, if any, for damage to Third Parties caused by the Consultant or any person or firm acting on behalf of the Consultant in carrying out the Services; (ii) be construed as providing the Consultant with any limitation or exclusion from liability which is prohibited by the Applicable Law (c) The maximum liability shall be twice the total cost of Remuneration of the Consultants under the Contract. (d) The liability of the Consultants expires after one (1) year from the expiry of defect liability period.
24.1	**The insurance coverage against the risks shall be as follows:** a) Professional liability insurance, with a minimum coverage of twice the total remuneration of the Consultants in accordance with the terms of the Contract. b) Third Party motor vehicle liability insurance in respect of motor vehicles operated in Pakistan by the Consultant or its Personnel or any Sub-Consultants or their Personnel, with a minimum coverage as per applicable law of Rs. One (1.000) Million Pak Rupees; c) Third Party liability insurance with a minimum coverage as per applicable law of Rs. Three (3.000) Million Pak Rupees. d) Employer's liability and workers' compensation insurance in respect of the Personnel of the Consultants and of any Sub-Consultants, in accordance with the relevant provisions of the Applicable Law, as well as, with respect to such Personnel, any such life, health, accident, travel or other insurance as may be appropriate; and

	e) Insurance against loss of or damage to (i) equipment purchased in whole or in part with funds provided under this Contract, (ii) the Consultants' property used in the performance of the Services, and (iii) any documents prepared by the Consultants in the performance of the services. Above insurances shall be taken out for Assignment A and Assignment B separately;
25.1	The following is added: The Consultants; (i) shall keep accurate and systematic accounts and records in respect of the Services, in accordance with internationally accepted accounting principles and in such form and detailed as will clearly identify all relevant time charges and cost and the basis thereof, and (ii) shall permit the Client or its designated representatives periodically and up to one year from the expiration or termination of this Contract, to inspect the same and make copies thereof as well as to have them audited by auditors appointed by the Client.
27.1	Exceptions to propriety rights*: N/A*
27.2	The Consultant shall not use any of the **maps, reports and design documents/other documents** for purposes unrelated to this Contract without the prior written approval of the Client.
35.1 (f)	Provide the foreign Consultant with official security from Government of Pakistan when they are required to access to the Project site.
41.2	The ceiling in foreign currency or currencies is for: Assignment: A 964,620US$ and Tax 232, 288.47 US$ Assignment: B 3,208,520US$ and Tax 866,220.24 US$ and The ceiling in local currency is for: Assignment A: 47,209,500 PKR and Tax 2,913,520 PKR Assignment B: 114,518,800 PKR and Tax 10,259,008 PKR This cover all costs related to Assignment A and Assignment B. Although the payments shall be time based, in case of delays in the deliverables, would be reviewed by the Client before making the payments. If the delays are not found to be justified the Client may

	take appropriate action as provided under Contract.
42.3	**Price adjustment on the remuneration applies:** Payments for remuneration made in accordance with clause 42.3 shall be adjusted as follows: (1) Remuneration paid in foreign currency on the basis of the rates set forth in **Appendix C** shall be adjusted every 12 months (and, the first time, with effect for the remuneration earned in the 13th calendar month after the date of the Contract Effectiveness date) by applying the following formula: $R_f = R_{fo} \times \left[0.1 + 0.9 \frac{I_f}{I_{fo}} \right]$ where R_f is the adjusted remuneration; R_{fo} is the remuneration payable on the basis of the remuneration rates (**Appendix C**) in foreign currency; In case of US Dollars, , "I_f" is the annual earnings index (Relevant category for Wage for Engineers of Industry Plant) for Korea published by Government Agency of Korea for the year being examined. "I_{fo}" is the same index at the date of signing of the contract. (2) Remuneration paid in local currency pursuant to the rates set forth in **Appendix C** shall be adjusted every 12 months (and, for the first time, with effect for the remuneration earned in the 13th calendar month after the date of the Contract) by applying the following formula: $R_l = R_{lo} \times \left[0.1 + 0.9 \frac{I_l}{I_{lo}} \right]$ where R_l is the adjusted remuneration; R_{lo} is the remuneration payable on the basis of the remuneration rates (**Appendix C**) in local currency; I_l is the Consumer Price Index of General for Government Employees as published by Government of Pakistan Statistics Division, Federal Bureau of Statistics. I_{lo} is the same index at the date of signing of the contract.
42.3	The rates for Foreign & Local Personnel are set forth in Appendix C,

	For local personnel, remuneration for period of less than one month shall be calculated on hourly basis for the time spent by the Head Office (Home Office) staff and directly attributable to the Services (one hour being equivalent to $1/176^{th}$ of a month), and on calendar day basis for time spent by the Field Office staff. For foreign personnel, remuneration for period of less than one month shall be calculated on hourly basis for the time spent by the Head Office (Home Office) staff and directly attributable to the Services, and on daily basis for time spent at the Field Office. The hourly rate shall be equal to $1/176^{th}$ of the month and the daily rate shall be equal to $1/30^{th}$ of the month.
43.1 and 43.2	The client warrants that the Consultant, the Sub-Consultants and the Personal shall not be exempted from any taxes, duties, fees, levies and other impositions imposed, under the Applicable Law, on the Consultant, the Sub-Consultants and the Personnel however, the same shall be reimbursed to the Consultant, the Sub-Consultants and the Personnel for: **(**a) any payments whatsoever made to the Consultant, Sub-consultants and the Experts (other than nationals or permanent residents of the Client's country), in connection with the carrying out of the Services; (b) any equipment, materials and supplies brought into the Client's country by the Consultant or Sub-consultants for the purpose of carrying out the Services and which, after having been brought into such territories, will be subsequently withdrawn by them; (c) any equipment imported for the purpose of carrying out the Services and paid for out of funds provided by the Client and which is treated as property of the Client; (d) any property brought into the Client's country by the Consultant, any Sub-consultants or the Experts (other than nationals or permanent residents of the Client's country), or the eligible dependents of such experts for their personal use and which will subsequently be withdrawn by them upon their respective departure from the Client's country, provided that: (i) the Consultant, Sub-consultants and experts shall follow the usual customs procedures of the Client's country in importing property into the Client's country; and (ii) if the Consultant, Sub-consultants or Experts do not withdraw but dispose of any property in the Client's country upon which customs duties and taxes have been exempted,

	the Consultant, Sub-consultants or Experts, as the case may be, (I) shall bear such customs duties and taxes in conformity with the regulations of the Client's country, or (II) shall reimburse them to the Client if they were paid by the Client at the time the property in question was brought into the Client's country.
44.1	**The currencies of payment shall be the following:** Assignment A: US Dollars and Pak Rs. Assignment B: US Dollars and Pak Rs
45.1(a)	The following provisions shall apply to the advance payment and the advance payment bank guarantee: **For assignment A:** (1) An advance payment equal to ten percent (10%) of the total contract amount denominated in the currencies of the contract in the same proportion shall be made within **45 days** after the receipt of an advance payment bank guarantee from any schedule bank of Pakistan acceptable to the Client. (2) The advance payment will be set off by the Client in equal portions against the first three payments. (3) The advance payment bank guarantee shall be in the amount and in the currency (ies) of the advance payment. (4) The bank guarantee will be released when the advance payment has been fully set off. **For assignment B:** (1) An advance payment equal to ten percent (10%) of the total contract amount denominated in the currencies of the contract in the same proportion shall be made within **45 days** after the receipt of an advance payment bank guarantee from any schedule bank of Pakistan acceptable to the Client.

	(2) The advance payment will be set off by the Client in equal portions against the first six payments. (3) The advance payment bank guarantee shall be in the amount and in the currency (ies) of the advance payment. (5) The bank guarantee will be released when the advance payment has been fully set off.
45.1(e)	The accounts are for: **Assignment: A** for foreign currency: *ACCOUNT NAME : KOREA ELECTRIC POWER CORPORATION* *ACCOUNT MUMBER : 077-JCD-700001* *BENEFICIARY BANK : KOREA EXCHANGE BANK* *BANK ADDRESS : 181, 2-Ka, Eulji-Ro, Chung-Ku, Seoul, Korea* *BANK SWIFT : KOEXKRSE* *ATTENTION : KEPCO Branch* for local currency: *[To be later]* **Assignment: B** for foreign currency: *ACCOUNT NAME : KOREA ELECTRIC POWER CORPORATION* *ACCOUNT MUMBER : 077-JCD-700001* *BENEFICIARY BANK : KOREA EXCHANGE BANK* *BANK ADDRESS : 181, 2-Ka, Eulji-Ro, Chung-Ku, Seoul, Korea* *BANK SWIFT : KOEXKRSE* *ATTENTION : KEPCO Branch* for local currency: *[To be later]*.
46.1	The interest rate is: 4% per annum for delayed payment
49.	**Disputes shall be settled by arbitration in accordance with the following provisions:** 1. Selection of Arbitrators. Each dispute submitted by a Party to arbitration shall be heard by a sole arbitrator or an arbitration panel composed of three (3) arbitrators, in accordance with the following provisions: (a) Where the Parties agree that the dispute concerns a technical matter, they may agree to appoint a sole arbitrator or, failing agreement on the identity of such sole arbitrator

	within thirty (30) days after receipt by the other Party of the proposal of a name for such an appointment by the Party who initiated the proceedings, either Party may apply to the Federation Internationale des Ingenieurs-Conseil (FIDIC) of Lausanne, Switzerland for a list of not fewer than five (5) nominees and, on receipt of such list, the Parties shall alternately strike names therefrom, and the last remaining nominee on the list shall be the sole arbitrator for the matter in dispute. If the last remaining nominee has not been determined in this manner within sixty (60) days of the date of the list, FIDIC shall appoint, upon the request of either Party and from such list or otherwise, a sole arbitrator for the matter in dispute. (b) Where the Parties do not agree that the dispute concerns a technical matter, the Client and the Consultant shall each appoint one (1) arbitrator, and these two arbitrators shall jointly appoint a third arbitrator, who shall chair the arbitration panel. If the arbitrators named by the Parties do not succeed in appointing a third arbitrator within thirty (30) days after the latter of the two (2) arbitrators named by the Parties has been appointed, the third arbitrator shall, at the request of either Party, be appointed by International Chamber of Commerce (ICC). (c) If, in a dispute subject to paragraph (b) above, one Party fails to appoint its arbitrator within thirty (30) days after the other Party has appointed its arbitrator, the Party which has named an arbitrator may apply to the ICC to appoint a sole arbitrator for the matter in dispute, and the arbitrator appointed pursuant to such application shall be the sole arbitrator for that dispute.
	2. Rules of Procedure. Except as otherwise stated herein, arbitration proceedings shall be conducted in accordance with the rules of procedure for arbitration of the ICC as in force on the date of this Contract. 3. Substitute Arbitrators. If for any reason an arbitrator is unable to perform his/her function, a substitute shall be appointed in the same manner as the original arbitrator. 4. Nationality and Qualifications of Arbitrators. The sole arbitrator or the third arbitrator appointed pursuant to paragraphs 1(a) through 1(c) above shall be an internationally recognized legal or

	technical expert with extensive experience in relation to the matter in dispute and shall not be a national of the Consultant's home country or of the Government's country. For the purposes of this Clause, "home country" means any of: (a) the country of incorporation of the Consultant; or (b) the country in which the Consultant's principal place of business is located; or (c) the country of nationality of a majority of the Consultant's shareholders; or (d) the country of nationality of the Sub-consultants concerned, where the dispute involves a subcontract.
	5. Miscellaneous. In any arbitration proceeding hereunder: (a) proceedings shall, unless otherwise agreed by the Parties, be held in a place decided by ICC Court; (b) the English language shall be the official language for all purposes; and (c) the decision of the sole arbitrator or of a majority of the arbitrators (or of the third arbitrator if there is no such majority) shall be final and binding and shall be enforceable in any court of competent jurisdiction, and the Parties hereby waive any objections to or claims of immunity in respect of such enforcement.

Terms of Reference

1. Background:

a. National Transmission and Despatch Company (NTDC) has conducted a feasibility study for evacuation of power from 26 planned hydro power plants (HPPs) in the mountainous northern areas through International Consultants. The study covered the power system analysis, interconnection scheme for dispersal of power, preliminary survey, geotechnical assessment, environment study, etc. The same interconnection scheme has been adopted in the National Power System Expansion Plan of Pakistan up to 2030.

b. Dasu Hydropower Project (DHP) is one of the 26 hydropower plants included in the feasibility study. DHP is located on the Indus River with a total capacity of 4,320 MW and its first stage of 2,160 MW is expected to be online by June 2019. Interconnection scheme for evacuation of this power i.e. Dasu Hydropower Transmission Project (DHTP) is required to be completed six months before commercial operation date (COD) of first unit.

c. The feasibility study carried out to determine transmission system requirements for evacuation of power from proposed 26 hydro power projects has laid down following scope of interconnection scheme for dispersal of power from DHP to Islamabad North:

- Two double circuit 500 kV quad bundle T/line from Dasu to Mansehra.
- One double circuit 500 kV quad bundle T/line from Mansehra to Islamabad North
- Load flow study is based on 3400 MVA maximum rating per circuit.
- ACSR Martin Conductor has been proposed for transmission lines that need further optimization studies before final selection of the conductor.
- The approximate length of Dasu-Mansehra section is 140km and that of Mansehra-Islamabad section is 150km.

No feasibility studies have been done for the transmission line from Mansehra to Islamabad, which is likely to pass through urbanized and densely populated areas.

d. The preliminary level route alignment, survey & profile, geotechnical assessment, and environment study of transmission lines from Dasu to Mansehra have been carried out under the feasibility study. The topography of the project area (Dasu to Mansehra) is dominated by high mountains, varying in elevation from 1,000 to 3,200 m.a.s.l. which entails geological/environmental hazards of active fault and earthquake shaking, landslides (rock and/or soil), rock falls, river scouring and flooding, snow drifts and avalanches, etc. On the other hand, no feasibility studies have been done for the transmission line from Mansehra to Islamabad, which is likely to pass through urbanized and densely populated areas.

2. Objective:

The prime objective of the proposed Consulting Services is to support NTDC (the client) in project preparation studies and construction supervision of proposed transmission lines for evacuation of power from Dasu Hydro power plant. The consultancy services shall aim to achieve technically viable, optimal, and cost effective transmission line design, route, and construction plan including safeguards studies for the challenging Dasu Hydropower Transmission Line Project (DHTP)which envisages and counters the multifarious issues posed by the difficult and inaccessible terrain, high altitude, environment hazards, harsh weather, etc.

These works will be carried out according to international standards & good engineering practices which will form the basis for project appraisal by the Government of Pakistan, the World Bank, and/or other international financial institutions.

3. Scope of Consultancy Services.

The consulting services are divided into following two assignments.

Assignment-A : It covers the route alignment, preliminary survey, preparation of tender level engineering design, bidding documents and support of the client during tendering. This assignment will be completed in 12 months.

Assignment-B: It covers Design Review, construction supervision and contract management during Engineering, Procurement, and Construction (EPC) execution of DHTP i.e. Two 500 kV double circuit transmission lines from Dasu to Islamabad via Mansehra . This assignment will be completed in 48 months including 12 months of defect liability period.

For clarity and ease detailed scope of each Assignment is given below:

Assignment A: Preparation of Tender Level Engineering Design, Tender Document, and Support to Client during Tendering;

The Assignment A is likely to consist of, but not limited to, the following:

- Task A1: Review of Feasibility Study & Use of Existing Information/Data
- Task A2: Transmission line Route Alignment and Survey
- Task A3: Preparation of Tender Level Engineering Design
- Task A4: Preparation of Tender Documents
- Task A5: Tendering and Contracting
- Task A6: Technical Training.

Task A1: Review of Feasibility Study &Use of Existing Information.

The Consultant will review all of the existing information, data, reports, maps, studies conducted so far including feasibility study for evacuation of power from 26 planned hydro power plants (HPPs) and make use of physical data of the site and available information, data, literature regarding transmission line projects carried out in similar terrain, altitude and weather conditions worldwide in preparation & performance of the Consulting Services.

Task A2: Transmission line Route Alignment and Survey

The Consultant would carry out route alignment, geotechnical assessment, estimation of BOQ including tender level foundation design and tower protection (including special foundation viz. pile foundation requirements). Functional requirements for survey are attached as Annex 'I'. This task inter-alia shall include but not limited to the following:

(i) Review of the route alignments of transmission line from Dasu-Mansehra (already carried out under the feasibility study) and re-alignment of routes wherever required or developing alternate route alignments, if necessary.

(ii) Route alignment for Mansehra – Islamabad section.

(iii) Route alignment task inter-alia will include but not limited to the following:

   (a) Identification of three alternative route alignments & selection of optimized route alignment. This shall be done through walk over survey, using low resolution satellite imageries and Survey of Pakistan maps. The output shall be in the form of final route alignment on digitized topographical map with latest details/features up to 8 km on both sides of selected route alignment.
   (b) Digital terrain modeling of line route using high resolution satellite imageries.
   (c) Associated field verifications.

(iv) Preparation of survey reports including estimation of Bill of Quantities (BOQ), identification and explanation of route constraints, and infrastructure details available en-route and required to be created for construction/execution of the Project.

Task A3: Preparation of Tender Level Engineering Design

This assignment is likely to consist, but not limited, to the following sub-tasks:

- Sub-Task A3.1: Collection of Data and Investigations
- Sub-Task A3.2: Design Criteria
- Sub-Task A3.3: Project Implementation Planning and Procurement Packaging
- Sub-Task A3.4: Preparation of BOQ and Costing

- Sub-Task A3.5: Environmental Assessment, Management Plan, EIAs etc.
- Sub-Task A3.6: Social Impact Assessment and Beneficiary Participation Aspects and Preparation of Resettlement Action Plan

Sub-Task A3.1: Collection of Data and Investigations

The Consultant shall collect and analyze weather/climate data and estimate weather loading factors for design of transmission line. Based on meteorological data, altitude & terrain; the Consultant shall determine various loading zones/develop maps along the transmission line route which shall pass through terrain with varying elevation and climatic conditions, to optimize design of various transmission line elements/components.

The Consultants shall indicate the need for further investigations and surveys at this stage of the studies in order to be able to fully establish the engineering design of the project. The Consultants shall be responsible for all surveys and investigations required to prepare the tender level design.

The Consultants would be responsible for all investigation required for the Assignment: A. These investigations should be mentioned in the methodology as well as in the financial proposals.

Sub-Task A3.2: Design Criteria

The Consultant shall perform necessary engineering & design to meet the objective/scope of the Project. The Consultant's design shall include but not limited to providing: a) design criteria, calculations, drawings, specifications for steel towers, conductor, Extra High Strength (EHS) shield wire, optical ground wire (OPGW), dampers, disc insulators, etc., including accessories and associated hardware/equipment and grounding materials; b) foundation design; c) construction and installation methods; and d) other documentation necessary to complete the Consulting Services. Functional requirements for transmission line design are attached as Annex 'II' as a guideline. This task inter-alia shall include but not limited to the following

(i) Selection of design parameters/criteria for the transmission line and preparation of design document as per IEEE Standard 1724: Guide for the Preparation of a Transmission Line Design Criteria Document. The Consultant shall perform and document necessary investigations, studies, design simulation, etc., to support the selection of design parameters/criteria.

(ii) Preparation of engineering and technical specifications and engineering drawings for transmission line material and construction. The consultant shall carry out technical analysis to determine and demonstrate that the specifications of technical parameters chosen are realistic and optimal.

(iii) Based on loading zones, line route, ground profile & contours, terrain and other relevant parameters; develop families of towers and determine their suitability for the Project.

(iv) Select one or more suitable conductors for the transmission line as per guidelines given in Annex-II.

(v) Develop specifications and work requirements for EPC execution of the project viz. civil works, erection, stringing, testing & commissioning of transmission line.

The Design Criteria shall be detailed enough to serve as basis for development of construction design and drawings by the Contractors later on during Assignment-B of the project.

Sub-Task A3.3: Project Implementation Planning and Procurement Packaging

The Consultant shall prepare a critical path schedule for the Project, starting with the tendering and contracting for International Competitive Bidding (ICB) and the construction period up to final completion, using appropriate project scheduling software. At a minimum, the schedule shall:

(i) demonstrate the sequence and interdependency of all work and identify activity durations and resources;
(ii) identify all key permitting, environmental commitments and assurances, and construction activity limitation including time for the submission, review and acceptance of submittals;
(iii) identify all critical and significant material or equipment deliveries; and
(iv) illustrate available float for all major schedule activities, etc., considering weather conditions.

The schedule shall be used as basis for the calculation of interest during construction and monitoring construction activities.

Considering the availability and performance of the contractors, and the construction industry as a whole, size and nature of works based on the detailed designs, and available technology, propose suitable construction scenarios for the project. In the context, identify packages of works which are to be implemented on the basis of International Competitive Bidding (ICB), National Competitive Bidding (NCB) or through other appropriate methods.

For each type of package identified above outline the basis for engineering, supervision and administration arrangements including implementation arrangements Environment Management Plan (EMP) and Resettlement Action Plan (RAP) etc.

Prepare a workable and cost effective overall implementation plan and plans for each component of the project as well as operation plan for the project consulting services for supervision and administration, and monitoring and evaluation of the project impact; civil works; equipment/plant, vehicles and other goods required for the project implementation.

Implementation plan inter alia should indicate the activity-wise phasing of construction. Quantitative information on the phasing of material and labor requirements during construction should be specified. The physical specifications of phasing given in this plan should be co-related with the capital cost requirements and cash flow. To the extent possible the plan should indicate the targeted levels for pouring of concrete and erection of equipment structures per month. Steps that could be taken to reduce the period of construction should also be indicated. Identify the critical factors that could affect construction/implementation of the project and develop a detailed management plan and recouping strategies for such factors.

Propose project implementation arrangements including institutional structure clearly defining the role of the Client, and its various entities, Ministry of Water and Power (MOWP), Finance, Planning, Potential need for a special Project Management Unit (PMU), field officers, contractors, and supervisory engineers. Layout and appropriate work flow using the proposed institutional setup such as technical approvals, approvals for technical design, approval for changes in technical designs during construction, for effective implementation, measurements and verification of works undertaken, payment procedure, flow of funds, etc. Also prepare a staffing plan for the PMU and other staff required for the project implementation;

Sub-Task A3.4: Preparation of BOQ and Costing

Based on the identified requirements of the various project components, the Consultant shall prepare a detailed BOQ for the project including plant, civil works, installation services, security services, vehicles, computers and softwares to be provided to NTDC for its capacity building etc.

The Consultant shall prepare a detailed cost estimate for each stage separately including: mobilization, infrastructure, surveys, investigations, civil works, Equipment& Plant (manufacturing/purchase, transport, erection, installation, testing, commissioning, etc.), environmental, land acquisition and resettlement (if any), taxes, import duties, administration and legal costs, engineering and supervision, insurance's, physical contingencies, price contingencies, etc.

The construction costs shall be determined on the basis of a quantity survey and adopting unit costs calculated by using material, equipment and labor costs valid for the project region. The cost figures established in this manner shall be broken down into local and foreign costs. The capability of the local industry to carry out civil works, installation portions of the project shall be investigated in order to arrive at a realistic estimate of the local component.

The major assumptions forming the basis of the cost estimates should be listed in the text along with presentation in the relevant format. The major item wise details of cost estimates should be given. It is desirable that following information should also be presented in the text:

a. Reference price level of estimates;
b. Currencies and exchange rates and their reference period ;
c. Level of contingencies;
d. The level of inventories of materials, spares, etc.;
e. Scaling factors used for scaling up/down cost of any major equipment/item;
f. Status of engineering and design on the basis of which the estimates have been framed;
g. Rates and quantities of items on the basis of value analysis;
h. Debt equity ratio and interest rates; and
i. Customs duty, excise duty, and sales tax, etc.

The cost on account of social and environmental safeguards, including implementation of the proposed environmental and social management plans etc.,should be shown in the capital cost estimates as a separate item.

Sub-Task A3.5: Environmental Assessment, Management Plan, EIA etc.

i) The Consultant will carry out the site specific Environmental Impact Assessments (EIAs) as required under the Pakistani Laws. The project is to be implemented following Pakistani Environmental Regulations and the World Bank Guidelines and Operational Policies (Operational Policies 4.01) and guidelines of other international financial institutions. Therefore, the Consultant will be responsible for preparing and updating these documents as needed to meet such requirements for environmental assessment of the Project as well as cumulative environmental impacts of the Project and development programs in the area. The details for this task are provided in Annex-III

ii) The Consultant will gather necessary data and information, and prepare all the environment related documents for all works to be implemented under the Project that may be necessary for getting location, construction, and other permits for constructing works, and carrying out activities for obtaining financing from the World Bank and other financial institutions.

iii) The Consultant will prepare EMP for any adverse impact of the construction activities, and operation of the Project may have on the environment. The EMP will consist of mitigation measures, monitoring program and institutional development/strengthening program for implementation of the EMP. Prepare cost estimates for implementation of the EMP, scope of work, terms of reference, and a plan of how various mitigating measures will be implemented either through modification of construction contracts for project facilities, or through additional works for which the Consultants will prepare designs, or through technical assistance

and training which the Consultant will carry out for operation of the project facilities.

Sub-Task A3.6: Social Impact Assessment and Beneficiary Participation Aspects and Preparation of Resettlement Action Plan

The Consultant will carry out a social assessment as needed by the GOP and the World Bank Guidelines for various project activities/works. Based on findings of the social assessment, if land or property is to be acquired permanently or temporarily and/or people are affected in a significant way, the Consultant will determine, on the recommendations of the Social Assessment, the mitigation approach and planning instruments necessary before the World Bank's appraisal of the Project. These possible instruments include Social Impact Management Framework, Resettlement Policy Framework, and RAP. These instruments need to be developed according to the timeframe and arrangements of the project. For the works to be included in the first year's contract for which basic designs would be prepared, a full RAP would be necessary. In this context major activities to be carried out are given in Annex-III**.**

Task A4: Preparation of Tender Documents

The preparation of the Tender Documents will begin after the acceptance of the Design Report and the instruction of the client and World Bank to continue the services. The tender documents shall be based on the Bank's standard documents.

The following parts of the tender documents shall be prepared by the Consultants for each lot separately:

- General conditions of contract
- Special conditions of contract
- Bid and Appendices
- Special provisions
- Specifications
- Bills of Quantities and equipment lists
- Drawings

Furthermore, an independent international expert for construction/erection contracts of transmission lines shall be nominated in the technical offer, who will check, scrutinize and, if required, revise the tender documents in consultation with the Consultants. Furthermore, the expert shall analyse and give professional advice on all matters concerning insurances to be included in all contracts in order to limit the risks involved with the construction design and execution of the works and to make sure that all liabilities/ responsibilities are clearly defined throughout all execution stages.

The Consultant shall particularly: a) Prepare, review or revise to a high professional standard the technical specifications of the Project, ensuring that they are in line with the Client's and World Bank's rules and permit submission of clear, complete and competitive tenders; b) Ensure that the environmental performance criteria, as set out by the environmental assessments and approved by the Panel of Experts are met by the specifications;

Assist the Client in following the procedures for international tendering (including pre-qualification, if required) and in complying with World Bank's procurement rules;

Task A5: Tendering & Contracting

The Consultant will provide as and when required basic professional services to assist the Client in tendering for the project, the evaluation of offers and the award of contracts. This comprises, but is not limited to:

i. Assist the Client in setting up the logistics for the tender process and in any pre-tendering meeting;
ii. Assist the Client in any clarifications requested by the prospective bidder/suppliers (drafting replies to written requests for additional information);
iii. Advise the Client during tender evaluation and assist in the preparation of tender evaluation reports in a format suitable for submission to World Bank, and make recommendations concerning the overall technical adequacy and reasonableness of the proposed contract price;
iv. Provide professional services as required to review the winning tenderer's drawings and detailed designs for compliance with all specified contract requirements and technical specifications, industry standards and good engineering practise and
v. Assist the Client in the preparation of contract documents, the contract negotiations with the winning tenderer and the award of the Contracts.

The consultants will provide the above services during tendering and contracting stage.

Employer and World Bank reserve the right to stop the implementation of the project after the evaluation of the offers of the contractors.

Task A6: Technical Training

The Consultant will also provide technical assistance and training to at least six personnel of the Client. The Consultant shall develop a comprehensive training program that enables the Client's engineers (trainees) to perform the Design & engineering services for similar projects of the Client in future. At minimum the training program should cover:

i. Preparation of basic design of T/line as per task A3 according to international standards.

ii. Training for using the softwares employed by the consultant for the Project to enable the Client to verify the results of the computer aided design software.

iii. Preparation of profile and plan, PLS tower model for the towers using PLS-CADD and provide one licensed copy of PLS-CADD software with 3 years validity to the Client.

iv. Computation of transverse, longitudinal and vertical wind loads for various loading cases and development of loading trees for Conductors, Overhead Shield wire & Insulators for the design of towers.

v. Complete description of GPS equipment regarding the use of it for the GPS Survey work and getting the GIS Co-ordinates to and from the PC.

vi. Integrating the GIS Co-ordinates in the GIS Software for 500kV T/Lines.

vii. Training for collecting co-ordinates using GPS receiver methodology used by the surveyor for collecting co-ordinates. The training shall also cover database creation, modification & addition in database, using software, digitization of map, upload & download techniques & all other relevant features of software.

viii. Satellite Image Processing and Digital Terrain Modeling.

ix. Preparation of Project cost estimate and implementation plan.

x. Any other training relevant to the scope of work of the Project.

Based on the training program developed and agreed with the Client, the training shall be arranged at the works of the Consultant/Client, the project sites or any other appropriate location/ training institute, etc.

Assignment-B: Design Review, Construction Supervision & Contract Management

The Commencement of Consultancy Services for Assignment-B will be contingent upon satisfactory performance by the Consultant during Assignment-A and clearance and approval from the World Bank and the GOP to proceed with the Assignment-B. The tasks and activities in Assignment-B shall include, but not limited to, the following:

- Task B1: Design Review
- Task B2: Contracts Management
- Task B3: Construction Supervision
- Task B4: Support in implementation of EMP and RAP.

Task B1: Design Review

The assignment is likely to consist of, but not limited to, the following tasks:

- Sub-Task B1.1: Review and Approval of Contractor's Design.
- Sub-Task B1.2: As Built Drawings.
- Sub-Task B1.3: Punch List.

Sub-Task B1.1: Review and Approval of Contractor's Design

The Consultant shall review and approve the Turnkey Contractor's basic and detailed design and drawings, technical data/schedules, line routes, plan and profile drawings, construction methodologies and procedures, design of tower foundation & protections, sag & tension charts, equipment and material specifications/ drawings, operation and maintenance manuals, as built data & drawings, etc., covering all the aspects of the construction/installation & commissioning of DHTP to ensure conformance with applicable codes, standards, manuals, work requirements and provisions of the contract.

If the Consultant is not satisfied with the design proposed by the contractor(s), may recommend changes necessary to meet the project overall requirements.

Sub-Task B1.2: As Built Drawings

The Consultants will review, comment and finally approve the drawings, which include revisions made during the construction. These drawings will be prepared by Contractor upon completion of the project. The As-built documents will include the following.

- As-built route alignment, profile & plan table drawings, structure list of transmission lines prepared and revised by the contractor.
- As-built drawings for tower foundations and protective structures.

The As-built drawings and documents shall be prepared in four set of hard and soft copies, three of them to be handed over to the Client, and another one for the Consultants files. The Consultant will store one complete set of the as-built drawings and documents as a hard copy and a softcopy in a safe manner in his files.

Sub-Task B1.3: Punch List

Towards the end of the construction and installation of transmission line the Consultant will prepare a punch list recording all works and installations which may require completion or remedial measures for incorrect execution/defects, and will monitor the completion of all incomplete or defect works and installations to ensure that a fully completed project of high quality is achieved.

Task B2: Contracts Management

The assignment is likely to consist, but not limited to the, following tasks:

- Sub-Task B2.1: Monitoring of Project Progress
- Sub-Task B2.2: Quality Assurance
- Sub-Task B2.3: Monitoring of Project Budgets
- Sub-Task B2.4: Development of a Financial Management System.
- Sub-Task B2.5: Project Coordination and Communication
- Sub-Task B2.6: Review and Reporting

Sub-Task B2.1: Monitoring of Project Progress

The Consultant will assist the Client in the preparation of the reports required for the reporting to World Bank, insofar as they are directly linked to progress in implementing and in administrating the Project. Upon request, the Consultant will provide those parts of the report to a Panel of Experts (PoE), which is relevant for the assessment of technical issues and monitoring of progress in project implementation.

The Consultants will prepare and update the procurement, implementation, delivery and disbursement schedules for the Contractors employed on site, to be used as basis for reporting on the progress of project implementation;

The Consultants will prepare appropriate progress reports and cost reports and assist the Client in the preparation of those reports necessary to comply with the reporting requirements of World Bank.

Sub-Task B2.2: Quality Assurance

The Contractors will establish systems of Quality Monitoring (QM), Control (QC) and Assurance (QA). The Consultant will check the implementation and performance of these QM, QC and QA procedures and realize quarterly QM/QA audits of the Contractor. Audit reports will be prepared and submitted to Clients for information and approval.

At the start of the supervision services the Consultant will prepare a Project Manual in accordance with ISO 9000 in order to establish an efficient Quality Management System for this Project. This manual will be binding to all parties involved with the execution of the project.

Sub-Task B2.3: Monitoring of Project Budgets

The Consultant will maintain a computerized cost control system within the Project Management System (PMS) for budget monitoring and cash flow forecasting. The system will be linked with the MS PROJECT CPM network to provide information on project costs for the periodic reports required by Client.

Payments to be made to the Contractor will either be on the basis of monthly invoicing for the construction works and will be linked to relevant milestones for delivery of plants or a combination of various construction/installation activities.

Sub-Task B2.4: Development of a Financial Management System

The Consultants will assess the financial management capacity of the institutions proposed for implementation of the project. Develop a computerized project and financial management system which is appropriate for the size and scope of the project. The computerized project financial management system should be capable of generating periodic progress reports required for project management. The Financial Management System would be documented in a Project Financial Management Manual which would include the following: (a) flow of funds process; (b) accounting system including internal controls; (c) financial and accounting policies; (d) planning and budgeting system; (e) financial forecasting system; (f) procurement and control of administration monitoring system; (g) financial reporting (quarterly reports and annual financial statements); (h) auditing arrangements; and (i) Organization, staffing, training, and technical support for financial management.

Sub-Task B2.5: Project Coordination and Communication

The Consultant will establish and coordinate communication between the various entities involved in the Project as there are:

- Employer/Owner
- Consultants head office
- Consultant's supervision team
- Consultants EMP and RAP team
- Contractors
- Government Authorities

The Consultant will be in charge of the coordination of all communication of the project team, i.e. defining the responsibilities of each project unit. A database comprising the contact data, such as telephone numbers, e-mail addresses, fax numbers etc. of each project party and their concurring key personnel will be established, regularly updated and made available to all project parties.

The Consultant will organize and document regular project meetings with all involved parties in order to assure a good working communication between the project parties and to assure that everyone works based on the latest information. The Consultant will prepare respective Minutes of Meeting of all meetings and subsequent agendas which will be distributed to the parties involved. All protocols, Minutes of Meetings, agendas and so on will be filed according to the filing system introduced through the Quality Management System.

Sub-Task B2.6: Review and Reporting

At the start and during the course of construction, the Contractor will have to submit details working methods, equipment, plant and temporary works design details in shop drawing before the start of construction /installation of the relevant part of the works. These revised methods and equipment/plant will be checked for conformance with the design concepts, the technical specifications and applicable national and international standards to ensure quality and safe working methods. The Consultant will check the Contractor operations against the approved revised method statements.

Task B3: Construction Supervision

The assignment is likely to consistof, but not limited to, the following sub-tasks:

- Sub-Task B3.1: Mobilization of Contractors
- Sub-Task B3.2: Site Instructions and Co-ordination with the Contractor
- Sub-Task B3.3: Supervision of the Construction/InstallationWork
- Sub-Task B3.4: Workshop Inspection and Acceptance
- Sub-Task B3.5: Testing & Commissioning
- Sub-Task B3.6: Transmission LineOperation & Maintenance
- Sub-Task B3.7: Know-How Transfer and Training
- Sub-Task B3.8: Environmental and Safety Issues

The Consultants will be responsible for construction supervision of all contracts. They will be designated as the Engineer in the civil works, goods, plants,and other equipment supply and installation works, and will be responsible for inspection and supervision of the construction works, installation of equipment, and testing of construction material, in order to ensure that the works are implemented and goods supplied in accordance with the deigns, specifications, and terms, and conditions of the relevant contracts and standards. As mentioned above, the Consultants shall ensure that procurement of plant, services, civil works, and installationare in accordance with the World Bank Polices and guidelines; and the contracts are signed and managed properly including any changes or variation orders during implementation.

Sub-Task B3.1: Mobilization of Contractors

The Consultant will provide highly qualified and experienced staff for the supervision of the construction activities, which will be responsible for:

- Implementing all the contract and inspection procedures to ensure that the design, specifications, drawings, and general contractual requirements are being met.
- Notifying the Contractor when he is failing in his obligations, and agreeing on necessary remedies.
- Maintaining adequate records.
- Quality control & assurance site testing.
- Checking the setting out of the work.

- Checking and approving the Contractor's work schedule.
- Checking and approving the Contractor’s construction material.
- Checking that the Contractor is providing sufficient qualified personnel and adequate construction equipment.
- Verification of the work performed and supply delivered to the sites.

Sub-Task B3.2: Site Instructions and Co-ordination with the Contractor

The Consultant will hold regular site and co-ordination meetings with the Contractor, to review the progress of the works, identify possible difficulties in the construction procedures and to discuss and agree the necessary remedial measures. The Consultant will establish, together with the Contractor, the necessary priorities at site and keep records of the issues discussed in the meetings, and informing the Client on any action to be taken.

Depending on the situation, the Consultant will issue either written or verbal site instructions. Verbal site instructions will be given, whenever quick decisions are required in order not to delay the ongoing works, to prevent ongoing works from being performed incorrectly or in an unsafe manner. This may particularly be the case during excavation and support works in critical ground conditions, concreting works and equipment erection and embedment works. Verbal site instructions will always require a written confirmation afterwards.

The Consultant will discuss and agree with the Contractor on the modus of provisional measurements and estimates for the Contractor’s monthly invoicing, with the aim to shorten the time for invoice checking and to avoid overpayments. However, exact figures will be established for the final measurements.

***Sub-Task B3.3:* Supervision of the Construction/Installation Work**

The Consultant shall supervise the construction of the DHTP and shall make a diligent effort to ensure the expeditious and economical construction thereof in accordance with the Plans and Specifications and the terms of the contract or contracts and ensure that all specified environmental criteria are followed. The Consultant shall carefully inspect all materials and equipment/plants prior to their incorporation in the Project and shall promptly reject those not in compliance with the Specifications. The Consultant shall also supervise and inspect the incorporation of the materials in the Project and the workmanship with which such materials are incorporated. The Consultant, as representative of the Employer/Client, shall have sole responsibility for requiring the Contractor to perform the Construction/installation work in accordance with its terms and the Plans and Specifications; and, in performing the duties incident to such responsibility, the Consultant shall issue to the Contractor such directives and impose such restrictions as may be required to obtain reasonable and proper compliance by the Contractor with

the terms of the Construction Contract and the Plans and Specifications, in construction of the DHTP.

The Consultant shall maintain at the site of the Project during the entire period of construction a competent resident Consultant with full authority to act for the Consultant, unless specifically directed otherwise by the Employer/Client in writing. When necessary to assure adequate inspection, one or more competent inspectors shall also be maintained when construction/installation or corrective work is being performed. The Consultant shall report, in writing, defects in workmanship or materials to the Contractor and the Employer/Client and shall instruct the Contractor to correct such defects immediately, in accordance with the terms of the Construction Contract. A resident Consultant shall be present during the final inspection of completed construction.

Sub-Task B3.4: Workshop Inspection and Acceptance

As far as practicable, quality of materials, workmanship and performance of the major items of the work and equipment to be furnished under the Contract shall be inspected at the places of manufacture. The Consultant will carry out the joint inspections with the Client in the Contractor workshops, assigning experts for these services who are conversant with the job and cognizant in the relevant codes and standards as well as acquainted with international practices for testing. The compliance of the materials, workmanship and erection methods with the specifications will be checked and any deviations will be recorded. Decisions will be made in co-ordination with the Client whether to accept or reject the deviations.

In addition to the quality and production tests, shop assembly work and tests shall be made to check measurements, fittings and functioning. As far as possible, equipment to be furnished shall be shop assembled to a status sufficient to prove that the design and workmanship have been executed in accordance with the specifications, that the delivery is complete, and that no work remains to be done at site which reasonably can or should be done in the shop. Where applicable, each item of the equipment shall be assembled completely prior to shipment.

Before shipment, tests will be performed by the Contractors as may be required to demonstrate the adequacy of the equipment and its component parts to the satisfaction of the specifications. All tests will be simulated to normal operating conditions. Tests at the manufacturers shop will be performed in accordance with the requirements of the specifications and approved test procedure. The Consultant will witness inspections and tests at the manufacturer's shops on major equipment. He will prepare the respective Inspection Certificates and Inspection Reports for above activities.

Sub-Task B3.5: **Testing &Commissioning.**

The Consultant will assist the Client in monitoring and reviewing testing & commissioning and taking over of Dasu Hydropower Transmission line upon completion.

This shall include the inspection of civil works, proper installation of plant and recommendation to the Client for issuance or not of the acceptance certificate.

The Consultant will further review test and initial operation procedures and recommend acceptance of or changes to them. The Consultant will particularly witness tests (electrical, mechanical, instrumentation and civil works, if applicable) of contractors to verify that components and systems are ready for initial operation; provide assistance to the Client personnel in the initial operation and maintain close contact with the contractors' personnel to resolve problems which may arise during initial operation; review the training programme and plant operation manuals, prepared by the contractors in English, for adequacy; witness and review performance testing to all equipment and systems in order to verify test results.

The Consultant will assist the Client in performing final inspection and testing of facilities and in determining final acceptability of them, and provide the same assistance with regard to the civil works, as far as applicable.

Sub-Task B3.6: Transmission Line Operation & Maintenance.

The Consultant will recommend to the Client the procedure to ensure a satisfactory management and operation of the project during the entire life time. Advise the Client on the staffing pattern and organisation of the transmission line operation & maintenance team.

The Consultant will advise the Client to establish maintenance, repair works and operation procedures to ensure the safe and reliable operation of the transmission line during its life time, including the setting-up of maintenance services. The Consultant will advise the Client on any additional training programme for the Client's staff which are necessary to ensure successful operation of the DHTP and outline those requirements (i.e. prepare terms of reference).

Sub-Task B3.7: Know-How Transfer and Training

The Consultant will assist the Client in establishing suitable procedures and in the capacity building for contract management, monitoring and administration, including monitoring of costs, accounting and certification of contractors' invoices and statements; detailed engineering activities; environmental management procedures; quality assurance and effective communication lines between the Client and the contractors operating on site.

The Consultant will provide and/or recommend training to this effect. A training program for the on-the-job training will be established for the Client personnel integrated in the construction management team.

Sub-Task B3.8: Environmental and Safety Issues

The Consultant will make sure that appropriate measures are taken to ensure adequate health, safety and environmental standards during construction and commissioning and

will monitor the compliance of all construction activities with the EMP and RAP. Furthermore the Consultant shall elaborate the detailed and final EMP for the operation stage of the project and shall prepare and submit the respective report.

Task B4: Support in Implementation of EMP and RAP

The Consultants would provide support in the implementation of the EMP. The EMP activities would be incorporated in the main construction contracts to the extent possible. The EMP activities which cannot be included in the main construction contract would be implemented through additional construction contracts, management, institutional, or technical assistance. The Consultants would carry out the designs of such program and help, monitor and supervise their implementation.

Provide support in implementation of RAP activities during the project implementation, including support to purchase of properties, for that purpose, preparation of documentation to help in negotiations with the beneficiaries, in obtaining of local permits, etc. This may also include identification of alternative sites for resettling people and related assets and cultural properties, development of the sites for resettlement, including planning, infrastructure, utilities, and replacement houses etc.

The Consultants will write Terms of Reference (TORs) for any additional work that have to be carried out under the project for which additional services will be required and identified during project implementation.

4. Completion Period

Assignment-A will be completed in 12 months. Assignment-B will be completed in 48 months including Defect Liability Period of 12 months after issuance of taking over certificate and completion of work by the Contractors.

5. Implementation Arrangement

The Consultant will work closely with office of Chief Engineer Design NTDC, Dasu PMU and coordinate with other relevant units of NTDC, local administration and relevant ministries and agencies. The Consultant's team leader will be the principal contact and will be expected to be readily available during the assignment period. The Consultant will be responsible for all aspects of performance of services set forth in the TORs. The Client will be responsible for providing the existing data and information and supporting arrangement for the necessary field investigations

After the review of feasibility study/inception stage the Consultants shall prepare a detailed schedule and task-flow diagram, which depicts the interrelationship of various tasks in the assignment which lead to the completion works and mechanism of coordination with the client and other related entities. This will be kept updated throughout the Project duration.

5.1 Method Statements:

Prior to commencing any section of the work, the Consultant shall submit method statement in accordance with the requirement of the Consulting Services. When requested by the Client, the Consultant shall provide additional method statements related to specific item of work.

5.2 Progress Meetings:

The Consultant shall hold periodic progress meetings with the Client, at a minimum of monthly and following receipt of the Client comments on the submission of the Consultant. Additional meetings shall be scheduled as required by the Consultant's design. The intent of these meetings will be for the Client to provide input and to discuss options for addressing the Client comments. The Consultant shall fully cooperate with the Client in scheduling and attending such meetings as requested by the Client. These meetings shall be held in Lahore. The Client will be responsible to take meeting minutes during each of these meetings. Minutes will be distributed to participants for review and comment.

5.3 Monthly Progress Reports:

The Consultant shall furnish the Client with a written Monthly Progress Report that summarizes all aspects of the completed month and cumulative work progress. The objectives of the Monthly Progress Report are to:

(i) Provide a reliable and readily accessible summary record of the project activities and progress during the just completed month.
(ii) Provide a detailed description of all work actually completed to date and revision to the project schedule required, which shall reflect changes in the critical path since the date of the last revision.
(iii) Identify issues and problems requiring action by the Client or Consultant, including issues of conflicting priorities.
(iv) Provide a forecast of the work to be accomplished in the next month.
(v) Provide information to help substantiate Consultant's pay request.

5.4 Submittal Protocol:

No later than thirty (30) days after the Contract Date**,** the Consultant shall submit a submittal protocol for the Consulting Services. The submittal protocol shall identify the submittal packages to be prepared by the Consultant, including a detailed listing of the content, the expected dates of the submittals, number of copies, and distribution of the submittals by the Consultant based on distribution information provided by the Client. The submittal protocol shall include the time allowed for the Client's review, which at a minimum shall be twenty (20) days. For large or complex submittals, the Client may

require a submittal review period longer and the Consultant shall coordinate with the Client inclusion of these review periods in the submittal protocol. The submittal protocol shall avoid the simultaneous submittal of a large number of submittals for the concurrent Client review.

5.5 Consultant's Obligations:

The Consultant shall provide submittals for review consistent with the submittal dates. The Consultant shall acknowledge that the Client's review will often involve input from, or consultation with, a number of individuals. Therefore, should submittal dates to the Client be delayed, the Consultant shall provide prompt notice to the Client of the delay. In no case shall this notice be given less than ten (10) calendar days prior to the scheduled submittal date for that submittal. The submittals shall identify any proposed change to the requirements, or the design concept, project delivery approach, or the project schedule provided in the consultant's proposal, accompanied by the rationale behind the proposed change. No changes shall be implemented without Client's acceptance. Such acceptance shall not, however, relieve the Consultant of any of its obligations under the contract.

5.6 Form of Submittals:

Each submittal shall be transmitted electronically and in hard copy, unless otherwise required by other sections of the contract, with a cover letter. Unless otherwise specified in the contract, the Consultant shall prepare up to Four (04) hard & soft copies of each submittal for distribution. The Consultant is responsible for distribution of submittals to reviewers based on distribution information provided by the Client. Electronic submittals shall be in the original file format. The Consultant is responsible for the accuracy and completeness of the information submitted.

The Consultant shall make submittals far enough in advance of subsequent activities to allow time for reviews, consultations with other entities, for securing necessary acceptance, for possible revisions and re-submittals. The Client intends to process Consultant's submittals as quickly as practical.

5.7 Client's Review:

The Client will review submittals for consistency with the design concept presented in the Consultant's proposal. The primary purpose of the Client's review is to satisfy itself that the submittals generally conform to the intent of the contract. The Client's review shall not relieve the Consultant of the sole risk and responsibility for all defects, errors or omissions, or of sole responsibility for meeting all requirements of the contract. The Consultant shall not proceed with implementation of any work affected by a submittal until review by the Client is complete and the submittal is returned with review comments as below:

5.7.1 Approved Indicates submittal has been reviewed and appears to be in conformance with requirements of the contract. The Consultant may proceed.

5.7.2 Approved as Noted Indicates submittal appears to be in conformance with requirements of the contract, with the exception of noted corrections. The Consultant shall incorporate the corrections noted but does not need to resubmit. The Consultant may proceed.

5.7.3 Returned for Correction Indicates submittal does not appear to be in conformance with the Contract. The Consultant shall make necessary revisions and resubmit. The Consultant shall not proceed.

5.7.5 Rejected Indicates submittal is incomplete or insufficient information has been provided to conduct a review. The Consultant shall provide the missing information. The Consultant shall not proceed.

5.7.6 Not Reviewed Indicates that the submittal is not called for by the contract and that no action was taken by the Client. The Consultant may proceed.

All submittals related to Engineering & Design of the project shall be reviewed by Chief Engineer (Design) NTDC, Lahore and those related to environment and social aspects shall be reviewed by Project Management Unit (Environment) WAPDA House, Lahore.

6. Distribution List:

The reports and documents would be distributed to the following or as instructed by the Client:

(i) Chief Engineer (Design) NTDC, WAPDA House, Lahore
(ii) Chief Engineer (MP&M) NTDC, WAPDA House, Lahore
(iii) Chief Engineer (EHV-I) NTDC, NTDC House, Lahore
(iv) Project Manager(EHV) NTDC, Islamabad
(v) Project Management Unit (Environment) WAPDA House, Lahore
(vi) Project Management Unit (DHTP) WAPDA House, Lahore

7. Staff Requirements:

The Consultants are free to propose a staffing plan and skill mix necessary to meet the objectives and scope of services. If all the required skills are not available within the consulting firms, they are encouraged to make joint ventures with other firms. Following is an indicative list of personnel and skills required for carrying out the Assignment-A and Assignment-B:

Assignment- A: Preparation of Tender Level Engineering Design, Tender Document and Support to Client during Tendering;

The following key staff is required for this assignment.

#	Staff Skills	Minimum Qualification / Experience	Man Months
1	**Project Manager / Team Leader**	He should have at least a Bachelor Degree in Electrical Engineering from a recognized University. He should have specific experience of working in senior Techno-managerial positions on similar projects including but not limited to designing of transmission lines in hilly and snowy terrains. The incumbent should have overall minimum experience of 20 years with at least 12 years in project related activities and 10 years as Team Leader. He should be able to lead the team of consultants and assist NTDC in timely implementation of the project with a quality output. In case of Masters in Electrical Engineering, the overall experience should be 20 years with at least 10 years in Project related activities and 6 years as Team Leader.	12
2	**Transmission Line Design & Engineering Expert.**	He should have a Bachelor Degree in Electrical Engineering from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. He should be able to decide/ review on the type of transmission line to be laid in a particular area considering all technical and economic factors etc. especially the designing of towers depending upon insulation zones and selection of appropriate conductor & accessories. In case of M.Sc. in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	6
3	**Transmission Line survey & Mapping Expert.**	He should have a Bachelor Degree in Electrical/ Civil Engineering from a recognized University and/or an MS in degree in GIS/RS with 15 years overall experience and minimum 10 years exposure to Project related activities. He should have a thorough understanding in dealing with satellite imagery and mapping of transmission lines using latest softwares & techniques and have experience	4

		in detailed plans and profiling along with contour markings. In case of M. Sc in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	
4	**Geotechnical Expert.**	He should have a Bachelor Degree in Engineering/masters in geology from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. The incumbent should have thorough understanding of following tasks: i. Snow avalanche assessment & protection ii. Snow creep assessment & protection iii. Earth movement assessment & Protection iv. Scour and flood assessment & protection In case of a higher degree in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	3
5	**Project Planning & Contract Expert.**	He should be an Engineering graduate from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. He should have experience of developing project implementation planning, procurement planning & packaging and overall project execution planning using the latest softwares and techniques. In case of M.Sc. in relevant field the overall experience should be 10 years with 7 years exposure to similar activities.	2
6	**Environment/Social/ Resettlement Expert.**	He should have Master's degree in Environmental Engineering/Social Sciences from a recognized university. He should have at least overall experience of 15 years with 10 years working on similar assignments/projects. He must have experience of developing EMPs and RAPs as per international standards and guidelines. In case of a higher degree in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	3
7	**Unallocated**		15
8	**Total**		45
9	**Technical & Support Staff.**		30

Assignment B: Design Review, Construction Supervision & Contract Management;

The following key staff is required for this assignment.

#	Staff Skills	Minimum Qualification / Experience	Man Months
1	**Project Manager/ Team Leader**	He should have at least a Bachelor Degree in Electrical/Mechanical Engineering from a recognized University. He should have specific experience of working in senior Techno-managerial positions on similar projects. The incumbent should have overall minimum experience of 20 years with at least 12 years in the relevant field and 10 years as Team Leader. In case of Masters in Electrical Engineering, the overall experience should be 15 years with at least 10 years in Project related activities and 8 years as Team Leader. He should be able to lead the team of Consultant and 'assist NTDC in timely implementation of the project with a quality output. He shall be the Leader of the Team and act as the consultant's representative on all matters related to the assignment. He shall keep close contacts with NTDC and shall be responsible for timely submission of reports and other documents. He should not be more than 65 years of age.	40
2	**Design & Engineering Expert (Transmission Line)**	He should have a Bachelor Degree in Electrical Engineering from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. He should be able to decide/ review on the type of transmission line to be laid in a particular area considering all technical and economic factors etc. especially the designing of towers depending upon insulation zones and selection of appropriate conductor & accessories. In case of M.Sc. in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	18

3	**Transmission Line Survey Expert.**	He should have a Bachelor Degree in Electrical/ Civil Engineering from a recognized University and/or an MS in degree in GIS/RS with 15 years overall experience and minimum 10 years exposure to Project related activities. He should have a thorough understanding in dealing with satellite imagery and mapping of transmission lines using latest softwares & techniques and have experience in detailed plans and profiling along with contour markings. In case of M.Sc. in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	6
4	**Engineering Expert (Civil /Structures)**	He should have a Bachelor Degree in Civil Engineering from a recognized University with 15 years overall experience and minimum 10 years' experience in engineering & designing of foundations and protection structures for EHV transmission line towers including hilly & hazardous terrain similar to this project area. In case of M.Sc. in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	8
5	**Geotechnical Expert.**	He should have a Bachelor Degree in Engineering/masters in geology from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. He should have thorough understanding of following tasks: i. Snow avalanche assessment & protection ii. Snow creep assessment & protection iii. Earth movement assessment & Protection iv. Scour and flood assessment & protection In case of a higher degree in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	4
6	**Project Planning & Contract Expert.**	He should be an Engineering graduate from a recognized University with 15 years overall experience and minimum 10 years exposure to Project related activities. He should have experience of developing project implementation planning, procurement planning & packaging and overall project execution planning using the latest softwares and techniques. In case of M.Sc. in relevant field the overall experience should be 10 years with 7 years exposure to similar activities.	4

7	**Financial Management Expert.**	He should have a master's degree in finance from a recognized University with 10 years' experience in the field of accounting and financial management from international, private or public organization. He should have sufficient experience with bookkeeping system management recording of financial information, and preparation of financial reports.	4
8	**Construction Supervision Expert (Electrical/Civil)**	He should have at least bachelor degree in Electrical/Civil Engineering from a recognized university with 15 years of overall experience and minimum 10 years of exposure to relevant field. He should not be more than 55 years of age.	24
9	**Environment/ Resettlement Expert.**	He should have Master's degree in Environmental Engineering/Social Sciences from a recognized university. He should have at least overall experience of 15 years with 10 years working on similar assignments/projects. He must have experience of developing EMPs and RAPs as per international standards and guidelines. In case of a higher degree in relevant field the overall experience should be 12 years with 8 years exposure to similar activities.	6
10	**Testing&Commissioning Expert.**	He should have at least bachelor degree in Electrical Engineering from a recognized university with 15 years over all experience and minimum 10 years exposure to relevant field. He should not be more than 55 years of age.	3
11	**Unallocated**		40
12	**Total**		157
13	**Technical & Support Staff.**		90

The estimated total man-month input of key personnel, technical and support staff for Consultants are 75 for Assignment-A. This estimation of man months includes 45 man months for key personnel's.

The estimated total man-month input of key personnel, technical and support staff for Consultants are 247 for Assignment-B. This estimation of man months includes 157 man months for key personnel.

8. Deliverables and Submission of Reports.

The schedule for various reports the Consultants are likely to prepare is given below. The Consultant will prepare reports in English and provide copies as mentioned below of the reports to the Government and the World Bank. Additional, reports may have to be prepared as needed by the investor, project authorities or based on needs.

	Assignment A
1.	Review of Feasibility Study/Inception Report
2.	Progress Reports including monthly and Quarterly Reports
3.	Preparation of special reports/memoranda to elicit Client's consent/approval on specific issues (as and when required)
4.	Report on loading zone including loading wind maps
5.	Route alignment and survey report
6.	Engineering Design Report consisting of a report for design of the project works, definition of the project, cost estimates, economic and financial, environmental and social analysis, implementation plan etc.
7.	**Bidding documents.** Bidding documents for all civil works and electrical and mechanical equipment, BOQs, tender cost estimates and technical specification for each lot separately.
8.	**Engineers Estimate** for EPC contract(s)
9.	**Contractors Bids Evaluation Report**
10.	Other deliverables as mentioned in Functional specifications.
	Assignment B
1.	Construction reports, quarterly reports, annual work plans
2.	Project implementation status reports quarterly, annual
3.	Implementation status report for EMP, SAP, RAP and work plans etc. On regular basis
4.	Minutes of Weekly/bimonthly/monthly meetings held with Contractors
5.	Correction/Amendment in Drawing submitted by Contractor
6.	As built Drawings
7.	Testing & Commissioning Report
8.	Maintenance & Operation Manual
9.	Completion Reports

9. Procurement of Goods:

Purchase of Equipment / Tools & Plants (T&P) will be carried out according to the need of Consultants. The Consultants shall maintain inventory list of all purchased equipment, vehicles and other T&P items etc. and submit details in quarterly reports. All these items shall be returned to WAPDA upon completion of contract with fair wear & tear condition.

ANNEX-I: FUNCTIONAL REQUIREMENTS FOR SURVEY

1. General:

i. This section sets forth minimum requirements for survey including route alignment, tower spotting and optimization of locations and check survey for the transmission lines. It is not intended to be all-inclusive of the requirements necessary to meet Project needs and objective. Nothing in this section relieves the Consultant of its obligation to perform the works to meet all of the requirements of the Contract.

ii. The work shall be carried out by using modern techniques. The bidder shall indicate in his offer, the detailed description of the procedure to be deployed. The details of the equipment & facilities including softwares for image processing etc. available with the bidder or his associates shall also be furnished; the technical literature& specifications thereof shall also be submitted with the proposal.

iii. The Client will not furnish the satellite imageries or topographical maps prepared by survey of Pakistan but will make available assistance that may be required in obtaining these by providing letters of recommendation to the concerned authorities. However, the Consultant may use the satellite imageries already available with the Client to the extent applicable. Further, in case the consultant opts for use of ALTM techniques for survey, he shall be responsible for obtaining necessary clearances / permissions by providing letters or recommendations to the concerned authorities.

iv. The Consultant shall also engage services of a reputed geo-technical consultant or experts from independent educational/research institutions for examining stability aspects of the selected transmission line route/locations in hilly terrain wherever required.

v. The Consultant shall study and review existing documents, studies, maps, etc. relating to the Project for better understanding and performance of the work.

vi. The consultant shall estimate BOQ of the transmission lines and submit the same to the Client.

2. Route Alignment:

i. Route alignment shall be most economical from the point of view of construction and maintenance. The consultant shall identify & examine alternative route alignments and suggest to the Client the optimal route alignment.

ii. Route alignment shall be done using satellite imageries (minimum resolution corresponding to scale 1:25,000) and survey of Pakistan topographical maps (scale 1:50,000). The consultant shall identify & examine three alternative route alignments and suggest to the Client the optimal route alignment between the terminal points.

iii. For examination of the alternatives & identification of the most appropriate route, besides making use of information/data/details available/extracted through Survey of Pakistan Topographical maps and computer-aided processing of satellite imagery, the consultant shall also carryout reconnaissance/preliminary walk over site survey for verification & collection of additional information /data /details.

iv. The consultant shall submit his preliminary observations & suggestions along with various information/data /details collected and also processed satellite imagery data, scanned topographical map data marked with the alternative routes etc. The optimal route alignment shall be proposed by the consultant and final evaluation of the alternative routes shall be conducted by the consultant in consultation with the Client.

v. Digital terrain modeling using contour data from topographical maps as well as processed satellite data shall be done by the consultant for the selected route. A fly through perspective using suitable software(s) shall be developed for further refinement of the selected route.

vi. Site visit and field verification shall be conducted by the consultant jointly with the Client's representatives for the proposed and selected route alignments.

vii. Final digitized route alignment drawing with latest topographical and other details/features including all rivers, railway lines, canals, roads, etc. up to 8 kms on both sides of selected route alignment shall be submitted by the consultant for the Client's approval along with report containing other information/details as mentioned above.

viii. Changes in the route alignment, if any, during detail survey, shall be incorporated in the final digitized route alignment drawings.

3. Requirement of Transmission Line Routing:

i. The route should have minimum crossings of Major river, Railway lines, National/State highways, overhead EHV power line and communication lines.

ii. The number of angle points shall be kept to minimum. The distance between the terminal points specified shall be kept shortest possible, consistent with the terrain that is encountered.

iii. Areas of active faults and earth quake shaking, landslides (rock and/or soil), rock falls, river scouring and flooding, avalanche paths, marshy areas, etc.; shall be avoided.

iv. It would be preferable to utilize level ground for the alignment.

v. Crossing of power lines shall be minimum. Alignment will be kept at a distance from power lines to avoid induction problems on the lower voltage lines.

vi. Crossing of communication line shall be minimized and it shall be preferably at right angle. Proximity and parallelism with telecom lines shall be eliminated to avoid danger of induction to them.

vii. Restricted areas such as civil and military airfield shall be avoided. Care shall also be taken to avoid aircraft landing approaches.

viii. The route alignment should be easily accessible both in dry and rainy seasons to enable maintenance throughout the year.

ix. Areas which will present problems in acquisition of right of way and way leave clearance during construction and maintenance should be avoided.

x. Preferably, angle points should be selected such that shifting of the point within 100 m radius is possible at the time of construction of the line.

xi. The line routing should avoid large habitations, densely populated areas, Forest, Animal/Bird sanctuary, reserve coal belt areas, oil pipe line/underground inflammable pipe lines etc. to the extent possible.

xii. Routing of transmission line through protected/reserved forest area should be avoided. In case it is not possible to avoid the forests or areas having large trees completely and then keeping in view of the overall economy, the route should be aligned in such a way that cutting of trees is minimum.

xiii. The areas requiring special foundations and those prone to flooding should be avoided.

4. Route Marking:

i. The route of the transmission line shall be recorded using GPS of positional accuracy less than 1m.

ii. The co-ordinates of all the angle points as well as other important crossings, landmarks, etc. shall be recorded using GPS for easy relocating.

iii. At the starting point of the commencement of route survey the co-ordinates shall be recorded. A punch mark on the top section of the angle iron shall be made to indicate location of the survey instrument. The co-ordinates of location of the survey instrument shall also be recorded. Further, the co-ordinates at prominent position at intervals of not more than 750 meter along the transmission line to be surveyed up to the next angle point shall also be recorded. Teak wood peg 50 x50 x 650mm size shall also be driven at prominent position at intervals of not more than 750 meters along the transmission line to be surveyed up to the next angle point. Wire nails of 100mm length should be fixed on the top of these pegs to show the location of instrument. The pegs shall be driven firmly into the ground only leaving 100 mm above ground level. Wherever the line alignment crosses EHT line, Railway line, T&T line or roads, the consultant shall record co-ordinates on the points of crossing. Wherever line route alignment passes over permanent land marks such as rock, boulders, culverts etc. suitable white paint marks with directional and NTDC markings shall be made and co-ordinates recorded. At angle position stone/concrete pillars of 150x150x1000mm marked with NTDC shall be embedded into the ground for easy identification.

5. Access Plan:

i. The consultant shall prepare an Access Plan to identify the access required for clearing of the transmission line corridor and construction of the transmission line. The Access Plan shall identify the location, type of access (e.g., temporary versus permanent, road versus helicopter), design and routing of access along the entire transmission line route.

ii. The Consultant shall evaluate the condition of existing access to and along the ROW, and determine the applicable requirements and facilities necessary for access upgrades and constructing new access. The Consultant shall determine the total quantity and type of access and access facilities required such as roads, culverts, bridges, fabric-reinforced gravel sections and helipads; and also prepare a cost estimate for the access plan.

iii. The access facilities for the Project shall meet the construction requirements of the Project and the needs of the Client for continued operation and maintenance of the transmission line. The Consultant's access plan shall balance the construction, and operation and maintenance needs with environmental protection and land owner needs, consider site conditions (e.g., terrain, drainage) and weather conditions etc.

6. Special Considerations for Dasu- Manshera Section viz-a-viz Transmissinn Line Route.

6.1 Snow Avalanche Assessment & Protection:

The Consultant shall complete an assessment of avalanche hazard for the Proposed Alignment. Avalanche hazard assessment shall be completed by a Professional Engineer who has a demonstrated background in avalanche assessment and design of avalanche protection structures. The results of the Consultant's avalanche assessment shall be submitted for review as part of survey report.

6.2 Snow Creep Assessment & Protection:

The Consultant shall complete an assessment of snow creep hazard for the Proposed Alignment. Snow creep hazard assessment and loading recommendations shall be completed by a Professional Engineer who has a demonstrated background in snow creep assessment and design. The results of the Consultant's snow creep assessment shall be submitted for review as part as part of survey report.

6.3 Earth Movement Assessment & Protection:

The Consultant shall complete an assessment of earth movement (rock fall and landslide) hazard for the transmission alignment. Earth movement hazard assessment shall be completed by a Professional Engineer or Professional Geologist who has a demonstrated background in earth movement assessment and design of protective structures. The results of the Consultant's earth movement assessment shall be submitted for review as part of survey report.

6.4 Scour and Flood Assessment & Protection:

The Consultant shall complete assessments of scour and flood hazard for portions of the Proposed Alignment which cross active rivers and flood plains. Scour and flood hazard assessment and the resulting scour force and flood level loading recommendations shall be completed by a a Professional Engineer who has a demonstrated background in river mechanics and scour and flood protection design. The results of the Consultant's scour and flood hazard assessment shall be submitted for review as part of survey report.

7. Survey Report:

i. Complete BOQ of the transmission lines shall be furnished in the survey report.

ii. Each angle point locations shall be shown with detailed sketches showing existing close by permanent land marks such as specific tree(s), cattle shed, homes, tube wells, temples, electric pole/tower, telephone pole, canal, roads, railway lines etc. The relative distance of land marks from the angle points and their bearings shall be indicated in the sketch. These details shall be included in the survey report.

iii. Information w.r.t infrastructure details available route, identification and explanation of route constraints etc. shall also be furnished in the Survey report and shall inter-alia include the following:

   a. Information regarding infrastructural facilities available along the final route alignment like access to roads, railway stations, construction material sources (like quarry points for stone, sand and availability of construction water), labor, existing transport facilities, fuel availability etc. shall be furnished in the survey report.

   b. All observations which the Consultant thinks would be useful to the construction of the transmission lines mentioned under scope of work are to be reported.

   c. Suggestions regarding the number of convenient zones (line segments / portions) in which the entire alignment can be divided keeping in view the convenience of construction are to be given.

   d. Suggestions regarding location for setting up stores during line construction in consultation with the Client's representative shall also be provided by the consultant.

   e. Working months available during various seasons along the final route alignment, with period, time of sowing & harvesting of different type of crops and the importance attached to the crops particularly in the context of way leave problems and compensation payable shall be stated by the Consultant.

   f. Availability of labor of various categories and contractors of civil works shall also be reported.

   g. Some portions of the line may require clearance from various authorities. The Consultant shall indicate the portion of the line so affected, the nature of clearance required and the name of concerned organizations such as local bodies, municipalities, Inland navigation, Irrigation Department, Electricity Boards and Zonal railways, Divisional Forest Authorities etc.

iv. All the requisite data for processing the case for statutory clearances such as Forest and Railway shall be provided along with the report.

v. The consultant shall also collect & report details pertaining to pollution levels envisaged along the transmission line.

vi. Access plan as per section 3.11 shall be submitted with the report.

vii. Avalanche, Snow creep, Earth Movement, Scour and Flood assessment as per section 3.12 shall be reported.

8. The following documents shall be submitted with the Proposal:

i. Methodology proposed to be adopted for selecting most suitable route.
ii. Details of maps and software to be used for marking route and obtain 3-D view and profile terrain.
iii. Methodology proposed for survey along with list of equipment, name of software for optimization of tower quantity.
iv. List of work done and sample documents.
v. Name of geotechnical consultants proposed.

ANNEX-II: FUNCTIONAL REQUIREMENTS FOR TRANSMISSION LINE DESIGN.

1. General

i. This section sets forth minimum requirements for design of the Project and is not intended to be all-inclusive of the design requirements necessary to meet Project needs and objective. Nothing in this section relieves the Consultant of its obligation to perform the works to meet all of the requirements of the Contract.

ii. The design, materials, construction and workmanship shall conform to the applicable standards, codes and regulations, or revisions thereto, which are in force as of the Contract Date. Where similar or common design requirements fall under more than one of the applicable standards, codes, regulations, and guidelines, the design shall comply with the more stringent requirement.

iii. The consultant shall under take necessary studies, investigations, develop various design models, carry out simulations, etc.; in order to select the design parameters/criteria for the transmission line and prepare a Design Document as per IEEE std1724: Guide for the Preparation of a Transmission Line Design Criteria Document.

iv. The Consultant shall perform all investigations necessary to support design of the Project and production of tender level drawings and specifications. The Consultant shall identify the type of investigations to be conducted and intimate the schedule for investigations to provide the Client's representative the opportunity to accompany and observe the investigation.

v. The Consultant shall study and review NTDC's design standards , practices and methods for 500 kV transmission lines for better understanding of NTDC system network and performance of the work.

vi. The Client reserves the right to request any integration or modification in the design as deemed necessary without the Consultant is being entitled to the payment of additional costs.

2. Standards

The list of standards used by NTDC for 500 kV transmission lines is given below for reference and guidance. The consultant may modify/update this list to meet the requirements of the project. However, the proposed standards should be at least equivalent and comparable to the specified standards. The consultant shall submit the proposed standards and all other information required in this respect for review of the Client. All submission must be made in the English language.

2.1 Steel Tower

(a) ASTM A6: General Requirements for Rolled Steel Plates, Shapes, Sheet Piling and Bars for Structural use.

(b) ASTM A36: Standard Specification for Structural Steel.

(c) ASTM A572: Standard Specification for High Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality.

(d) EN 10025: EURONORM Standard Specification for Structural Steel; Rolled Steel, Steel Sheets and Plates etc.

(e) ASTM A123: Zinc (Hot Galvanized) coatings on products fabricated from rolled, pressed, and forged steel shapes, plates, bars and strip.

(f) ASTM A143: Safe-guarding against embrittlement of hot-dip galvanized structural steel products and procedure for detecting embrittlement.

(g) ASTM A563M: Standard Specification for Carbon and Alloy Steel Nuts.

(h) ASTM A153: Standard Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware.

(i) ASTM A239: Standard Method of Test for Locating the Thinnest Spot in Zinc (Galvanized) Coating on Iron or Steel Articles by the Preece Test (Copper Sulphate Dip).

(j) ASTM A325: High-strength bolts for structural steel joints including suitable nuts and plain hardened washers.

(k) ASTM A370: Standard test methods and definitions for mechanical testing of steel products.

(l) ASTM A384: Standard practice for safe-guarding against warpage and distortion during Hot-dip Galvanizing of Steel Assemblies.

(m) ASTM A394: Standard specification for Steel Transmission Tower Bolts, Zinc-coated and Bare.

(n) ASTM F436: Standard specification for hardware steel washers.

(o) ASTM B201: Chromatic treatment test.

(p) ASTM E 94: Radiographic Testing.

(q) ASTM E109: Dry powder magnetic particle inspection.

(r) ISO 898-1: Mechanical Properties of Fasteners made of carbon steel and alloy steel. Bolts, screws and studs

(s) DIN 267: Fasteners technical delivery conditions steel spring washers for bolt/nut assemblies.

(t) DIN 7990: Hexagon head bolts for structural steel bolting for supply with nut.

(u) DIN 555: M5 to M100 x 6 hexagon nuts.

(v) DIN 127: Spring lock washers with square ends or tang ends.

(w) DIN 128: Spring lock washer curved and wave.

(x) ASCE Manual No. 74: "Guidelines for Transmission Line Structural Loading" prepared and published by the American Society of Civil Engineers in 2010 (3rd Edition).

(y) ASCE Standard 10-97: "Design of Latticed Steel Transmission Structures".

2.2 Conductor and Shield wire

(a) ASTM A90: Standard Method of Test for Weight of Coating on Zinc Coated (galvanized) Iron or Steel Articles.

(b) ASTM A153: Standard Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware.

(c) ASTM B230: Standard Specification for Aluminum Wire, EC-H19 for Electrical Purposes.

(d) ASTM B232: Standard Specification for Aluminum Conductor Concentric-Lay-Stranded Coated Steel Reinforced (ACSR).

(e) ASTM A239: Standard Method of Test for Locating the Thinnest Spot in Zinc (Galvanized) Coating on Iron or Steel Articles by the Preece Test (Copper Sulphate Dip).

(f) ASTM A363: Standard Specification for Zinc Coated (Galvanized) Steel Overhead Ground Wire Strand.

(g) ASTM B498: Standard Specification for Zinc Coated Galvanized Steel Core Wire for Aluminum Conductor Steel Reinforced (ACSR).

(h) IEC-61089: Round wire concentric lay overhead electrical stranded conductors.

(i) IEC-61395: Overhead electrical conductors – creep test procedures for stranded conductors

2.3 OPGW and Associated Hardware

(a) ITU-T G.652: Characteristics of a single-mode optical fiber cable

(b) ITU-T G.654: Characteristics of a cut-off shifted single-mode optical fiber cable

(c) IEC 60793: Optical fibers

(d) IEC 60794: Optical fiber cables

(e) EIA 598A: Color coding of optical fibers

(f) ASTM B415: Standard specification for hard-drawn aluminum-clad steel wires

(g) ASTM B416: Standard specification for hard-drawn aluminum-clad steel wires

(h) ASTM B398: Standard Specification for Aluminum-Alloy 201-T81 Wire for electrical Purposes

(i) IEEE Std. 1138: Construction of Composite Fibre Optic Overhead Ground Wire (OPGW) for use on Electric Utility Power Lines

(j) IEC 61300-2-36: Fiber Optic Interconnection Devices & Passive component-Basic Test and Measurement Procedures – Part 2-36: Tests-Flammability (Fire hazards)

(k) IEC 68-2-14: Joint boxes/hardware fittings

(l) EN 60529: Protection class of cabinets/cubicles

2.4 Insulators and Hardware

(a) ASTM A47: Specifications for Ferritic Malleable Iron Castings.

(b) ASTM A143: Standard Recommended practice for safeguard against embrittlement of Hot-dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement.

(c) ASTM A153: Standard Specification for Zinc coating (Hot dip) on Iron & Steel Hardware.

(d) ASTM A220: Specifications for Pearlitic Malleable Iron Castings.

(e) ASTM A384: Standard practice for safeguarding against warpage and distortion during Hot-dip Galvanizing of Steel Assemblies.

(f) ASTM A239: Standard Method of Test for Locating the Thinnest Spot in Zinc (Galvanized) Coating on Iron or Steel Articles by the Preece Test (Copper Sulphate Dip).

(g) ASTM A536: Specifications for Ductile Iron Casting.

(h) ASTM A668: Specifications for Steel Forgings, Carbon and Alloy, for general industrial use.

(i) ASTM C151: Test Method for Autoclave Expansion of Portland Cement.

(j) ANSI C29.1: American National Standard Test Methods for Electrical Power Insulators.

(k) ANSI C29.2: American National Standard for Wet Process Porcelain and toughened Glass Insulators.

(l) IEC 575: Thermal mechanical performance test and mechanical performance test on string insulator units.

(m) IEC 61284: Overhead Lines – Requirements and Tests for fittings.

(n) BS 3288: Insulators and Conductor fittings for overhead power lines Part I. Performance and General Requirements.

(o) Other relevant ANSI, ASTM and IEC standards.

2.5 Accessories and Dampers

(a) ASTM A153: Standard Specification for Zinc coating (Hot Dip) on Iron and Steel Hardware.

(b) ASTM A164: Standard Specification for Electrodeposited Coatings of Zinc on Steel.

(c) ASTM A239: Standard Method of Test for Locating the Thinnest Spot in Zinc (Galvanized) Coating on Iron or Steel Articles by the Preece Test (Copper Sulphate Dip).

(d) ASTM D1149: Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber.

(e) ANSI B.1.1: "Unified Screw Threads" class 2A.

(f) A-8025 C: U.S Military Specifications MIL A-8025 C.

(g) IEEE Paper No.
31 TP 65-156: Recommended Method of IEEE Task Force on Conductor Vibration.

(h) IEEE Paper
T74 061.8: Decay Test Evaluation

(i) IEEE Paper 31
TP 65-707: An Investigation of the Forces of Bundle Conductor Spacers under Fault Conditions.

(j) CISPR 18-1: Radio interference characteristics of overhead power lines and high voltage equipment.

(k) CISPR 18-2: Methods of measurement and procedure for determining limits.

(l) CISPR 18-3: Code of practice for minimizing the generation of radio noise.

(m) IEC 61284: Overhead Lines – Requirements and Tests for fittings.

(n) BS3288 : Insulators and Conductor fittings for overhead power lines Part I. Performance and General Requirements.

(o) IEC 61897: Overhead Lines – Requirements and Tests for Stockbridge type Aeolian vibration dampers.

(p) IEC 61854: Overhead Lines – Requirements and Tests for spacers.

(q) Other relevant IEEE, IEC and ASTM Standards.

2.6 Transmission Line Construction

(a) ASTM A615: Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement.

(b) ASTM C33: Standard Specification for Concrete Aggregates

(c) ASTM C39: Test Method for Compressive Strength of Cylindrical Concrete Specimens.

(d) ASTM C150: Standard Specification for Portland Cement.

(e) ASTM D1556: Test Method for Density and Unit Weight of Soil in-place by Sand Cone Method.

(f) ASTM D1557: Test Method for Laboratory Compaction Characteristics of soil Using Modified Effort.

(g) ASTM D1586: Standard Penetration Test

(h)	ASTM D2167:	Test Method for Density and Unit Weight of Soil in-place by the Rubber Balloon Method.
(i)	ASTM C494:	Chemical Admixtures for Concrete.
(j)	ACI Building Code:	Building Code Requirements for Reinforced Concrete (ACI-318M, latest edition) and Commentary (ACI 318RM, latest edition).

3. Calculations

i. The Consultant shall prepare all design calculations and submit to the Client for information and review. The design calculations shall be detailed, well elaborated and include all necessary analysis, inputs, controls, requirements, output and other miscellaneous calculations to demonstrate the accuracy of the design.

ii. The calculations shall be drawn up adopting the standard calculation methods and according to easy-to-consult formats and typologies. Code or standard references should be quoted and equations be written out in full. The design manuals, papers or other aids referred to in design and calculations shall be submitted with the calculations.

iii. If produced using automatic calculating methods, the calculation program shall be one that is internationally renowned and routinely used. Where a computer program is used for design calculations a full explanation in the English language shall be provided to assist the Client approval of the calculations for each and every program used. Details must include name of program author source, comprehensive description of theoretical basis including all references to relevant documentation, checks undertaken on program and list of projects on which the program has been used.

iv. The Consultant shall submit for review and approval of the Client, information identifying all computer design tools intended for use in the Project.

v. The Consultant shall provide the Client with an electronic version of source files used in any computer design or drafting tool.

4. Tender Level Drawings

i. Tender level drawings and specifications prepared for the Project shall be complete and in reasonable detail to guide the procurement of materials and construction in order to provide a fully functional, reliable and safe transmission line.

ii. The drawings and specifications shall be prepared in a form suitable for issue in tender documents.

iii. Each drawing submitted by the Consultant shall be clearly marked with the name of the Client, the specification title, the specification number and the name of the Project. All titles, noting, markings and writings on the drawing shall be in English. All the dimensions should be to the scale and in S.I.units.

iv. All drawings shall be prepared using AutoCAD software latest version. After final approval all the drawings shall be submitted to the Client in CDs**.**

v. All rights of the design/drawing prepared for the project shall be strictly reserved with the Client only and any designs/drawings/data sheets submitted by the consultant from time to time shall become the property of the Client. Under no circumstances, the Consultant shall be allowed to user/offer above designs/drawings/data sheets to any other authority without prior written permission of the Client. Any deviation to above is not acceptable and may be a cause for rejection of the proposal.

vi. The Consultant shall prepare drawings that adequately describe the works design, to be fabricated and installed. The scope of drawings prepared and submitted to Client's representative for information, approval shall include but not limited to the following:

4.1 ROW Preparation Drawings

- Access alignment details and routes
- Clearing standards and criteria
- Clearing boundary and danger tree mapping
- Access road details
- Culvert location details
- Bridge location details

4.2 Right-of-Way

- ROW cross sections
- ROW requirements
- ROW structure location plans

4.3 Route

- Route plans

4.4 Design

- Conductor clearance charts (conductor dip and deflection clearance to tower head)
- Structure specifications
- Leg extension ranges
- Foundation specifications

4.5 Towers

- Tower outline (elevation and plan) drawings

The outline drawings shall show the design information and dimensions. The outline drawings shall also include the following information related to the design of structures:

- Description of phase conductors, overhead ground wires/OPGW, controlling sag and tension conditions, etc.

- Design spans (ruling span, wind and weight spans).
- Clearance diagrams (may be on separate drawings).
- Maximum foundation reactions (along the vertical and along the leg slope) with and without structure overload capacity factors.

• Loading tree and calculations

4.6 Foundation

• Tender Level Foundation drawings

4.7 Insulator Suspension and Dead-end Hardware

• Suspension assembly specification/ drawings.
• Dead-end assembly specification/ drawings.
• Jumper assembly drawings

4.8 Spacers and Dampers

• Drawings for bundle spacers/ spacer dampers
• Drawings for bundle jumper spacer
• Specifications for other vibration control devices
• Marker ball support on bundle conductor, specifications and installation details

4.9 Conductor Fittings

• Specification drawings for compression fittings and assemblies
• Conductor die details

4.10 Miscellaneous Conductor Fittings /data

• Shield wire /OPGW installation arrangement
• Shield wire /OPGW vibration damper details
• Shield wire /OPGW suspension and dead-end hardware/Joint box details
• Conductor/Shield wire/OPGW Sag data

5. Transmission Line Optimization

For the transmission line, its design process includes at first the electrical requirements such as power transfer capability and voltage from which the tower geometry, the electric field effects, the corona effects, the over voltage and insulation coordination and the required right of way are established. Then the mechanical design of the towers and foundations, the determination of conductors and shield wire stresses are carried out; finally the economics including direct cost, cost of losses, operation and maintenance cost

along line life is evaluated. The design process is iterative as the electrical parameters can be met with variety of solutions. The optimum solution is to be derived.

For selection of optimum transmission line alternatives; electrical, mechanical, civil and environmental aspects (as above) shall be taken into account as a whole set simultaneously so as to develop transmission line design for satisfactory performance and reasonable cost.

6. Insulation coordination study

The consultant shall carry out insulation coordination study to determine basic electrical stresses and clearances.

For line insulation coordination, the insulation strength depends on the geometry of towers and properties of the insulator string. The selection of the parameters defining tower geometry such as tower height, strike distance, phase spacing and shielding angle should be based on the specified degree of reliability vis-a-vis the over voltages/stresses which can appear on the system under anticipated environmental/climatic conditions. These parameters and their basis shall be clearly defined/documented and shown to meet the system reliability criteria under the most unfavorable conditions.

The determination of phase spacing should be such as to avoid clashing of conductors at mid span due to ice drop, galloping and differential wind speed.
The type of insulator string i.e. V-string vs I-string shall be determined considering the reliability requirements and economy of structure/Transmission line.

The shielding requirements (number of shield wires, shielding angle) shall be determined while taking into consideration, inter alia, the terrain i.e. plain or hilly, isokeraunic level, altitude, etc.;

The determination of electrical clearances has a direct impact on the size of structure and economy of transmission line and shall be optimal.

7. Determination of Rights-of-Way

The Consultant shall identify the ROW necessary to accommodate the Consultant's design and construction as well as future operation and maintenance by the Client. The Consultant shall determine the ROW required for the transmission line for the parameters identified below and state the calculation method and results for the following minimum set of parameters:

a. conductor swing;
b. ROW edge distances;
c. circuit separations;
d. characteristic and long span lengths;
e. total statutory ROW width;

f. construction and maintenance considerations;
g. existing and future use of adjacent land;
h. zoning and published official community plans;
i. existing buildings, planned developments, and other facilities;
j. road allowance & aerial rights and permits at major transportation corridors;
k. clearance over navigable watercourses;
l. simultaneous line outages caused by lightning;
m. radio interference;
n. audible noise;
o. electric and magnetic fields;
p. electric and magnetic induction;
q. geotechnical hazards & topography; and
r. clearances required for public safety and system reliability.

8. Steel Towers

8.1 Type of Towers

The following types of towers are proposed for the transmission lines:

i. Double Circuit Suspension Type Tower for line angles up to 2°. The tower shall consist of a basic body, body extension of 6.0 m and leg extensions of 2.0 m, 4.0 m, 6.0m &8.0 m.
ii. Double Circuit Light Angle Strain Type Tower for line angles up to 20°. The tower shall consist of a basic body, body extension of 6.0 m and leg extensions of 2.0 m, 4.0 m, 6.0m &8.0 m.
iii. Double Circuit Medium Angle Strain Type Tower for line angles up to 40° The tower shall consist of a basic body, body extension of 6.0 m and leg extensions of 2.0 m, 4.0 m, 6.0m&8.0 m.
iv. Double Circuit Heavy Angle/Terminal/deadend Strain type Tower for line angles up to 60° The tower shall consist of a basic body, body extension of 6.0 m and leg extensions of 2.0 m, 4.0 m,6.0 &8.0 m.
v. Double Circuit Strain Type Tower for line angle 60° shall be modified for use as transposition tower. The tower shall consist of a basic body, body extension of 6.0 m and leg extension of 6.0 m.

Notes:

a. Special towers for long spans, river crossings, multi circuit towers; if required by site conditions; shall be designed as well.
b. It is proposed to use different families of towers at elevations up to 1000 masl, at elevations between 1000 and 2000 masl, at elevations between 2000 and 3000 masl and at elevations above 3000masl. The consultant shall carry out necessary studies, investigations and techno-economic analysis to optimize the families and types of towers including body/leg/uneven leg extensions.

c. All towers shall be designed with reference to the environmental data and design characteristic to be agreed by the Client.
d. The number of tower types and height of towers shall be optimized to provide the most cost-effective line design considering reliability, materials, construction and maintenance solutions.
e. Normal ruling span and the tower height shall be determined by the Consultant for the most economical line construction according to the prudent engineering practices.

8.2 Reliability Requirement

A transmission line shall be designed to withstand the defined weather loads during its projected service life. The reliability requirement is also intended to provide service continuity under additional loading conditions derived from these defined weather loads, such as unbalanced ice loads due to long spans or rugged, mountainous terrain. The reliability requirement is expressed in terms of a return period, T, of the design weather loads which the transmission line is to be designed to withstand. For reference and further deliberations the following criteria is suggested for the transmission line:

The transmission line shall be designed for weather load events based on a return period of 1-in-100 years and a seismic event based on a return period of 1-in-475 years. The line shall be designed for no major damage under the 1-in-100 year weather-related loads and no service interruption (i.e., maintain electrical clearances) under the 1-in-50 year return weather loads.

8.3 Security Requirement

The Consultant's design shall demonstrate that, at a minimum, the security considerations listed below were evaluated and addressed in the transmission line design.

Security Design Considerations

Condition	Application	Location
Flood Events	Safe Electrical Clearances and Structure Location	Site Specific
Snow Creep	Structure Loads (Snow Legs)	Site Specific
Snow Avalanches	Structure Location	Site Specific
Rock Fall, Debris Flows, Slides	Structure Location	Site Specific
Seismic, Soil Liquefaction	Structure Location	Site Specific

Differential Ice Loads	Structure Loads Conductor Sags	Site Specific

8.4 Safety Requirement

The transmission line shall be designed to provide electrical safety clearance and to have the necessary strength to support weather loads under construction and maintenance conditions, including stringing tackle loads and wire snubbing loads. The design shall also accommodate energized live line and bare hand maintenance methods and equipment.

9. Conductor Selection

The selection of conductor is one of the major factors in optimization of design/economy of the transmission line. The Conductor selection study shall take into consideration a defined set of criterion e.g. project load, line voltage, line length, power factor, load factor, system loss factor, generation reserve factors, corona/noise performance, EMF performance, ice loading and galloping conductor performance, Spacer/vibration damper or spacer-damper requirements, sag-tension of conductor vis-à-vis tower cost for designed spans along with overall transmission line cost per km, etc. The techno-economic comparison shall be carried out for a range of conductors (i.e. different size of conductors & number in bundles) that are likely to satisfy the minimum criterion.

An outline for conductor optimization study is proposed as below:

- Define the main evaluation parameters and criteria;
- Determine the minimum conductor size based on meteorological (ice/wind) loading, mechanical requirements, electrical performance and thermal capacity (to be determine in accordance with IEEE std 738& CIGRE Technical Brochure 299.
- Develop a range of conductor of increasing size starting from the minimum;
- Evaluate the transmission line capital cost and annual cost of losses at the design loading level for each conductor considered;
- Combine the transmission line capital cost and the capital equivalent of the annual cost of losses to give the total economic cost of the line for each conductor by using present worth of revenue required method (PWRR);
- Select the minimum cost alternative as the optimum conductor size.

Note: As per limitations imposed by the feasibility study, the ampacity of the proposed conductor at the design operating temperature shall not exceed 4000 Amperes.

10. Conductor Motion

The consultant shall study and determine the requirements for Aeolian vibration control and conductor motion due to galloping.

In case of galloping, the clearances between phases and phases to shield wire should be specified to account for expected conductor motion. The maximum span that can be allowed before galloping ellipses touch should also be specified.

11. Corona Effect and Electric Fields

Corona losses, Radio Noise, Audible Noise, Electric and Magnetic fields shall be calculated and determined as per applicable codes, standards, guidelines.

12. Deliverables

The following is the general list of the Engineering documents and drawings that are to be submitted by the Consultant for review/approval by the Client.

a) Route alignment
b) Survey Report
c) Soil Resistivity Report
d) Design Criteria Document
e) Tower loading trees and calculations
f) Tender Level Specifications and drawings of Transmission Line material and towers.
g) Construction methodology including but not limited to clearing ROW, Access plan, Erection, Stringing, Civil works, location wise specific mitigation measures/design wherever Required
h) Basic Foundation Design
i) Estimated Cost of the Project
j) Project execution and Implementation Plan & Schedule
k) Project Completion Report
l) Documents, reports, drawings, calculation etc.; other than above, as mentioned in the Functional Requirements for Survey and Design,
m) Any other document/information necessary for the complete design & installation of the transmission line.

ANNEX-III: DETAILED TORS FOR ENVIRONMENTAL AND SOCIAL ASSESSMENT (ESA)

1. Objectives of ESA Studies

The specific objectives of ESA are to:

a. Establish the social environmental baseline conditions in and along the transmission line corridor
b. Screen and identify adverse environmental and social impacts associated with the proposed transmission line.
c. Develop an Environmental Management and Monitoring Plan (EMMP) to address the adverse environmental impacts.
d. Undertake Social Assessment (SA) and recommend mitigation interventions to address the social impacts
e. Ensure that the EMMP and social interventions are complying with relevant governing national laws and World Bank environmental and social safeguard policies given under **Item 'G'** of this annexure.

2. The ESA Studies

Through the proposed study, NTDC desires to ensure safeguards' compliance for the Project, including its pre-construction, construction, operation and maintenance, and de-commissioning phases, and to assess environmental and social consequences in line with the World Bank's operational policies given in Annex-I as well as the national, provincial regulations on environment and social aspects. The proposed study is aimed at screening and assessing the proposed project activities for potential adverse environmental and social impacts and recommending, where necessary, appropriate mitigation and enhancement measures, and course of action for implementation.

The ESA will take into account the natural environment (air, water, and land); human health and safety; social aspects (involuntary resettlement, indigenous peoples, and physical cultural resources); trans-boundary and global environmental aspects including climate change and its implications, and also induced impacts as well as the cumulative impacts of other development projects in the area. The ESA will consider environmental and social aspects in an integrated way. It will also take into account variations in the project and country conditions; the findings of previous environmental studies; national environmental action plans; the country's overall policy framework on environment, national legislation, and institutional capabilities particularly that of WAPDA and NTDC related to the environment and social aspects; and obligations of the country pertaining to project activities, under relevant international environmental treaties and agreements.

3. Specific Tasks for the Consultants

To achieve the broader aim of ensuring safeguards' compliance, the consultants will carry out the following activities:

3.1 Review the Project Details

- Review the proposed project and its geographic, ecological, economic, social, and temporal context, including any offsite investments that may be required.
- Define the Impact Area and the "Corridor of Influence" of the project on the basis of the project scope and extent.

3.2 Review of the Legislative and Regulatory Framework

- Review the policy, legal, and administrative framework within which the ESA is to be carried out.
- Review the national environmental requirements and those of possible co-financiers (international donors) of the project.
- Indicate relevant international environmental agreements to which the country is a party.
- Review the World Bank Operational Policies on Safeguards with an analysis of which of these will be triggered due to the proposed Project activities, and identify possible gaps between relevant national and World Bank policies;

3.3 Scoping

Scoping is the first step of the ESA and is essentially the process of identifying the significant issues related to the proposed project and of determining the scope of the issues to be addressed in the ESA. The key tasks include: i) carry out reconnaissance field visit(s); ii) hold initial stakeholder consultations; iii) identify the key aspects to be studied during the detailed ESA, iv) finalize ESA ToRs in consultation with the stakeholders; v) prepare detailed work plan for the subsequent ESA tasks; vi) prepare the Scoping Statement compiling the process and outcome of the scoping tasks described above; and vii)review the definition of Area of Impact and the Corridor of Influence and revise (if necessary).

3.4 Analysis of Alternatives

The consultants will analyze alternative sitting of transmission lines in terms of their environmental and social impacts and benefits. The analyses should include the sitting process, based on the description of the selected route, the alternative routes and the environmental justification for the choice. The analysis of alternatives will also

avoid/minimize involuntary resettlement (including involuntary acquisition of land and impact on livelihood of PAP). The consultant will carry out consultations with local communities over the alignment, particularly the angle tower locations, and take into consideration their views and suggestions in the analysis. Based on the selected route, the consultants will identify and prepare maps showing the corridor of impact (Col).

3.5 Baselines Studies

The consultants will review the available literature, visit the project area and consult NTDC and other line departments to establish the baseline conditions in the project areas, in terms of i) physical and biological environment, ii) climatic and socio-economic conditions, iii) demographic and ethnic profiles, iv) social economic development status, challenges and needs, v) employment and income patterns, vi) land resources, ownership, tenure and holding size, registration, vii) public services, viii) institutions (formal and informal), ix) women status and x) security situation. The consultants will prepare maps of suitable scale to highlight the socio-environment resources of the project area. The baseline studies should conclude by providing the trend analysis of key environmental assets in the project under 'without project scenario' and identify environmental hotspots.

3.6 Public/stakeholders' Consultations

The consultants will hold at least two rounds of consultations on the environmental front. The consultations will help to identify project stakeholders and hold consultations with them to delineate the appropriate boundaries of the environmental assessment and to screen potential adverse environmental and social issues. The consultations will also receive feedback on the expected social issues related to the project impact and suggested mitigation measures. The stakeholders will include the primary stakeholder (those directly affected by the intervention) and secondary stakeholders (those indirectly affected and those who have an interest in the project) the consultants will document the proceedings of the consultations along with the list of participants and photographs. The consultants will provide systematic response to the concerns shown by the stakeholders in the ESA report. The final round of consultation will be used to inform the stakeholders on the study outcomes and the way project will address and the approach it has adopted to attend their concerns.

On the social side, the consultant will follow a participatory fashion. Stakeholder consultations are an important input in the baseline surveys, impact screening and identification and development of the proposed mitigation measures. The consultant will screen and identify the stakeholders, prepare for necessary project info disclosure and hold consultations with the stakeholders in appropriate formats. The consultation approach, organization, process, substance discussed, participation and feedback will be documented in detail in the social assessment report.

3.7 Screening And Identification of Environmental/Social Impacts of Transmission Lines

The consultants will identify potential environmental and social issues of transmission lines in terms of their nature, magnitude, extent and location, and timing and duration. These impacts may relate to the project design stage, construction stage and/or the project operation and decommissioning stage. Based on impact prediction methods and as the result of public consultations, the consultants will screen adverse environmental impacts for inclusion in mitigation measures and environmental management plan. The same process will be followed for identification of social impacts. The consultants will carry out consultations with local communities in impact screening and identification. Experiences show that feedback from such consultations has always been helpful and can provide important input in fully assessing project impacts. This is particularly so on the social side. These impacts will be documented in the environmental and social assessment reports.

3.8 Mitigation measures

The consultants will propose appropriate mitigation measures for the adverse environmental and social impacts. These measures could base on exploring the ways to achieve the project objectives by alternative ways, proposing changes in the project design (size of the right of way / height of the power line towers, sitting of facilities), through improved monitoring, and management practices (storage of construction materials, labor camps, waste disposal, disposal of construction debris, etc.) and / or through monitory compensation (resettlement). The consultants will also explore ways and means to enhance the environmental and social benefits rather than merely focusing on 'do no harm approach'. Social mitigation measures will follow from the alternative design implication (i.e. size, row, sitting of grid stations and towers, location of high voltage transmission lines particularly through crowded urban areas, need for land, project's impact on land prices, if any, etc.) and stakeholders consultation.

3.9 Environmental Management & Monitoring Plan (EMMP) and Mitigation Approach on Social Impacts

The consultants will prepare site specific and comprehensive environmental management and monitoring plan to ensure the adequacy and effectiveness by clearly identifying the roles and responsibilities of the contractor, supervisory consultant and the client. The consultants will also elaborate on the monitoring mechanism (both internal and external) and the reporting frequency. The consultants will prepare the cost estimates for the proposed EMMP for its inclusion in the project cost. Similarly, the consultants will also prepare the costs of the Social component / social mitigation measures as a part of the project costs / benefits. The Consultants will also develop environment performance indicators to monitor, audit, evaluate and supervise negative and positive project environmental impacts. The Consultants will also suggest monitoring, auditing and evaluation tools and formats of a monitoring plan including

frequency and methodology of monitoring as well as allocate institutional responsibility and costs.

Based on the findings of the social screening and Social Assessment, the consultant will determine and propose a mitigation approach to address the social impacts under the project. Where the alignment is fixed with the final power towers, a resettlement action plan may need to be developed. In situations where the alignment towers are not finalized and the alignment corridor is still subject to change before finalization, a phased approach may be appropriate. In this case, a resettlement policy framework or a social impact management framework as needed, can be developed before project appraisal and detailed resettlement action plans can be developed during implementation when the alignment is finalized. Once the proposals are endorsed, the consultant will need to develop the detailed mitigation plan. Outlines of a resettlement action plan and a resettlement policy framework are given under item 'H' and 'I' respectively.

4. Methodology

The assignment will be carried out through desk reviews, field visits and stakeholder consultations through interviews and focus group discussions.

5. Time Schedule.

The consultants will carry out the ESA study in 14 weeks.

6. Deliverables

The following deliverables will be required from the consultants.

6.1 Environmental Assessment (EA)

- Inception Report (2 weeks after the commencement of work)
- Draft EA Report (12 weeks after the commencement of work)
- Final EA Report after incorporating the comments by the Client, stakeholders and the Bank (14 weeks)
- Executive Summary both in English and Urdu languages (14 weeks)

6.2 Suggested Structure of the EA Report

- Cover Page
- Executive Summary
- Table of Contents
- Introduction/ Background
- Policy, Legal and Administrative Framework
- Description of the Project
- Description of Baseline Environment
- Analysis of Project Alternatives

- Environmental Impacts and Mitigation measures
- Environmental Management and Monitoring Plan (EMMP)
- Grievance Redressal Mechanism
- Environmental mitigation, management and auditing Costs
- Conclusions and recommendations

6.3 Social Assessment (SA)

- Inception Report (2 weeks after the commencement of work)
- Draft SA Report (11 weeks after the commencement of work)
- Final SA Report after incorporating the comments by the Client and Bank (12 weeks)
- Executive Summary both in English and Urdu languages (12 weeks)

6.4 Suggested Structure of the SA Report

The Social Assessment (SA)would be a standalone document and contain the followings:

- Table of Contents
- Executive Summary
- Introduction/ Background
- Objective and scope of assignment
- Socio economic baseline
    - demographic characteristics of the affected area and affected persons
    - ethnic profiles and communities
    - employment patterns and income structures
    - access to and use of natural resources, particularly land and forest
    - women and vulnerable groups
    - social infrastructure and services
    - land tenure, ownership and registration
    - development needs, challenges and status
    - security situation
- Screening and assessment of adverse impact and mitigation approach
    - Approach and methodology
    - Impact zones and areas
    - Adverse social impacts (physical and non-physical)
    - Recommendations on mitigation interventions (RAP or policy framework) and rational
- Policy, Legal and Administrative Framework in addressing adverse impacts
    - Relevant laws and policies (government and World Bank)
    - Existing institutions and decision-making mechanism (formal and informal)

- Recommended institutional mechanism and process for the project implementation

- Stakeholder identification and consultation
  - Government agencies
  - Local administration
  - Non-government organizations
    Community organizations
  - Consultation approach and methodology
  - Process and consultation organizations
  - Key feedback from the consultations
  - Matrix of consultation sessions (time, place, participants and issues)
- Security and conflict situation

6.5 Appendices

- List of ESA Team Members
- List of participants (stakeholders)
- Personal communications/ minutes of public meetings
- Other information (Photographs, rate lists obtained from forest and agriculture departments, etc.)
- Outline for a resettlement action plan
- Outline for a resettlement policy framework

6.6 Resettlement Action Plan

6.7 Resettlement Policy Framework

7. List of the World Bank Operational Policies and Guidelines to be considered by the consultant for the preparation of reports[Also list key Government laws and acts as well that the ESA needs to comply with]

Operational Policies (OP) / Bank Practices (BP) / Operational Directives (OD)

OP / BP 4.01	Environmental Assessment
OP / BP 4.04	Natural Habitats
OP 4.10	Indigenous Peoples
OP 4.11	Physical Cultural Resources
OP / BP 4.12	Involuntary Resettlement
OP / BP 4.36	Forestry
OP / BP 4.37	Safety of Dams
OP / BP 7.50	Projects on International Waterways
OP / BP 7.60	Projects in Disputed Areas
OP 4.09	Pest Management

8. Basic Elements of a Resettlement Action Plan

8.1 Description of the project.

General description of the project and identification of the project area

8.2 Potential impacts.

Identification of

i. the project component or activities that give rise to resettlement;
ii. the zone of impact of such component or activities;
iii. the alternatives considered to avoid or minimize resettlement; and
iv. the mechanisms established to minimize resettlement, to the extent possible, during project implementation.

8.3 Socioeconomic studies.

The findings of socioeconomic studies to be conducted in the early stages of project preparation and with the involvement of potentially displaced people, including

i. the results of a census survey covering
    1. current occupants of the affected area to establish a basis for the design of the resettlement program and to exclude subsequent inflows of people from eligibility for compensation and resettlement assistance;
    2. standard characteristics of displaced households, including a description of production systems, labor, and household organization; and baseline information on livelihoods (including, as relevant, production levels and income derived from both formal and informal economic activities) and standards of living (including health status) of the displaced population;
    3. the magnitude of the expected loss--total or partial--of assets, and the extent of displacement, physical or economic;
    4. information on vulnerable groups or persons as provided for in OP 4.12, para. 8, for whom special provisions may have to be made; and
    5. provisions to update information on the displaced people's livelihoods and standards of living at regular intervals so that the latest information is available at the time of their displacement.
ii. Other studies describing the following
    1. land tenure and transfer systems, including an inventory of common property natural resources from which people derive their livelihoods and sustenance, non-title-based usufruct systems (including fishing, grazing, or use of forest areas) governed by local recognized land allocation

mechanisms, and any issues raised by different tenure systems in the project area;

2. the patterns of social interaction in the affected communities, including social networks and social support systems, and how they will be affected by the project;
3. public infrastructure and social services that will be affected; and
4. social and cultural characteristics of displaced communities, including a description of formal and informal institutions (e.g., community organizations, ritual groups, nongovernmental organizations (NGOs)) that may be relevant to the consultation strategy and to designing and implementing the resettlement activities.

8.4 Legal framework.

The findings of an analysis of the legal framework, covering

i. the scope of the power of eminent domain and the nature of compensation associated with it, in terms of both the valuation methodology and the timing of payment;
ii. the applicable legal and administrative procedures, including a description of the remedies available to displaced persons in the judicial process and the normal timeframe for such procedures, and any available alternative dispute resolution mechanisms that may be relevant to resettlement under the project;
iii. relevant law (including customary and traditional law) governing land tenure, valuation of assets and losses, compensation, and natural resource usage rights; customary personal law related to displacement; and environmental laws and social welfare legislation;
iv. laws and regulations relating to the agencies responsible for implementing resettlement activities;
v. gaps, if any, between local laws covering eminent domain and resettlement and the Bank's resettlement policy, and the mechanisms to bridge such gaps; and
vi. any legal steps necessary to ensure the effective implementation of resettlement activities under the project, including, as appropriate, a process for recognizing claims to legal rights to land--including claims that derive from customary law and traditional usage

8.5 Institutional Framework.

The findings of an analysis of the institutional framework covering

i. the identification of agencies responsible for resettlement activities and NGOs that may have a role in project implementation;
ii. an assessment of the institutional capacity of such agencies and NGOs; and
iii. any steps that are proposed to enhance the institutional capacity of agencies and NGOs responsible for resettlement implementation.

8.6 Eligibility.

Definition of displaced persons and criteria for determining their eligibility for compensation and other resettlement assistance, including relevant cut-off dates.

8.7 Valuation of and compensation for losses.

The methodology to be used in valuing losses to determine their replacement cost; and a description of the proposed types and levels of compensation under local law and such supplementary measures as are necessary to achieve replacement cost for lost assets.

8.8 Resettlement measures.

A description of the packages of compensation and other resettlement measures that will assist each category of eligible displaced persons to achieve the objectives of the policy (see OP 4.12, para. 6). In addition to being technically and economically feasible, the resettlement packages should be compatible with the cultural preferences of the displaced persons, and prepared in consultation with them.

8.9 Site selection, site preparation, and relocation.

Alternative relocation sites considered and explanation of those selected, covering
i. institutional and technical arrangements for identifying and preparing relocation sites, whether rural or urban, for which a combination of productive potential, locational advantages, and other factors is at least comparable to the advantages of the old sites, with an estimate of the time needed to acquire and transfer land and ancillary resources;
ii. any measures necessary to prevent land speculation or influx of ineligible persons at the selected sites;
iii. procedures for physical relocation under the project, including timetables for site preparation and transfer; and
iv. legal arrangements for regularizing tenure and transferring titles to resettlers.

8.10 Housing, infrastructure, and social services.

Plans to provide (or to finance resettlers' provision of) housing, infrastructure (e.g., water supply, feeder roads), and social services (e.g., schools, health services); plans to ensure comparable services to host populations; any necessary site development, engineering, and architectural designs for these facilities.

8.11 Environmental protection and management.

A description of the boundaries of the relocation area; and an assessment of the environmental impacts of the proposed resettlement and measures to mitigate and manage these impacts (coordinated as appropriate with the environmental assessment of the main investment requiring the resettlement).

8.12 Community participation.

Involvement of resettles and host communities in the following

i. a description of the strategy for consultation with and participation of resettlers and hosts in the design and implementation of the resettlement activities;
ii. a summary of the views expressed and how these views were taken into account in preparing the resettlement plan;
iii. a review of the resettlement alternatives presented and the choices made by displaced persons regarding options available to them, including choices related to forms of compensation and resettlement assistance, to relocating as individuals families or as parts of preexisting communities or kinship groups, to sustaining existing patterns of group organization, and to retaining access to cultural property (e.g. places of worship, pilgrimage centers, cemeteries);
iv. institutionalized arrangements by which displaced people can communicate their concerns to project authorities throughout planning and implementation, and measures to ensure that such vulnerable groups as indigenous people, ethnic minorities, the landless, and women are adequately represented.

8.13 Grievance procedures.

Affordable and accessible procedures for third-party settlement of disputes arising from resettlement; such grievance mechanisms should take into account the availability of judicial recourse and community and traditional dispute settlement mechanisms.

8.14 Organizational responsibilities.

The organizational framework for implementing resettlement, including identification of agencies responsible for delivery of resettlement measures and provision of services; arrangements to ensure appropriate coordination between agencies and jurisdictions involved in implementation; and any measures (including technical assistance) needed to strengthen the implementing agencies' capacity to design and carry out resettlement activities; provisions for the transfer to local authorities or resettlers themselves of responsibility for managing facilities and services provided under the project and for transferring other such responsibilities from the resettlement implementing agencies, when appropriate.

8.15 Implementation schedule.

An implementation schedule covering all resettlement activities from preparation through implementation including target dates for the achievement of expected benefits to resettlers and hosts and terminating the various forms of assistance. The schedule should indicate how the resettlement activities are linked to the implementation of the overall project.

8.16 Costs and budget.

Tables showing itemized cost estimates for all resettlement activities, including allowances for inflation, population growth, and other contingencies; timetables for expenditures; sources of funds; and arrangements for timely flow of funds, and funding for resettlement, if any, in areas outside the jurisdiction of the implementing agencies.

8.17 Monitoring and evaluation.

Arrangements for monitoring of resettlement activities by the implementing agency, supplemented by independent monitors as considered appropriate by the Bank, to ensure complete and objective information; performance monitoring indicators to measure inputs, outputs, and outcomes for resettlement activities; involvement of the displaced persons in the monitoring process; evaluation of the impact of resettlement for a reasonable period after all resettlement and related development activities have been completed; using the results of resettlement monitoring to guide subsequent implementation.

9. Basic Elements of a Resettlement Policy Framework (RPF)

a) a brief description of the project and components for which land acquisition and resettlement are required, and an explanation of why a resettlement plan cannot be prepared by World Bank appraisal;
b) principles and objectives governing resettlement preparation and implementation;

c) description of the process for preparing and approving resettlement plans;
d) estimated population displacement and likely categories of displaced persons, to the extent feasible;
e) eligibility criteria for defining various categories of displaced persons;
f) a legal framework reviewing the fit between borrower laws and regulations and Bank policy requirements and measures proposed to bridge any gaps between them;
g) methods of valuing affected assets;
h) organizational procedures for delivery of entitlements, including, for projects involving private sector intermediaries, the responsibilities of the financial intermediary, the government, and the private developer;
i) a description of the implementation process, linking resettlement implementation to civil works;
j) a description of grievance redress mechanisms;
k) a description of the arrangements for funding resettlement, including the preparation and review of cost estimates, the flow of funds, and contingency arrangements;
l) a description of mechanisms for consultations with, and participation of, displaced persons in planning, implementation, and monitoring; and
m) arrangements for monitoring by the implementing agency and, if required, by independent monitors.

Note:
When a resettlement policy framework is the only document that needs to be submitted as a condition of the loan, the resettlement plan to be submitted as a condition of subproject financing need not include the policy principles, entitlements, and eligibility criteria, organizational arrangements, arrangements for monitoring and evaluation, the framework for participation, and mechanisms for grievance redress set forth in the resettlement policy framework. The subproject-specific resettlement plan needs to include baseline census and socioeconomic survey information; specific compensation rates and standards; policy entitlements related to any additional impacts identified through the census or survey; description of resettlement sites and programs for improvement or restoration of livelihoods and standards of living; implementation schedule for resettlement activities; and detailed cost estimate.

Form TECH-5

Work Schedule and Planning for Deliverables

(Assignmant-A) (Rev.1)

No.	Deliverables (D-..)	Months												
		1	2	3	4	5	6	7	8	9	10	11	12	Total
D-1	**Review of Feasibility Study/Inception Report**													1.5
	i) Project Office Open													
	ii) Data Collection													
	iii) Inception Report													1st week
D-2	**Rprogress Reports**													10
	i) Monthly Progress Report													1st week
	ii)Quarterly Progress Report													1st week
D-3	**Preparation of Special Report/Memoranda (as and when required)**													
D-4	**Report on Loading Zone**													1
D-5	**Route Alignment and Survey Report**													6
	i) Acquisition of Satellite Imageries of Potential Corridors													
	ii) Indentification of three possible route													
	iii) Prelimary route selection recommendation to NTDC													
	iv) Approval of NTDC for recommended route													
	v) Walk over detailed survey fixing angle point and straight run marking													
	vi) Joint verification with NTDC													
	vii) Submission of draft route alignment Survey Report													1st week
	viii) Reviw by NTDC for the route Survey Report													
	ix) Final Survey Report incorporting comments of NTDC													last week

Form TECH-5

Work Schedule and Planning for Deliverables

(Assignmant-A) (Rev.1)

No.	Deliverables (D-..)	Months												
		1	2	3	4	5	6	7	8	9	10	11	12	Total
D-6	**Engineering Design Report**													5.5
	i) Collection of Data and Investigations													
	ii) Review of the Load flow study													
	iii) Conductor Optimization Draft Report													1st week
	iv) Reviw by NTDC for Conductor Optimization Draft Report													
	v) Design Criteria Draft Report													1st week
	vi) Reviw by NTDC for Design Criteria Draft Report													
	vii) Design Criteria Final Report													1st week
	viii) Project Implementation Planning and Procurement Packaging													
	ix) Preparation of BOQ and Costing													1st week
D-7	**EIA Report**													4
	i) Draft EIA Report													1st week
	ii) Comments of NTDC/World Bank													
	iii) Final EIA Report													1st week
D-8	**RAP Report**													3
	i) Draft RAP Report													1st week
	ii) Comments of NTDC/World Bank													
	iii) Final RAP Report													1st week

Form TECH-5

Work Schedule and Planning for Deliverables

(Assignmant-A) (Rev.1)

No.	Deliverables (D-..)	Months												
		1	2	3	4	5	6	7	8	9	10	11	12	Total
D-9	**Preparation of Bidding Documents**													3
	i) Preparation of Tender Documents (Draft)													1st week
	ii) Review by NTDC													
	iii) Final Tender Documents incorporting comments of NTDC													1st week
D-10	**Contractor's Bids Evaluation Report**													3.5
	i) Clarification of Pre-bid Inquiries													
	ii) Evaluation Bids													1st week
	iii) Approval and Award of Contract													
D-11	**Technical Training**													1
	i) Arrangement Trnainning Schedule													
	ii) Training													

Form Tech - 5
Work Schedule And Planning for Deliverables
(Assignment B) (Rev.1)

No.	Deliverables (D..)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	Total
D-1	**Construction Reports**																																					36
	i) Comments on Monthly Progress Report submitted by EPC Contractor																																					2nd week
	ii) Quarterly Progress Report																																					2nd week
	iii) Annual Work Plan																																					2nd week
D-2	**Project Implementation status report**																																					35
	i) Approval of Geological Investigation Report																																					
	ii) Approval of Line Route Drawings																																					
	iii) Budget & Financial Implementation Activity																																					
	iv) Approval of Construction Material																																					
	v) Approval of Profile Drawing																																					
	vi) Inspection of T/Line Material																																					
	vii) Foundation Work																																					
	viii) Erection of Tower																																					
	ix) Stringing Work																																					
D-3	**Implementation Status Report for EMP, SAP &RAP Work Plan etc. on regular basis**																																					33
	i) Preparation of Report																																					
	ii) Review by PMU (Envirnoment) WAPDA House Lahore																																					
	iii) Funds allocation by Client representative																																					
	iv) Compensaton to affect																																					

Form Tech - 5
Work Schedule And Planning for Deliverables
(Assignment B) (Rev.1)

No.	Deliverables (D..)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	Total
D-4	**Minutes of Weekly/monthly meeting held with Contractor**																																					36
D-5	**Correction/Amendment in Drawings submitted by Contractor**																																					35
D-6	**As-Built Drawings**																																					3
	i) Contractor's Submittals																																					
	ii) Reviewed and approved by Consultant																																					
	iii) Final as-built drawings																																					
D-7	**Testing & Commissioning Report**																																					5
	i) Joint patrolling of T/Line with staff of Consultant, Client & Contractor																																					
	ii) Resolve Line Problems																																					
	iii) Initial Testing of the line																																					
	iv) Final Testing & Comminssioning																																					
	v) Preparation of Punch list																																					
D-8	**Maintenance & Operation Mannuals**																																					3
	i) On job Training for Construction of T/Line, Operation & Maintenance																																					
	ii) Setting of work shop equipped with line maintenance																																					
D-9	**Completion Report**																																					2
	i) Draft Completion Report																																					1st week
	ii) Final Completion Report																																					1last week
D-10	**Training**																																					1

Continuous Intermittent

APPENDIX-B (KEY EXPERTS)

The consultant to attach the original CV's of the key experts with their signatures as a confirmation of their availability here.

Form TECH-6(B)
Assignment – A & B

CURRICULUM VITAE (CV)

Position Title and No.	Project Manager / Team Leader (K-1)
Name of Expert	Park, Jae-Heun
Date of Birth	January 7, 1968
Country of Citizenship / Residence	Republic of Korea

Education:

1990, Hongik University, Korea, B.E in Electrical Engineering
Overseas Construction Global Expert, International Contractors Association of Korea (2009)

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2009 to date	KEPCO	Senior Manager	Overseas Project Development Department
2002 to 2008	KEPCO	Senior Manager	Suwon Power Transmission District Office
1995 to 2002	KEPCO.	Transmission Engineer	Power System Construction Office
1992 to 1995	KEPCO	Transmission Engineer	Daegu Power Transmission District Office.

Membership in Professional Associations and Publications

Member of KEEA (Korea Electric Engineers Association)
Member of KECA (Korea Electrical Contractors Association)

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task A1: Review of Feasibility Study & Use of Existing Information Task A2: Transmission line Route Alignment and Survey Task A3: Preparation of Tender Level Engineering Design Sub-Task A3.1: Collection of Data and Investigations Sub-Task A3.2: Design Criteria Sub-Task A3.3: Project Implementation Planning and Procurement Packaging Sub-Task A3.4: Preparation of BOQ and Costing Sub-Task A3.5: Environmental Assessment, Management Plan, EIA etc. Sub-Task A3.6: Social Impact Assessment and Beneficiary Participation Aspects and Preparation of Resettlement Action Plan Task A4: Preparation of Tender Documents Task A6: Tendering and Contracting Task A7: Technical Training	**Name of assignment or project:** Consultants' Services for Construction of Bibiyana - Kaliakoir 400kV Transmission Line **Year**: 2011 ~ Present **Location**: Bangladesh **Client:** PGCB (Power Grid Company of Bangladesh Limited) **Main project features**: Consultancy service for 400kV Transmission Line and two 400kV Conventional Substations Construction Project **Positions held**: Project Manager (Transmission Line Expert) **Activities performed**: PBCB's Data reviewing, Conducting necessary site survey, Ascertaining the scopes of works, Planning and Concept Design the Projects, Preparing specifications for various equipment, materials & services, Preparing a Bill of Quantity (BOQ), Preparing relevant documents for invitation of tenders, Assist the Bid evaluation and Carrying out Construction Supervision
	Name of assignment or project: Consultants' Services for Construction of Aminbazar - Meghnaghat 400kV Transmission Line **Year**: 2008 ~ 2013 **Location**: Bangladesh **Client:** PGCB (Power Grid Company of Bangladesh Limited) **Main project features**: Consultancy service for 400kV Transmission Line Construction Project **Positions held**: Project Manager (Transmission Line Expert) **Activities performed**: PBCB's Data reviewing, Conducting necessary site survey, Ascertaining the scopes of works, Planning and Concept Design the Projects, Preparing specifications for various equipment, materials & services, Preparing a Bill of Quantity (BOQ), Preparing relevant documents for invitation of tenders, Assist the Bid evaluation and Carrying out Construction Supervision

	Name of assignment or project: Feasibility study of Bangladesh Sylhet T/Lines Project **Year**: 2008 ~ 2008 (3 months) **Location**: Bangladesh **Client:** PGCB (Power Grid Company of Bangladesh Limited) **Main project features**: Feasibility of the 230kV transmission system and 400kV basic designs to the Bangladesh. The consultant carried out demand forecasting, a master plan, a feasibility study, and a basic design etc. **Positions held**: Project Manager (Transmission Line Expert) **Activities performed**: Feasibility study, data collection and review, system improvement plan, establish 400kV line and tower design criteria, basic design of the 400kV transmission system, review equipment specification and technical drawings.
	Name of assignment or project: Line Route and ESIA Study for Han - Bobo Dioulasso – Sikasso - Bamako Interconnection Project in WAPP **Year:** 2008 ~ 2010 **Location**: Ghana, Burkina Faso, Mali **Client:** WAPP (West African Power Pool) **Main project features**: The project is to provide the Line Route and ESIA study of Han-Bobo Dioulasso-Sikasso-Bamako Interconnection Project in WAPP (Line length 742km, 5 substations) **Positions held**: Transmission Line Expert **Activities performed**: Line Route and ESIA study for the 225/330kV high voltage transmission line (about 750km) from Han in the northern part of Ghana to the capital of Mali, Bamako, through Bobo Dioulasso in Burkina Faso, and Sikasso in Mali with associated high voltage substation
	Name of assignment or project: The development study on the power system operation and protection scheme in Myanmar **Year:** 2006 ~ 2008 **Location**: Yangon, Myanmar **Client:** MEPE (Myanmar Electric Power Enterprise)

	Main project features: For the 230kV Transmission system, Field survey and data collection, strategies and action plan for system operation, system statistics management system, Transmission line protection system review and reinforcement action plan **Positions held**: Transmission Line Expert **Activities performed**: Field survey and data collection, strategies and action plan for system operation, system statistics management system, Transmission line design review, on-the-job training for the transmission line protection system
	Name of assignment or project: Power Development Master Plan and Institutional Strengthening in Cambodia **Year:** 2005 ~ 2006 **Location**: Cambodia **Client:** EDC **Main project features**: Establishment the Power development master plan and 230kV transmission system design **Positions held**: Transmission Line Expert **Activities performed**: Feasibility study, data collection and review, power system analysis in Cambodia, 230kV Transmission system engineering design, preparation of bid documents between two (2) Cambodian cross-border transmission systems, review of technical design and drawings substation
	Name of assignment or project: Feasibility study, basic design of 500kV Transmission system in Myanmar **Year:** 2003 ~ 2005 **Location**: Yangon, Myanmar **Client:** MEPE (Myanmar Electric Power Enterprise) **Main project features**: The project is to provide the feasibility of the 500kV transmission system upgrade and 500kV basic designs to the Myanmar. The consultant carried out demand forecasting, a master plan, a feasibility study, and a basic design etc. **Positions held**: Transmission Line Expert **Activities performed**: Feasibility study, data collection and review, developing power master plan, system

	improvement plan, establish 500kV line and tower design criteria, basic design of the 500kV transmission system, review equipment specification and technical drawings, training of counterpart engineer
	Name of assignment or project: 765kV Uljin - Shintaebaek Transmission line Project **Year**: 2002 ~ 2003 **Client:** Ministry of Commerce, Industry and Energy **Location:** Korea **Main project features:** The project is to construct the 765kV transmission line from Uljin nuclear power plant to 765kV Sintaebaek substation (approximately 84km) **Positions held:** Transmission Line Expert **Activities Performed:** Detail line design survey, Environmental Impact Assessment, Preparation and evaluation of bidding document, review of engineering design and drawings, supervision of project implementation.
	Name of assignment or project: 765kV Sinnamwon - Shinansung Transmission line Project **Year:** 1998 ~ 2000 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 765kV transmission line from 765kV Sinnamwon Substation to 765kV Sinansung Substation (approximately 220km) **Positions held:** Transmission Line Engineer **Activities Performed:** Performed feasibility study (data collection, preliminary engineering design, detail line design survey, Environmental Impact Assessment, cost estimates)
	Name of assignment or project: 765kV Sinseosan - Sinansung Transmission line Project **Year**: 1995 ~ 1998 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features**: The project is to construct the 765kV transmission line from 765kV Sinseosan Substation to 765kV Sinansung Substation **Positions held**: Transmission Line Engineer

	Activities Performed: Performed preliminary engineering design, equipment specifications, cost estimates, Carried out basic design (conductor size, tower structure, tower, hardware, insulator)

Expert's Contact information:

Email: jaeheun.park@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Jae Heun Park		2014. 12. 22
Name of Expert	Signature	Date

Jae Heun Park		2014. 12. 22
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6(B)
Assignment - A & B

CURRICULUM VITAE (CV)

Position Title and No.	Transmission Line Design Expert (K-2)
Name of Expert	Park, Dug-Man
Date of Birth	January 28, 1956
Country of Citizenship / Residence	Republic of Korea

Education:`

- 1982, Bachelor's degree in Electrical Engineering, Korea National Open University

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2007 to date	KEPCO	General Manager	Overseas Project Development Department
1997 to 2007	KEPCO.	Senior Manager	Seoul Power Transmission District Office
1977 to 1997	KEPCO	Transmission Engineer	Power System Construction Office.

Membership in Professional Associations and Publications

Member of KEEA (Korea Electric Engineers Association)
Member of KECA (Korea Electrical Contractors Association)

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task B1: Design Review Sub-Task B1.1: Review and Approval of Contractor's Design Sub-Task B1.2: As Built Drawings Sub-Task B1.3: Punch List Task B2: Contracts Management Sub-Task B2.1: Monitoring of Project Progress Sub-Task B2.6: Review and Reporting Task B3: Construction Supervision Sub-Task B3.6: Transmission Line Operation & Maintenance. Sub-Task B3.7: Know-How Transfer and Training	**Name of assignment or project:** Consultants' Services for Construction of Bibiyana - Kaliakoir 400kV Transmission Line **Year**: 2011 ~ Present **Location**: Bangladesh **Client:** PGCB (Power Grid Company of Bangladesh Limited) **Main project features**: Consultancy service for 400kV Transmission Line and two 400kV Conventional Substations Construction Project **Positions held**: Transmission Line Expert **Activities performed**: PBCB's Data reviewing, Conducting necessary site survey, Ascertaining the scopes of works, Planning and Concept Design the Projects, Preparing specifications for various equipment, materials & services, Preparing a Bill of Quantity (BOQ), Preparing relevant documents for invitation of tenders, Design review and approval for Construction Supervision
	Name of assignment or project: 765kV Singori-Bukkeongnam Transmission Line Construction Project **Year:** 2009-2011 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features:** The project is to construct the 765kV transmission line from 765kV Singori Substation to 765kV Bukkeongnam Substation (approximately 90.5km) **Positions held:** Transmission Line Expert **Activities Performed:** Performed Detail design (data collection, Planning and Concept Design the Projects, Preparing specifications for various equipment, materials), Design review and approval for Construction Supervision
	Name of assignment or project: Consultants' Feasibility Study of Establishing 230kV Substation in Cambodia **Year**: 2007 ~ 2009 **Location**: Phnom Penh, Cambodia

	Client: EDC (Electricity of Cambodia) **Main project features**: Basic design of the 230kV transmission system **Positions held**: Transmission Line Expert **Activities performed**: · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Carried out basic design (conductor size, tower structure, tower, hardware, insulator) · Made drawings and implementation plan
	Name of assignment or project: 765kV Singapyeong-Sinansung Transmission Line Construction Project **Year:** 2004-2006 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features:** The project is to construct the 765kV transmission line from 765kV Singapyeong Substation to 765kV Sinansung Substation (approximately 80km) **Positions held:** Transmission Line Expert **Activities Performed:** Performed Detail design (data collection, Planning and Concept Design the Projects, Preparing specifications for various equipment, materials), Design review and approval for Construction Supervision
	Name of assignment or project: 345kV Songdo-youngheung,154kV Songdo transmission line **Year:** 2002-2003 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: The project is to construct the 345kV G/P power Transmission Line Extension. **Positions held**: Transmission Line Expert **Activities performed**: · Certification of design and drawings, Procurement, Construction schedule and supervision of project implementation

	Name of assignment or project:380kV Boogsong-Tamaret Transmission Line Project **Year:** 1999-2001 **Location**: Thailand **Client:** EGAT **Main project features**: · The project is to construct the 380kV transmission line from Boogsong to Tamaret. **Positions held**: Transmission Line Expert **Activities performed**: · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification , Design review and approval for Construction Supervision
	Name of assignment or project: 345kV Singapyeong - Uijeongbu Transmission line Project **Year**: 1998 ~ 1999 **Client:** Ministry of Commerce, Industry and Energy **Location:** Korea **Main project features:** · The project is to construct the 345kV transmission line from Singapyeong to Uijeongbu **Positions held:** Transmission Line Expert **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification , Design review and approval for Construction Supervision

	Name of assignment or project: 345kV Samcheonpo - Sinkimhae Transmission line Project **Year:** 1997 ~ 1997 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to design on the 345kV transmission line from Samcheonpo to Sinkimhae **Positions held:** Transmission Line Expert **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Carried out basic design (conductor size, tower structure, tower, hardware, insulator) · Made drawings and implementation plan
	Name of assignment or project: 154kV Seosan-Taean , 345kV Gwayang, Hadong - Gwangyang Transmission line Project **Year**: 1993 ~ 1997 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features**: The project is to construct the 154kV transmission line from Seosan to Taean and the 345kV transmission line from Gwagyang, Hadong to Gangyang **Positions held**: Transmission Line Engineer **Activities Performed**: · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification, Design review and approval for Construction Supervision

	Name of assignment or project: 345kV Ildo-Yangju Transmission line Project **Year:** 1990 ~ 1993 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 345kV transmission line from Ildo to Yangju **Positions held:** Transmission Line Engineer **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification, Design review and approval for Construction Supervision
	Name of assignment or project: 154kV Anseong-Jincheon Transmission line Project **Year:** 1987 ~ 1990 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 154kV transmission line from Anseong to Jincheon **Positions held:** Transmission Line Engineer **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification, Design review and approval for Construction Supervision
	Name of assignment or project: 345kV Gojeong, Golee - Sinpohang, Youngcheong, Jepo Transmission line Project **Year:** 1983 ~ 1987 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** · The project is to construct the 345kV transmission line from Gojeong, Golee to Sinpohang, Youngcheong, Jepo **Positions held:** Transmission Line Engineer

	Activities Performed: · Performed feasibility study (data collection, preliminary engineering design, equipment specifications, cost estimates) · Responsibility of engineering design, preparation for bid document and including technical specification, Design review and approval for Construction Supervision · Construction schedule and supervision of project implementation

Expert's Contact information:

Email: ydmpark@daum.net

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Dug Man Park		2014. 12. 26
Name of Expert	Signature	Date
Park Jae Heun		2014. 12. 26
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6(B) Assignment – B

CURRICULUM VITAE (CV)

Position Title and No.	Transmission Line Survey and Mapping Expert (K-3)
Name of Expert	Abdul Waheed Mir
Date of Birth	December, 6, 1941
Country of Citizenship / Residence	Pakistan

Education:

Name of Institution	**Year**	**Degree/Diploma**
University of the Punjab Lahore – Pakistan	1961	B.Sc. - Degree
University of East Africa, University College, Nairobi-Kenya	1967	B.Sc. Engineering in Surveying and Photogrammetry - Degree

Post Graduate Training

1. Short Training Course on Remote Sensing Applications, Lahore (March, 1991)
2. Training Course in Applied Remote Sensing, Lahore (June, 1994)
3. Short Training Course on Satellite Remote Sensing and Geographic Information System (GIS) Applications, SUPARCO, Karachi (30 October to 08 November, 1995).

Employment Record:

Period	**Employing organization and title/ position.**	**Country**	**Summary of activities performed relevant to the Assignment**
March 2012 to update (Intermittent input)	EnMasse Pvt. Ltd. / Survey and Mapping Expert	Pakistan	Checking of route line, profiles, Tower spotting & Bill of quantities of 132kV Transmission lines • Mehr-Dadu-Larkana (length: 128 kms) • Thull-Jacobabad (length: 30 kms) • Araian Road-Sukkur City (length: 8 kms) • Dadu New-Mehar-Larkana-3 (length: 65 kms) • Larkana-2 Larkana-Old (length: 5 kms) • T.M Khan-Badin (length: 60 kms) • Mirpurkhas-Kindairi 2nd Circle (length: 40 kms) • 132kV Transmission Line MEPCO – ADB (Tranche-III) (length: 147kms)
September, 2011 to – September, 2012 (Full Time)	Pakistan Engineering Services (Ltd) / Chief Engineer (Survey)	Pakistan	• Establishment of horizontal and vertical survey control network at the project site. • Surveying and mapping related to Dasu Dam Project. • KKH relocation, Right Bank Access Road – RAR-1, RAR-2 and RAR-3. • Site survey of resettlement villages • Assistance to Patwaris in land acquisition work. • Monitoring line and grade of exploration adits. • Survey and mapping work for geotechnical

			investigations. • Neelum Jhelum Hydro Project Transmission Line – to Rawat & Ghakhar –Joint Venture of G&ES and Barqaab.
Sep-2011 to Sep.- 2012	Pakistan Engineering Services Pakistan (Pvt.) Ltd (PES) / Chief Engineer (Survey) – DASU Hydropower Project	Pakistan	• Responsible for all the surveying and mapping activities related to this project; organizing field work and supervision of the same to assure quality of the data acquired. • Checking and verification of the project contractor's survey control and other related surveying and mapping work at site pertaining to KKH Relocation.
June-2008 to April 2009 (Full time basis) From May 2009 to July 2010 Part time basis.	National Engineering Services Pakistan (Pvt.) Ltd. (NESPAK) / Chief Surveyor – Neelum Jhelum Hydroelectric Project – Azad Jammu & Kashmir (Pakistan)	Pakistan	• Responsible for all the surveying and mapping activities related to this project; organizing field work and supervision of the same to assure quality of the data acquired. • Checking and verification of the project contractor's survey control and other related surveying and mapping work at site. • Checking of line and grade & setting out of works by the contractor.
June 2008 to July 2010	NESPAK Foundation Neelum Jhelum Hydropower Project / Chief Engineer (Survey)	Pakistan	• Establishment of survey control. • Supervision of the contractor's survey & mapping work and setting out of works. • Connection of site network to Rawalpindi SOP SBM for verification of site coordinates, using GPS.
2007-2008	Associated Consulting Engineers (Pvt.) Ltd., (ACE) Lahore / Head Survey & GIS	Pakistan	• Responsible for supervision and quality assurance of surveying and mapping including GIS activities related to the Mahmand (Munda) Dam Project (KPK) – Pakistan.
April 2004 to 2008	Geomatics and Engineering Services Pvt. Ltd. Lahore. (Pakistan) / Chief Survey and Mapping	Pakistan	Provided surveying and mapping services on a number of hydro power projects, and other major ones, a few are listed hereunder: ▪ Jinah Hydropower Project, Kalabagh ▪ Munda Dam Multipurpose Hydropower Project, NWFP ▪ Muzafargarh – Gati 500 kV power transmission line project (180 km). Pakistan. ▪ **Bostan Zob RailwayRoute Survey (295 km),Balochistan, Pakistan** ▪ RahimYar Khan – TMP – Lodharan (N-5) Highway (202 Km) Route Survey-Pakistan ▪ 800 km long Iran-Pakistan Gas Pipeline Project – GPS

December 2001 to April 2004	NESPAK Foundation – Lahore / Chief Surveyor Ghazi-Barotha and Basha Dam	Pakistan	• **Survey & Mapping Advisor to H&TE Division (January 2003 to April 2004),** • Supervisor surveying and mapping work of the Mangla Dam Raising Project & KKH Up-gradation Project. • **Chief Surveyor of the Ghazi Barotha Hydropower Project (Dec. 2001 to July 2002),** • Responsible for Survey and Mapping services applicable to this huge hydro power construction project. **Chief Surveyor** **Basha Dam Project (July 2002 to Dec. 2003).** • Responsible for establishment of credible horizontal and vertical survey control at project site and in the reservoir area plus detailed topographic survey of the project site. To provide mapping assistance to the site geologists and carry out other misc. survey and mapping work in the project area.
July 1985 – to – December 2001	National Engineering Services Pakistan (Pvt.) Ltd. / General Manager (Surveying & Mapping)	Pakistan	As Head of Survey and Mapping (S&M) responsible for providing surveying services on Company-wide basis and to outside agencies. S & M group had a staff of four survey engineers and eighty plus surveyors including four CAD men for computer aided mapping services. Surveying Mapping services provided on several projects like roads/ highways survey(3700kms), Motorway (M-3) Survey (110 Km), Survey of 12650Km of distribution digitalization and computer aided mapping of Mangla and Tarbela watershed project using GPS with AGRAR-UND TECHNIK GMBH. **Foreign Projects** - Djogou - Porga Road (195 Km.) in Benin Republic (West Africa). - Alkhwad Dam Project, Updation of Krookies in Almawaleh Area, Roads in Mussandam Province, Diba and Madha Ring Roads and Kasab Coastal road in Sultanate of Oman (length 45 kms). - Yemen Arab Republic Farm Roads Project (length 60 kms)
1978 to 1985	Montreal Engineer Company Limited, Montreal, (MONENCO) – Canada/ Surveying Engineer	Nigeria & Indonesia	• Resident Survey Engineer, Jebba Hydroelectric Development Project, Jebba, Nigeria. Head Survey Department at the Jebba Project, was responsible for surveying and mapping activities on the Geodetic control network and Topographical surveys. **Advisor on Lower Solo River Water Resources Project, Indonesia.** Surveying Specialist Adviser to the Lower Solo

			River Development Project, Indonesia, undertaken by Monenco on behalf of the Canadian International Development Agency. Responsible for reviewing methodology and quality control of topographic mapping & surveys and precise levelling surveys covering the 6000 sq. km. of lower Solo River drainage basin. The project included studies of five major water resources projects to feasibility level.
1975 to 1978	Dastoor Survey Company and Dastan computer Services, Teheran / Senior Engineer	Iran	Engaged on a variety of surveying jobs i.e. control surveys, roads/highways survey and engineering Surveys. Chief Surveyor on the following projects: - Roads Survey at different locations - Kermanshah New Township - Gilan Wood and Paper Industries Complex on the Caspian Coast.
1971 to 1975	M/s TAMS, New York, USA / Survey Engineer Tarbela Dam,	Pakistan	Responsible for: - Geodetic control network at the project site (Tarbela). - Topographic and Hydrographic surveys for the project. - Construction survey supervision of major civil works.
1967 to 1971	Hunting Surveys and Consultants Limited, Boreham-Wood, Herts, England / Field Surveyor	Tanzania, Iran & Libya	Engaged on a variety of land, geodetic and engineering survey jobs. Carried out survey and photogrammetry ground control work for road route mapping in Tanzania. Gas pipe line route survey, Oilwell location surveys, and boundary survey, Geodetic control and Topo surveys in Iran and Libya. Surveying services for different Oil Companies in Libya and Iran including: - Geodetic control network over very large area in Libya - Ground control for photogrammetric mapping of highway project in Tanzania (700 miles) - Topographical surveys (Libya and Iran) - Oil well location surveys (Libya) - Transmission line and gas line route surveys. (Iran)

Membership in Professional Associations and Publications:

1. Member, Pakistan Engineering Council
2. Member, Pakistan Engineering Congress.
3. Member, Institute of Engineers Pakistan
4. Fellow, Royal Institution of Chartered Surveyors-London-England.

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent
Punjabi	Excellent	Excellent	Excellent

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	**Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks**
Review of line route survey given in Feasibility Report and other available and relevant data. Transmission line Route alignment and survey	1) As Chief Engineer Survey at G&ES carried out route selection, study and physical verification of routes for power lines route surveys and associated topographic survey and computer aided mapping of the plan and profile drawings. 2) As Principal Engineer and later G.M. In-charge of Survey and Mapping Specialty Group (NESPAK) supervised the surveying and mapping services of all the projects. 3) As survey engineer engaged on Projects in Nigeria & Indonesia for MONENCO carried out surveying and mapping duties. 4) As survey engineer employed by Dastoor Survey Co. Teheran (Iran) carried out survey and mapping duties of a variety of projects all over Iran. 5) As survey engineer employed by Hunting Survey and Consultants Limited – Borehmawood - England carried out a variety of survey and mapping services on projects in Tanania, Libya and Iran.
Review of detailed survey, profile and Tower spotting to be carried out by the EPC Contractor.	Complete survey checking, review of plan and profile drawings, preparation of structure list, bill of quantities, preparation bidding documents for several 500kV Transmission lines in NTDC Network, 220kV & 132kV Transmission Line as described in Employment Record.

Expert's Contact information:

Email: enmasse@enmassepakistan.com

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Abdul Waheed Mir		30-12-2014
Name of Expert	Signature	Date
Park, Jae Heun		30-12-2014
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6(B)
Assignment-A

CURRICULUM VITAE (CV)

Position Title and No.	Transmission Line Survey and Mapping Expert (K-3)
Name of Expert	Choi, Byang Chur
Date of Birth	March 20, 1961
Country of Citizenship / Residence	Pakistan

Education:

1984 Bachelor's degree in electronics engineering, Inha University Korea

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2003 to date	KEPCO	Senior Manager	Overseas Project Development Department
1987 to 2003	KEPCO	Transmission Engineer	Power System Construction Office.

Membership in Professional Associations and Publications

Line design S/W using *PLS-CADD,*
Mapping S/W using *ARC-GIS* & *Global Mapping*

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task A1: Review of Feasibility Study & Use of Existing Information Task A2: Transmission line Route Alignment and Survey Sub-Task A3.5: Environmental Assessment, Management Plan, EIA etc. Sub-Task A3.6: Social Impact Assessment and Beneficiary Participation Aspects and Preparation of Resettlement Action Plan Task A7: Technical Training	**Name of assignment or project:** 330 kV Overhead TL. Iraq Ministry of Electricity **Year**: 2013 **Location**: Iraq **Client:** Samsung Engineering **Main project features**: Design for Power 330kV Transmission Line and Substations **Positions held**: Transmission Line Design **Activities performed**: Power lines planning, tower spotting, surveying and Concept Design the Projects, Preparing specifications for various equipment, materials & services
	Name of assignment or project: Design for Saudi Arabia Aramco 230kV Shaybah T/P Transmission Line **Year**: 2012 ~ 2013 **Location**: Saudi Arabia **Client:** Samsung Engineering **Main project features**: Consultancy service for Transmission Line and Substations **Positions held**: Transmission Line Design **Activities performed**: Power lines planning, tower spotting, surveying and Concept Design the Projects, Preparing specifications for various equipment, materials & services
	Name of assignment or project: Various 33kV Distribute line Design using PLS-Pole **Year**: 2011 ~ 2012 **Location**: Haiti **Client:** USAID **Main project features**: Consultancy service for Power Line **Positions held**: Transmission Line Design **Activities performed**: Power lines planning, tower spotting, surveying and FE tower analyze using PLS-tower

	Name of assignment or project: Power Transmission System Design using PLS-CADD for 110kV Bintan Transmission Line, 120km& new Substation Basic Design. **Year:** 2008 ~ 2009 **Location**: Indonesia **Client:** STX **Main project features**: Basic Design for 110kV Transmission Line and Substations **Positions held**: Project Manager **Activities performed**: Power lines planning, tower spotting, surveying and Concept Design the Projects, Preparing specifications for various equipment, materials & services
	Name of assignment or project: Feasibility study, basic design of 500kV Transmission system in Myanmar **Year:** 2003-2005 **Location**: Yangon, Myanmar **Client:** MEPE (Myanmar Electric Power Enterprise) **Main project features**: The project is to provide the feasibility of the 500kV transmission system upgrade and 500kV basic designs to the Myanmar. The consultant carried out demand forecasting, a master plan, a feasibility study, and a basic design etc. **Positions held**: Transmission Line Engineer **Activities performed**: Feasibility study, line route survey and mapping, developing power master plan, system improvement plan, establish 500kV line and tower design criteria, basic design of the 500kV transmission system, review equipment specification and technical drawings, training of counterpart engineer
	Name of assignment or project: 765kV Uljin - Shintaebaek Transmission line Project **Year**: 2002 ~ 2003 **Client:** Ministry of Commerce, Industry and Energy **Location:** Korea

	Main project features: The project is to construct the 765kV transmission line from Uljin nuclear power plant to 765kV Sintaebaek substation (approximately 84km) **Positions held:** Transmission Line Expert **Activities Performed:** Detail line design survey, Line route mapping, Environmental Impact Assessment, Preparation and evaluation of bidding document, review of engineering design and drawings, supervision of project implementation.
	Name of assignment or project: 765kV Sinnamwon - Shinansung Transmission line Project **Year:** 1998 ~ 2000 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 765kV transmission line from 765kV Sinnamwon Substation to 765kV Sinansung Substation (approximately 220km) **Positions held:** Transmission Line Expert **Activities Performed:** Performed feasibility study (data collection, preliminary engineering design, detail line design survey and mapping, Environmental Impact Assessment, cost estimates)
	Name of assignment or project: 765kV Singapyeng - Sintaebaek Transmission line Project **Year**: 1993 ~ 1998 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features**: The project is to construct the 765kV transmission line from 765kV Singapyeng Substation to 765kV Sintaebaek Substation **Positions held**: Transmission Line Engineer **Activities Performed**: Performed feasibility study (data collection, preliminary engineering design, detail line design survey and mapping, Environmental Impact Assessment, cost estimates)

	Name of assignment or project: 345kV Ildo-Yangju Transmission line Project **Year:** 1990 ~ 1993 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 345kV transmission line from Ildo to Yangju **Positions held:** Transmission Line Engineer **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, line route survey and mapping) · Carried out basic design (conductor size, tower structure, tower, hardware, insulator) · Certification of civil/foundation design and drawings, Procurement, Construction schedule and supervision of project implementation
	Name of assignment or project: 154kV Anseong-Jincheon Transmission line Project **Year:** 1987 ~ 1990 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features:** The project is to construct the 154kV transmission line from Anseong to Jincheon **Positions held:** Transmission Line Engineer **Activities Performed:** · Performed feasibility study (data collection, preliminary engineering design, line route survey and mapping) · Carried out basic design (conductor size, tower structure, tower, hardware, insulator) · Certification of civil/foundation design and drawings, Procurement, Construction schedule and supervision of project implementation

Expert's Contact information:

Email: Catchchoi@hanmail.net

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Choi, Byung Chur	Choi	2014.12.26
Name of Expert	Signature	Date

Park, Jae Heun		2014.12.26
Name of Authorized Representative of the Consultant	Signature	Date

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Form TECH-6(B)
Assignment-B
CURRICULUM VITAE (CV)

Position Title and No.	Engineer Expert (Civil) Structure (K-4)
Name of Expert	Kim, Donghyun
Date of Birth	31 August 1969
Country of Citizenship / Residence	Republic of Korea

Education:

Chung-Ang University, Korea, B.A in Civil Engineering, 1996
Advanced Project Management Training Course (KEPCO), 2004
Expert for Civil Engineer Training Course (KEPCO), 2008
The Advanced Course of Overseas Business Expert (Hankuk University of Foreign Studies, KEPCO E&C, KPLI, KERI), 2012
Project Management, Project Management Institute (PMI), KOREA, 2000

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2012 to date	KEPCO	Senior Manager	Overseas Project Development Department
2001 to 2012	KEPCO	Civil Engineer	Transmission &Substation Planning Department
1996 to 2001	KEPCO.	Civil Engineer	Young-Heung Power plant Construction Department

Membership in Professional Associations and Publications

Member of the Korea Society of Professional Engineer (KSPE), 2000
Member of the Project Management Professional (PMP), 2001

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task B1: Design Review Sub-Task B1.1: Review and Approval of Contractor's Design Sub-Task B1.2: As Built Drawings Sub-Task B1.3: Punch List Task B2: Contracts Management Sub-Task B2.1: Monitoring of Project Progress Sub-Task B2.6: Review and Reporting Task B3: Construction Supervision Sub-Task B3.7: Know-How Transfer and Training	**Name of assignment or project :** CONSTRUCTION OF BIBIYANA-KALIAKOIR 400KV TRANSMISSION AND ASSOCIATED SUBSTATION PROJECT **Year**: 2012~ present (18 months) **Location**: Bibiyana ~Kaliakoir in Bangladesh **Client:** PGCB (Power Grid Company in Bangladesh) **Main project features** : Project Implementation Consultant (PIC) for 400kV power transmission and substation project **Positions held**: Consultant / Civil engineer **Activities performed** : Completion of 400kV Transmission Line and Substations. Completion of Earth Filling and Foundation of Kaliakoir Substation.
	Name of assignment or project : SECOND POWER TRANSMISSION AND DISTRIBUTION PROJECT **Year**: 2013~ present (6 months) **Location**: Kampot, Sihanoukville in Cambodia **Client:** EDC (ELECTRICITE DU CAMBODGE) **Main project features** : Project Implementation Consultant (PIC) for second power transmission and distribution project **Positions held**: Consultant /Civil engineer **Activities performed** : Completion of 230kV Kampot-Stunghav Transmission Line and Substations. Completion of implementing for construction drawing for civil works
	Name of assignment or project : Consultancy services for Greater Mekong Power Network Development Project **Year**: 2011 ~ 2011 (5 months) **Client**: EDC (Electricite Du Cambodge) **Location**: Cambodia **Main project features**: Consulting project for Greater Mekong Power Network Development Project, 115/22 kV Substation and Associated Transmission Line are constructed at Sihanoukville area in Cambodia

	Positions Held: Civil Engineer **Activities performed**: Transmission Line: Approximately 13km long 115 kV double circuit, Substations: 230 kV substation at Sihanoukville (115kV bay extension) and 115 kV Stung Hav substation (newly constructed)
	Name of assignment or project: West African Power Pool Interconnection Transmission line Route Survey & ESIA project **Year**: 2012 ~ Present **Location**: Benin, Cote D'ivior, Sierra Leone, Liberia, Guinea **Client**: West African Power Pool (WAPP), Benin **Main project features**: 1,411km(length), Transmission Line Route Detail Survey, ESIA Study **Positions held**: Civil Engineer **Activities performed**: Transmission line route investigate, Detail surveys, and ESIA Study
	Name of assignment or project: 345kV Migum Substation Construction Project **Year**: 2011 ~ 2012 (8 months) **Location**: Korea, Namyangju **Client**: Ministry of Knowledge, Economy, Korea **Main project features**: 345kV 4Banks, 120,000M2 **Positions held**: Civil Engineer **Activities performed**: Construction Management, drawings (type of tower, conductor size, insulation strings), cost estimates
	Name of assignment or project: 345kV Shinbupyung~Youngseo Transmission Line Project **Year**: 2011~2011 (12 months) **Location**: Korea, Inchun **Client**: Ministry of Knowledge, Economy, Korea **Main project features**: Civil Works **Positions held:** Civil Engineer **Activities performed**: Equipment Specifications, Procurement, Construction schedule and budget control, determination of line route and substation location, Soil investigation, line route survey and Foundation design (transmission and Substation) review, supervision of project implementation

	Name of assignment or project: Line Route and ESIA Study for Han-Bobo Dioulasso-Sikasso-Bamako Interconnection Project in WAPP **Year**: 2007~2010 **Location**: Ghana, Burkina Faso, Mali **Client**: West African Power Pool (WAPP) **Main project features**: Line Route and ESIA Study for the 225kV and 330kV transmission system(742km, 5 substations) **Positions held**: Civil Engineer **Activities performed**: Transmission line route investigate, Detail surveys, and ESIA Study, Public consultation, Provisional environmental and social management plan
	Name of assignment or project: 345kV Sinyangsan ~Dongbusan Construction Project **Year**: 2005 ~ 2006 (12 months) **Clien**t: Ministry of Commerce, Industry and Energy **Location**: Korea, Busan **Main project features**: Civil Works The project is to construct 345kV GIS Foundation **Positions held**: Civil Engineering Expert **Activities Performed**: Review of engineering design and drawings, supervision of project implementation and test. Check the construction drawings and implementation plans.
	Name of assignment or project: 345kV Dong-Seoul Substation Rebuilding Project **Year**: 2004 ~ 2005 (11 months) **Clien**t: Ministry of Commerce, Industry and Energy **Location**: Korea, Seoul **Main project features**: Civil Works **Positions held**: Civil Engineering Expert **Activities Performed**: Review of engineering Re-design and drawings, supervision of project implementation and test. Check the construction drawings and implementation plans. Planning for the Substation re-location and implementation

	Name of assignment or project: 345kV Daegu Substation Construction Project **Year**: 2003 ~ 2004 (14 months) **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea, Daegu **Main Project Features**: Civil Work **Positions held**: Civil Engineering Expert **Activities Performed**: Review of engineering design and drawings, supervision of project implementation and test. Check the construction drawings and implementation plans.
	Name of assignment or project : 345kV Shinyangsan~ Dongbusan Transmission Line Project **Year**: 2002~2004 (24 months) **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea, Busan **Main Project Features**: Civil Work **Positions Held**: Civil Engineer **Activities Performed**: Equipment Specifications, Procurement, Construction schedule and budget control, determination of line route and substation location, Soil investigation, line route survey and Foundation design (transmission and Substation) review, supervision of project implementation
	Name of assignment or project : 154kV Namhae-Jinju Transmission Line Project **Year**: 2000~2002 (26 months) **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea, Seoul **Main Project Features**: Civil Work **Positions Held**: Civil Engineer **Activities Performed**: Equipment Specifications, Procurement, Construction schedule and budget control, determination of line route and substation location, Soil investigation, line route survey and Foundation design (transmission and Substation) review, supervision of project implementation

Expert's Contact information:

Email: donghyun.kim@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Dong Hyun Kim	Kim	2014.12.26
Name of Expert	Signature	Date

Park, Jae Heun		2014.12.26
Name of Authorized Representative of the Consultant	Signature	Date

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Form TECH-6(B) Assignment – A & B

CURRICULUM VITAE (CV)

POSITION TITLE AND NO.:	Geotechnical & Geological Expert [K-4(A) & K-5(B)]
NAME OF EXPERT:	Munir Ahmad Baig
DATE OF BIRTH:	December 27, 1951
COUNTRY OF CITIZENSHIP/ RESIDENCE:	Pakistan

EDUCATION:

- M.Sc. (Geology with specialization in (Engineering Geology & Geo-hydrology) from Punjab University, 1975.
- B.Sc. (Geology) from Punjab University, 1973.

EMPLOYMENT RECORD RELEVANT TO THE ASSIGNMENT:

Period	Employing organization and your title/ position. Contact information for references	Country	Summary of activities performed relevant to the Assignment
May 2013 – To-date (Intermittent)	Engineering and Management Advisory Services (EnMasse) - 239-P Model Town Ext: Lahore/ Geologist	Pakistan	Responsible to handle all the activities related to geotechnical engineering on the project.
Sep 2009 – Apr 2013	Associated Consulting Engineers – ACE (Pvt) Ltd./ Resident Engineer Geology	Pakistan	Managed / Supervised the excavation and installation of complete designed support works of Access Tunnels (3 Nos. of about 6 KM length) to Headrace Tunnels and sections of single Headrace Tunnel (100 m^2) about 2.5 Km and Headrace Twin Tunnels (55 m^2) Sections about 4 km.
Feb 2009 – Aug 2009	Associated Consulting Engineers – ACE (Pvt) Ltd./ Chief Engineering Geologist/ Dam Specialist	Malaysia	- Construction Material Studies, identification and selection of barrow areas for Main Dam core material (clayey silt/ silty clay). - Riprap material for the protection of upstream and downstream slopes of main dam and auxiliary dam.
Jan 2007 – Jan 2009	Associated Consulting Engineers – ACE (Pvt) Ltd./ Chief Engineering Geologist	Pakistan	- Planning and supervision of **Geological mapping** of 6 dam sites and two powerhouse sites Preparation/ supervision of **geological mapping** on Auto CAD drawings for pre-feasibility report. - Collection of geologic data/ maps. - Preparation of time & work schedules of geological/ geotechnical activities.

Jun 2004 – Nov 2006	Associated Consulting Engineers – ACE (Pvt) Ltd./ Chief Engineering Geologist	Pakistan	- Participation in the joint meetings between WAPDA, Project Consultants and contractors to discuss project oriented technical problems relating to geotechnical and geological studies/ investigations and give advice accordingly.
Jan 2004 – May 2004	Associated Consulting Engineers – ACE (Pvt) Ltd./ Chief Geology	Pakistan	- Preparation of sub-surface exploratory drilling programme for GTC-05 and GTC-06 components of the project. - Supervision of drilling operations, including field tests and collection of soil samples.
May 2003 – Dec 2003	NEAC Consultants/ Principal Geologist/ Material Engineer	Pakistan	- Detailed **Geologic/ Lithologic mapping** of potential borrow areas of concrete aggregate (coarse and fine) materials.
Aug 2002 – May 2003	Azad Jammu & Kashmir Hydroelectric Board, Muzaffarabad/ Project Site Incharge	AJK, Pakistan	- Supervision of **geological mapping** of project sites (alternative dam axis areas) and reservoir area. - Preparation of geologic x-sections of different alternative sites of dam and its appurtenant structures.
Mar 2001 – Jul 2002	Associated Consulting Engineers – ACE (Pvt) Ltd./ Principal Engineering Geologist/ Material Engineer	Pakistan	Responsible to handle all the activities related to engineering geology/ material on the project accordingly.
1994 – Oct 1995	Associated Consulting Engineers – ACE (Pvt) Ltd./ Site Incharge (Supervising Geologist)	Pakistan	- **Geological Mapping** of proposed weir site and intake area, along tunnel alignment, penstock route, powerhouse and switchyard area. - Construction material Investigation and studies. - Writing up "Report on Geo-technical Investigational Works" at site.
1993 – 1994	Associated Consulting Engineers – ACE (Pvt) Ltd./ Principal Engineering Geologist)	Pakistan	- Analysis & interpretation of surface and sub surface geological investigation carried out at Vaniar dam site, Tabriz (Iran). - Preparation and finalization of Geologic plans and sections of Vaniar Dam Project, Tabriz, Iran.
1992 – 1993	ACE and Ashnab Engineering Consultants, Tabriz, Iran/ Principal Engineering Geologist/ Material Engineer	Iran	- Surface **geologic mapping** of **Vaniar Dam Reservoir** and **Aji Chai Basin**. - **Geological mapping**, investigations and studies for construction materials for **Nahand Dam Project.** - Preparation of a report on preliminary Geologic Investigations and studies of proposed **Khuda Afrin Canal Alignment.**

Apr 1992 – Nov 1992	Associated Consulting Engineers – ACE (Pvt) Ltd./ Principal Engineering Geologist/ Material Engineer	Pakistan	- Planning of exploratory drilling, testpits, adits, shafts and studies for construction material for **Kurram Tangi Dam Project** and interpretation of field data for pre-feasibility studies report. - Preparation & finalization of exploratory boreholes and test pits logs.
1990 – 1992	Pakistan Hydro Consultants (PHC) [a joint venture of M/s ACE, NESPAK (Pakistan), EWBANK PREECE (UK), HARZA Engineering Co. (USA) and Binnie & Partners (UK)]/ Senior Engineering Geologist/ Material Engineer	Pakistan	- **Geological mapping** and reserve estimation of the selected borrow areas. - Strip **Geologic Mapping** along Power channel from Ghazi (Barrage Site) to Barotha village (Power Complex Site). - **Geological Mapping** of exploratory Adit and Shaft in Power Complex area.
Mar 1990 – Jul 1990	Associated Consulting Engineers – ACE (Pvt) Ltd./ Senior Geologist	Pakistan	- Preliminary geological strip mapping along proposed **Transmission Line route.** - Preparation of Boreholes/Test pit logs.
1989 – 1990	Associated Consulting Engineers – ACE (Pvt) Ltd./ Senior Engineering Geologist/ Material Engineer	Pakistan	- Reconnaissance and identification of potential Borrow Areas for concrete aggregate material for **Sarobi Dam in North Waziristan**. - Report on geological and construction materials investigations/ studies.
1988 – 1989	A joint venture of STV Lyons, U.S.A. and Engineering Consultants, Karachi/ Senior Engineering Geologist	Pakistan	- Strip geological mapping along the proposed road alignment.
1986 – 1987	Associated Consulting Engineers – ACE (Pvt) Ltd./ Senior Engineering Geologist	Pakistan	- Rip-rap and derrick stone studies which included **geo-mapping** of potential borrow area, supervision of controlled blasting and analysis of blast tests. - Kalabagh Dam reservoir studies which included geo-mapping, seepage studies and potential land slide areas.
1983 – 1986	Associated Consulting Engineers – ACE (Pvt) Ltd./ Senior Engineering Geologist	Pakistan	- Preparation and updating of Geologic Sections, bedrock contour maps and Geologic Plans etc. - Compilation/ finalization of reports based on data received from dam site.
1977 – 1983	Associated Consulting Engineers – ACE (Pvt) Ltd./ Junior Geologist	Pakistan	- Supervision of field tests. - Geological mapping of the dam site area. - Reconnaissance & identification of rip rap, derrick stone and other construction materials.

MEMBERSHIP IN PROFESSIONAL ASSOCIATIONS AND PUBLICATIONS:

- Member, Punjab Geological Society, Lahore.

LANGUAGE SKILLS:

	Speaking	**Reading**	**Writing**
English	Good	Good	Good
Urdu	Very Good	Excellent	Excellent

ADEQUACY FOR THE ASSIGNMENT:

DETAILED TASKS ASSIGNED ON CONSULTANT'S TEAM OF EXPERTS:	**REFERENCE TO PRIOR WORK/ ASSIGNMENTS THAT BEST ILLUSTRATES CAPABILITY TO HANDLE THE ASSIGNED TASKS**
	May 2013 – To-date: Working as **Geotechnical Engineer/ Geologist** on Daral Khwar Hydropower Project. September 2009 – April 2013: Worked as **Resident Engineer Geology** on Neelum Jhelum Hydroelectric Project. February 2009 – August 2009: Worked as **Chief Engineering Geologist & Dam Specialist** on Timah Tasoh Dam Raising Project, Kangar, Malaysia. January 2007 – January 2009: Worked as **Chief Engineering Geologist** on Palas Valley/ Spat Gah Hydropower Complex. May 2003 – December 2003: Worked as **Principal Geologist/ Material Engineer** on **Basha Diamer Dam Project** with NEAC Consultants. August 2002 – May 2003: Worked on **Gulpur Hydropower Project**, Kotli, AJK, a project of Azad Jammu & Kashmir Hydroelectric Board, Muzaffarabad. March 2001 – July 2002: As Principal Engineering Geologist/ Material Engineer provided home office support on various projects. April 1992 – November 1992: Worked on **Kurram Tangi Dam Multipurpose Project**, NWFP for Pre-feasibility Studies and Reconnaissance of two dam sites (**Nariab and Changoz in Kohat, and Bannu Districts, N.W.F.P**). 1990 – 1991: Worked for Pakistan Hydro Consultants (PHC) on **Ghazi Barotha Hydroelectric Power Project** – feasibility and detailed design stage. March 1990 – July 1990: Worked on **Lahore – Tarbela 500 KV Transmission Line Project** (Joint Venture of ACE-NESPAK.

	1989 – 1990: Worked on **Perspective Planning and Feasibility Studies of Small Dams Project in North Waziristan & F R Bannu, NWFP**. 1988 – 1989: Worked on **Awaran-Turbat Road Project, Baluchistan** (A joint venture of STV Lyons, U.S.A. and Engineering Consultants, Karachi. 1987 – 1988: Worked on **Rural Roads Development Project in Badin**, **Sanghar and Hyderabad Districts Sindh**. 1986 – 1987: Deputed to **Kalabagh Consultants**on **Kalabagh Dam Project**. 1983 – 1986: Worked at **Kalabagh Consultants Design Office, Lahore** (On deputation from ACE Ltd.). 1977 – 1983: Worked on Kalabagh Dam Project at site, responsible for supervision of sub surface investigations (land & river).

EXPERT'S CONTACT INFORMATION: Email: enmasse@enmassepakistan.com

CERTIFICATION:

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/ or sanctions by the Bank.

Munir Ahmad Baig		30-12-2014
Name of Expert	Signature	Date

Park Jae Heun		30-12-2014
Name of authorized Representative of the Consultant	Signature	Date

Form TECH-6(B)
Assignment – A & B

CURRICULUM VITAE (CV)

Position Title and No.	Project Planning and Contract Expert [K-5(A) & K-6(B)]
Name of Expert	Jung, Yongseok
Date of Birth	December 1, 1970
Country of Citizenship / Residence	Republic of Korea

Education:

1996, Korea University, Korea, B.Eng. in Electrical Engineering
The course of overseas regional expert in Russia, Hankuk University of Foreign Studies, Korea (2007)

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2009 to date	KEPCO	Senior Manager	Overseas Project Management Department
2005 to 2009	KEPCO	Senior Manager	Power System Construction Office
1996 to 2005	KEPCO.	Transmission Engineer	345kV Youngheung Transmission line project

Membership in Professional Associations and Publications

Member of KIEE (Korea Institute of Electrical Engineers)

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task A1: Review of Feasibility Study & Use of Existing Information Task A3: Preparation of Tender Level Engineering Design Sub-Task A3.1: Collection of Data and Investigations Sub-Task A3.3: Project Implementation Planning and Procurement Packaging Sub-Task A3.4: Preparation of BOQ and Costing Task A4: Preparation of Tender Documents Task A6: Tendering and Contracting Task A7: Technical Training	**Name of assignment or project :** SECOND POWER TRANSMISSION AND DISTRIBUTION PROJECT **Year**: 2013~ present (10 months) **Location**: Kampot, Sihanoukville in Cambodia **Client:** EDC (ELECTRICITE DU CAMBODGE) **Main project features** : Project Implementation Consultant (PIC) for second power transmission and distribution project **Positions held**: Residential Engineer/Transmission engineer **Activities performed** : Completion of 230kV Kampot-Stunghav Transmission Line and Substations. Completion of about 50km of Distribution Line
	Name of assignment or project : Moinak Electricity Transmission Project **Year**: 2010~2012 (18 months) **Location**: Shelek, Robot in Kazakhstan **Client:** KEGOC (Kazakhstan Electricity Grid Operating Company) **Main project features** : Engineering, Procurement and Construction for the 230 kV Moinak Transmission Line(325km) **Positions held**: Project Manager/Transmission engineer **Activities performed** : Field survey, Engineering, Procurement for the Material and Construction Management in Kazakhstan
	Name of assignment or project: Coal-fired IPP(Independent Power Producer) Project **Year**: 2009 (12 months) **Location**: Kalimantan, Indonesia **Client**: PLN (Perusahaan Listrik Negara) **Main project features**: Bidding preparation for Transmission Line and Power Plant Construction and O&M Project **Positions held**: Project Manager/Transmission engineer **Activities performed**: Data collection and review, profiling and determination of interconnection line route, identification & assessment of potential environmental impacts, resettlement plan

	Name of assignment or project: Windfarm Project in Gulf of Suez **Year**: 2010 (10 months) **Location**: Egypt **Client**: EETC (Egyptian Electricity Transmission Company) **Main project features**: Bidding Preparation for 250MW Windfarm and Interconnection Line Construction, O&M **Positions held**: Project Manager/Transmission Engineer **Activities performed**: Data collection and review, profiling and determination of interconnection line route, identification & assessment of wind quality
	Name of assignment or project: 345kV Youngheung Transmission line project **Year**: 1997 ~ 2004 (89 months) **Location**: Korea **Client**: Ministry of Commerce, Industry and Energy, Korea **Main project features**: 320 C-km (Line length), 137 (Towers), HACSR&TACSR 480 ㎟ 4-cond bundle. The Largest Sea Transmission Line in the world **Positions held**: Transmission Engineer **Activities performed**: Transmission line route surveys, basic design, drawings (type of tower, conductor size, insulation strings), equipment specifications, construction schedule, cost estimates, bid documents, procurement, construction supervision, capacity building
	Name of assignment or project: 765kV Sinanseong - Singapyeon Transmission System Project **Year**: 2007 ~ 2009 (24 months) **Location**: Korea **Client**: Ministry of Knowledge, Economy, Korea **Main project features**: 80 C-km (Line length), 155 (Towers), ACSR 480 ㎟ 6-cond bundle **Positions held**: Transmission Engineer **Activities performed**: Construction Management, drawings (type of tower, conductor size, insulation strings), cost estimates
	Name of assignment or project: 345kV Sinchungju Intersection transmission line project **Year**: 2008~2009 (12 months) **Location**: Korea **Client**: Ministry of Knowledge, Economy, Korea **Main project features**: 88 C-km (Line length), 63 (Towers), ACSR 480 ㎟ 4-cond bundle

	Positions held: Manager/Transmission Engineer **Activities performed**: Transmission line route surveys, basic line designs and drawings, equipment specifications, construction schedule, cost estimates, bid documents, environmental impact assessment, resettlement plan, procurement, construction supervision
	Name of assignment or project: 345kV Sinonyang – Sintangjeong transmission line Project **Year**: 2008 ~ 2009 (12 months) **Location**: Korea **Client**: Ministry of Knowledge, Economy, Korea **Main project features**: 28 C-km (Line length), 21 (Towers), ACSR 480 ㎟ 4-cond bundle **Positions held**: Manager/Transmission Engineer **Activities performed**: Transmission line route surveys, basic line designs and drawings, equipment specifications, construction schedule, cost estimates, bid documents, environmental impact assessment, resettlement plan, procurement, construction supervision
	Name of assignment or project: 154kV Jocheon Intersection transmission line project **Year**: 2005 ~ 2007 (22 months) **Location**: Korea **Client**: Ministry of Commerce, Industry and Energy, Korea **Main project features**: 13C-km (Line length), 20 (Towers), ACSR 330 ㎟ **Positions held**: Manager/Transmission Engineer **Activities performed**: Transmission line route surveys, basic line designs & drawings, equipment specifications, construction schedule, cost estimates, bid documents, procurement, construction supervision
	Name of assignment or project: 154kV Dongjeju Underground transmission line project **Year**: 2005 ~ 2006 (18 months) **Location**: Korea **Client**: Ministry of Commerce, Industry and Energy, Korea **Main project features**: 13C-km (Line length), XLPE 2,000 ㎟ **Positions held**: Manager/Transmission Engineer **Activities performed**: Transmission line route surveys, basic line designs & drawings, equipment specifications, construction schedule, cost estimates, bid documents, procurement, construction supervision

Expert's Contact information:

Email: yongseok.jung@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Yong Seok, Jung	Jung	2014. 12. 26
Name of Expert	Signature	Date

Park, Jae Heun		2014. 12. 26
Name of Authorized Representative of the Consultant	Signature	Date

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Form TECH-6(B) Assignment – A & B

CURRICULUM VITAE (CV)

POSITION TITLE AND NO.:	Environmental/Social/Resettlement Expert [K-6(A) & K-9(B)]
NAME OF EXPERT:	Afzal Hayat Khan
DATE OF BIRTH:	04-05-1964
COUNTRY OF CITIZENSHIP/ RESIDENCE:	Pakistan

EDUCATION:

MA Sociology	University of Peshawar	1990
MSc. Urban & Regional Planning	University of Peshawar	2000

EMPLOYMENT RECORD RELEVANT TO THE ASSIGNMENT:

Period	Employing organization and your title/ position. Contact information for references	Country	Summary of activities performed relevant to the Assignment
2008 to date (intermittently)	Engineering and Management Advisory Services (EnMasse)/239-P Model Town Ext: Lahore	Pakistan	• Environmental and Social Assessment (ESA) study for mitigation of negative impacts and enhancing the positive impact to improve livelihood of and Resettlement Action (RAP) for establishment 500 KV Grid Station Shikarpur and allied T/Lines, 220 KV Transmission line for Aai khwar, Khan Khwar and Daher khwar to Manseher. • Third party validation of Environmental Management and monitoring Plan and Resettlement Plan of 6th STG project under LESCO Electric Distribution and Transmission Improvement Project (EDTIP) • Third Party Validation of Environmental Management and Monitoring Plan for 220KV Grid Station Kassowal and allied 220KV Transmission lines (Inception and Midterm report submitted) • Initial Environmental Examination of 220kV Grid Station Ghazi Road and allied 220kV Transmission line
2011-2011	Capacity Building and Advisory Services (WCAP), Ministry of Water and Power, Pakistan/ World Bank/ Benefit sharing expert/ Sociologist	Pakistan	• Initiate and supervise survey on all the project sites, questionnaires formulation for conducting surveys, undertaking specific case studies and organizing stakeholder meetings in the proposed hydropower sites of varied stakeholder groups' iii. Report preparation and presentation of finding of the study with all the stake holder including the elected representatives, heads of various hydropower generation authorities/ companies, representatives IFIs and opinion leaders for advocacy of reforms in

			the present policy of one time compensation for PAP, resulting would result in enhanced social acceptability and creation of a sense of ownership amongst the PAP instead of being reactive and defiance, it would also reduce poverty and improve standard of living of poor people and hence would lead to sustainable development.
2008 - Present ((intermittently) onwards	Sarhad Hydro Development Organization (SHYDO), KPK Province, Pakistan/ ADB/ External Monitoring Expert	Pakistan	• Review and verify progress on timely and efficiently compensation payment for all the landed and non landed assets claims of the PAP outlined in the approved Resettlement Plan for all these Project up to the entire satisfaction of the PAP. ii. Monitoring implementation of the mitigation plan up to the entire satisfaction of the PAP with special focus on the marginalized fraction of community i.e. most vulnerable people, female folks, women headed households, old age people, people with special need etc. iii. Probe into various bottlenecks for grievance redress mechanisms at the policy and operational level, for adoption of corrective measures resolution of grievances with fairness, justice and transparency. iv. Capacity building of the concerned staff for efficiently implementation of Resettlement Plan, EMP and Health Safety Plans. v. Preparation of monitoring reports, manuals and responding to various quarries made by relevant ADB officials
2007-2008	National Transmission and Dispatch Company (NTDC)/ JBIC/ Resettlement Expert / Sociologist (Gender Specialist)	Pakistan	• Review of various studies findings, data, and all the available reports/ document of all the feature of all the projects ii. exploring the types and number of affected population in term of losses of their landed and non landed assets, undertaking field survey by using structured questionnaire for quantifying impacts on the landed assets of all the individual affected person (s); also undertook Strengths, Weaknesses, Threats and Opportunities (SWOT) analysis associated with these projects in a participatory manners, in order to overcome Weakness/ Threats and to convert them into Strengths and Opportunities by adopting corrective mitigation measure by all the stakeholders including the PAPs.
2006-2007	National Transmission and Dispatch Company (NTDC)/ WB/ Project Manager/ Senior Sociologist	Pakistan	• Activities performed in almost the same manners as give above for preparation of a need based ESA with the intentions of improvement of living standards of the project Affected Population and or at least maintaining their living conditions to the pre Project scenario.

2005-2006	Small Dams Organization, Government of KPK Province, Pakistan/ GTZ (Germany) Donor Agency/ Social Development Specialist	Pakistan	• Socioeconomic impact analysis surveys for probing into the negative and positive impacts of all the proposed Projects of dams construction on the landed and non landed assets (living conditions) of PAP and accordingly mitigation plans and enhancing the positive impacts were prepared respectively in consultation with the active involvement of all key player and the PAP themselves. ii. for assessment of these losses, an inventory of lost assets was made for perpetration of entitlement matrix in consultation with affected persons for compensation and rehabilitation of their livelihood. For resolution of grievances and a justifiable use of the water stored in these dams with giving rights to people living nearby on priority basis, Water User Associations (committees) were formed. Costs and budgets estimates were included for undertaking resettlement and environmental rehabilitation expenditures, besides occurrences of other costs.
2006-2007	Response International (a British charity)/ Big lottery Fund, UK/ Project Manager (PM)	Pakistan	• Studies were undertaken for probing into the root causes of the prevailing deteriorating mother child healthcare situation. Based on the finding of these study various activities were undertaken including mass awareness program for prevention of various communal disease amongst the dwellers with special focus on mother child health care, training of Traditional Birth Attendant (TBA) and provided with delivery kit in the far flung areas of the FATA, medical equipment and life saving drugs were provided to BHU and dispensaries in the deserving area of FATA.
2001 - 2002	Water And Power Development Authority/ ADB/ Relocation And Resettlement Engineer	Pakistan	• Undertaking Socioeconomic Impact Assessment for preparation of a need based Resettlement Plan (RAP) in consultation with PAP, elected representatives, elders & opinion leaders of the area. Coordination with all the relevant nation building departments of the Project area for incorporating their views in the Resettlement Plan
1998 - 2001	Water And Power Development Authority/ ADB/ Deputy Team Leader/ Sociologist	Pakistan	• Induction and management of the field staff, Preparation of questionnaire and supervision of farm level, and household socioeconomic impacts survey, case studies on various important aspects of the Project with the target population interpretation/ analysis of the data for preparation of all reports including the final Report.
1997-1998	Ministry of Health/ WHO/ Instructor for Community	Pakistan	• Based on TNA, Preparation of training module for conducting training to all the stakeholders involved in the BDN Program, with special emphasis on

	Development		social mobilization and participatory community development. ii. Conducting research studies for improvement in the BDN to meet the requirement of social mobilization and unification of BDN program in all the BDN sites for consolidation and expansion of the BDN Program.
1994-1996	ActionAid Pakistan (a British charity)/ Community Facilitator and HRD Coordinator	Pakistan	• Preparation of Community Interaction Strategy (CIS) and Human Resource Development (HRD) Strategy, for delegating of more operational role to community (iv) Developed and strengthen linkages of community organization (both males & females) with district line departments for undertaking community development initiatives in their respective sector (ii) Regular monitoring and evaluation of all activities of ActionAid in the Project area including, Natural Resource Management (NRM), Health, Education and credit disbursement for Income generation Activities.
1993-1994	Sungi Development Foundation/ HRD Coordinator	Pakistan	• Organizing TNA workshop for the training need assessment for the professional field force and based on these felt needs prepared of schedule of the training calendar for their capacity building.
1992-1993	Sarhad Rural Support Corporation/ Social Organizer	Pakistan	• Formation of CBO both for male and female for making collective efforts for their own development. • Carrying out various types of diagnostic surveys' e.g. Base Line Surveys, Rapid Target Appraisals and PRA techniques for assessing the prioritized needs of the marginalized fraction of community and for preparation of a comprehensive Community Development Plan CDP with clear objectives, role & responsibilities and time frame for implementation, monitoring and evaluation of activates by the community themselves. iii. Credit Need Assessment (CNA), feasibilities of small entrepreneurs, credit disbursement and recovery etc..
1991-1992	USAID/ Consultant Trainer	Pakistan	• TOT on; Participatory training module designing, • Deliver and organize trainings on undertaking Base Line surveys, PRA techniques, Conflict Resolution and team building for development professionals and government official of the nation building departments for adoption of participatory practice in development with special focusing on involvement of the most vulnerable fraction of community and to address disparities based on gender discrimination
1990-1991	UNHCR/ District/Agency Coordinator	Pakistan	• Coordination with various international NGOs/ donors' agencies for relief activities. Monitoring and evaluation of various welfare and developmental activities launched by various NGOs, donors agencies and UNHCR

MEMBERSHIP IN PROFESSIONAL ASSOCIATIONS AND PUBLICATIONS:

Pakistan Association of Social Scientists (PASS)

LANGUAGE SKILLS:

	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent

ADEQUACY FOR THE ASSIGNMENT:

DETAILED TASKS ASSIGNED ON CONSULTANT'S TEAM OF EXPERTS:	REFERENCE TO PRIOR WORK/ ASSIGNMENTS THAT BEST ILLUSTRATES CAPABILITY TO HANDLE THE ASSIGNED TASKS
1. Preparation of Initial Environmental Assessment of the whole project to identify the environmental concerns and mitigation measures to be taken and public consultations in this regard (Assignment-A)	1. Preparation of initial environmental Examination of 220 kV Grid Station Ghazi Road Lahore and the allied Transmission lines 2. Preparation of Environmental Impact Assessment of 500kV Grid Station Shikarpur and allied Transmission Lines and 220kV Transmission Lines for Allai Khawar ,Khan Khawar, Dubai Khawar to Mansehra
1. Preparation of ESA Studies taken into account Environmental and Social Aspects in an integrated way, Social Assessment (SA) report consider socio economic studies. Preparation of Environmental Management and Monitoring plan identify the roles and responsibility of the Contractor, Consultant and the Client. Resettlement action Plan to address the grievances of PAPs (Assignment-B)	1. Preparation of ESA studies of various NTDC Projects and Third Party validation of Environmental Management and Monitoring Plan and Resettlement plan of 6th STG project (EDTIP) LESCO and 220KV Grid Station Kassowal and allied Transmission line (NTDC)

CERTIFICATION:

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/ or sanctions by the Bank.



Afzal Hayat Khan		30-12-2014
Name of Expert	Signature	Date



Park, Jae Heun		30-12-2014
Name of authorized Representative of the Consultant	Signature	Date

Form TECH-6(B) Assignment – B
CURRICULUM VITAE (CV)

Position Title and No.	Financial Management Expert (K-7)
Name of Expert	Riaz Hussain Chaudhry
Date of Birth	April 02, 1951
Country of Citizenship / Residence	Pakistan

Education:

Fellow Member of Institute of Cost & Management Accountants of Pakistan (Qualified in 1988)
M.Sc from Punjab University 1971-72
B.Sc (Hons) from Government College Lahore 1970-71

Employment Record:

Period	Employing organization and your title/ position. Contact information for references	Country	Summary of activities performed relevant to the Assignment
Jan 2013 to date	Engineering and Management Advisory Services (EnMasse)/239-P Model Town Ext: Lahore / Financial Management Expert	Pakistan	• Financial Management of ongoing Projects with MEPCO, HESCO, FESCO and NTDC.
June 2008 to Dec 2012	Pakistan Electric Power Company Ltd. (PEPCO)/ Financial Consultant	Pakistan	• Represented PEPCO to World Bank's team for developing new KPI's for performance monitoring of its entities GENCOs, DISCOs and NTDC. • Developed "Incentive Reward" to boost up recovery and for detection of different malpractices. • Compiled report on "Good Governance to mitigate Energy Crisis" and suggested ways & means to reduce losses by +1% and improve recovery by +10% instantaneously within one month's time. • As a member of audit team, carrying out technical, operational and financial audit of DISCOs and also suggesting measures to plug in the loopholes in the system. My audit findings were well recognized by PEPCO and W & P ministry.

			• As a member of the team, compiled the Revenue Generation Report in a fortnight, which is considered to be a BIBLE on this subject by PEPCO's management. • Examined draft tariff petitions for FY: 2009-10 for all eight DISCOs and suggested improvements. • Updated the years old "Remittance Procedure" of DISCOs for prompt remittances to PEPCO. • Assisted CFO-PEPCO in appraising the performance of Finance Directors of DISCOs by highlighting their contributions in relation to over all objectives of corporate culture i.e. maximizing the return on share of each DISCO.
2007	ADB thru British Power International Ltd. / Financial Analyst	Pakistan	Analyzed the performance data for PESCO, HESCO and QESCO for appraisal of Asian Development Bank for grant of loans.
2001 to 2006	Peshawar Electric Supply Company Ltd. (PESCO) / Finance Director	Pakistan	• Responsible for all functions of accounts and finance like recording/maintenance of accounts, banking, payroll and costing. Ensure the accuracy of book keeping. • Prepare CAPEX (Investment Plan) with component-wise break-up and update the same quarterly keeping in view the project progress and availability of funds. • Preparation of OPEX budget and projected financial statements and monitor the actual results vs. budgeted one's through monthly variance analysis. • Compilation of financial statements i.e. Profit & Loss, Balance sheet and Cash Flow etc. for audit purposes. • Pre-audit of payments and ensure their in-time processing. • Supervision, control and management of 62 accounting units scattered over Khyber PK province. • Liaison with NEPRA for tariff determination of DISCO. • Acted as chief coordinator with financial/legal consultants to undertake due diligence for privatization of PESCO.

			• As Company's Secretary fulfill all corporate requirements for holding Board of Directors' meetings and filing of statuary's returns.
1991 to 2000	PAKTEL Limited / Sr. Manager / Manager (accounts)/ Sr. Reg. Accountant	Pakistan	• Evaluating management and financial systems/procedures & controls and suggest measures for improvements. • Being Custodian of PAKTEL's assets, develop internal audit function and remained its in-charge. • Establishing financial viability of all MIB's (marketing information bulletin) • Responsible for supervising, controlling and managing PAKTEL's accounts for 17 branches • Preparing and monitoring of monthly variance analysis of actual and budgeted figures. • Liaising with banks and financial institutions for credit ceilings and control the cash flow. • Procuring goods/services, involved in the whole process starting from negotiation to contract finalization.
1973 to 1991	Pakistan Engineering Company Limited LAHORE / Manager Accounts/ Assistant Data Officer	Pakistan	• Responsible for all functions of accounts and finance like recording/maintenance of accounts, banking, payroll and costing. • Supervise the accounts, prepare the budget and monitor its results. • Prepare financial statements for audit purposes. Computerize payroll, inventory and receivables systems.

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent
Punjabi	Excellent	Excellent	Excellent

Adequacy for the Assignment:

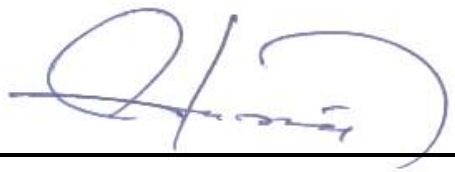
Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Monitoring of Project Budgets. The Consultant will maintain a computerized cost control system within the Project Management System (PMS) for budget monitoring and cash flow forecasting. The system will be linked with the MS PROJECT CPM network to provide information on project costs for the periodic reports required by Client. Payment to be made to the Contractor will either be on the basis of monthly invoicing for the construction works and will be linked to relevant milestones for delivery of plants or a combination of various construction installation activities.	I had supervised the finance function of following organizations during the period 1988-2012. 1. Peshawar Electric Supply Company (PESCO) as Finance Director. 2. PAKTEL Ltd (A subsidiary of Cable & Wireless, England) as National Manager (Audit & Accounts). 3. Pakistan Engineering Company Ltd. As Manager Accounts. Being the head of finance department in all above entities, I was responsible for all functions of accounts and finance like maintenance and projection of accounts, compilation of financial statements for audit purposes, costing, budgeting and it's monitoring against the actual one's expenses, pre-audit of payments and their processing.
Development of a Financial Management System. The Consultants will assess the financial management capacity of the institutions proposed for implementation of the project. Development a computerized project and financial management system which is appropriate for the size and scope of the project. The computerized project financial management system should be capable of generating periodic progress repots required for project management. The Financial Management System will would be documented in Project Financial Manual which would include the following (a) flow of funds process; (b) accounting system including internal controls; (c) financial and accounting policies; (d) planning and budgeting system (e) Financial forecasting System (f) Procurement and Control of administration monitoring system; (g) financial reporting (quarterly reports and annual financial statements); (h) auditing arrangements; and (i) Organization, staffing, training and technical support for financial management.	Having diversified experience in public and private sector in financial management, the undersigned is well equipped to assess the financial management capacity of any institution. Moreover being the internal auditor and financial consultant in power sectors entities for about 14 years, I not only checked the compliance of laid down system but also used to find out the loopholes in the system and suggested remedial actions to plug in those bugs. I would also like to mention that at the start of my career I computerized three systems i.e. Payroll, Inventory and Receivables in Pakistan Engineering Company Ltd. But now after 26 years of post-qualification experience and grooming in financial management in different sectors and cultures, I can quite easily develop a new financial management system or tailor an existing one to become suitable to the requirements as outlined by the client at serial (a) to (i). Another feather to my cap is different financial analysis carried out at PEPCO level which could have been possible only because of my possessing in-depth knowledge of financial activities and their treatment in financial books and statements.

Expert's Contact information:

Email: enmasse@emmassepakistan.com

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.



Riaz Hussain Chaudhry		30-12-2014
Name of Expert	Signature	Date



Park, Jae Heun		30-12-2014
Name of Authorized Representative of the Consultant	Signature	Date

FORM TECH-6 (A & B)

CURRICULUM VITAE (CV)

Position Title and No.	Project Coordinator [K-7(A) & K-11(B)]
Name of Expert	Ashiq Hussain Bhatti
Date of Birth	September 14, 1946
Country of Citizenship / Residence	Pakistan

Education:

B.E Engineering (Electrical) from University of Engineering & Technology, Lahore in 1968.

Employment Record:

Period	Employing organization and your title/position. Contact infor for references	Country	Summary of activities performed relevant to the Assignment
June 2007 to date	EnMasse Pvt. Ltd. / Executive Director	Pakistan	Transmission Lines and Substations Supervision of 220kV Grid Station Toba Tek Singh as Project Manager. Management and Technical Control of 132kV Grid Stations of MEPCO and SEPCO (ADB-Tranche-II) and allied Transmission Lines, Deposit Work of 132kV Grid Stations and Transmission Lines of Flying Cement Factory and Khadim Steel Mills Ltd. Faisalabad and EIA Reports of Different Projects.
October 2005 to September 2006	NTDC, WAPDA/ General Manager	Pakistan	Design & Engineering of 500/ 220/ 132/ 66 kV Transmission Lines & Grid Stations, Telecommunication and technical services.
January 2004 to October 2005	Grid System Construction (North), WAPDA, Lahore/ Chief Engineer	Pakistan	Construction of transmission system including extension of 220 / 132 kV Grid Station Sarfraz Nagar Lahore and Guddu.
January 2003 to January 2004	GSC, NTDC WAPDA, Islamabad./ Project Director	Pakistan	Construction of 220/ 132/ 33kV transmission system including extension of 220 kV Rawat Grid Station.

June 2002 to December 2002	FESCO, WAPDA, Faisalabad / Director (C&MO)	Pakistan	Construction, Operation, and Maintenance monitoring of distribution system.
February 2001 to May 2002	LESCO, Lahore/ Project Director Construction	Pakistan	Construction and Rehabilitation of HT/ LT system including Village Electrification.
July 1996 to January 2001	Area Electricity Board Lahore (LESCO)/ Project Director Construction	Pakistan	Construction and Rehabilitation of HT/LT system including Village Electrification.
July 1992 to June 1996	EHV, WAPDA, Lahore/ Project Director	Pakistan	Construction of EHV Project including 500kV Transmission Line Tarbela-Lahore, 220kV transmission line Multan – Vehari – Sahiwal and 500/ 200 kV Grid Stations Lahore, Peshawar and Rawat and 220 / 132 KV Grid Station Vehari and Sahiwal.
April 1992 to June 1992	Area Electricity Board Gujranwala/ Project Director Construction	Pakistan	Construction and Rehabilitation of HT/ LT system.
March 1991 to March 1992	Area Electricity Board, WAPDA, Fasialabad / Director Commercial	Pakistan	Implementation of commercial procedure of Distribution system.
April 1990 to February 1991	EHV, WAPDA, Lahore/ Director	Pakistan	Implementation and monitoring of EHV projects including 500kV Transmission line Multan – Guddu and 500 kV Grid Station Multan
March 1986 to March 1990	T&G Construction Div. Faisalabad / Executive Engineer	Pakistan	Construction of various 220/132kV Transmission lines and Grid Stations including erection, stringing and commissioning, extension of 220 kV Grid Station in Jaranwala Road and Nishatabad.
May 1984 to February 1986	Area Electricity Board Lahore / Executive Engineer “E” Chunian	Pakistan	Construction, Maintenance and Operation of HT/ LT Distribution system.

July 1979 to April 1984	EHV, WAPDA, Faisalabad / Executive Engineer Transmission Line Construction	Pakistan	Survey, Profile, Tower Spotting/ Stacking and Construction of 2nd Circuit 500kV transmission line Tarbela – Gatti Faisalabad.
June 1976 to May 1979	EHV, WAPDA, Faisalabad / Executive Engineer 500 kV Grid Station Construction	Pakistan	Construction, erection and commissioning of 500 kV Grid Station Gatti under the supervision of EHV Consultant Canada.
December 1975 to May 1976 & November 1968 to November 1975 &	EHV, WAPDA / 220kV Construction Sub Division Faisalabad & Rawalpindi Sub Divisional Officer S&I.	Pakistan	Survey of 500/ 220 kV Transmission line including profile preparation and tower spotting (500kV Transmission line Tarbela - Gatti, Gatti – Multan, Multan – Guddu) (220kV Transmission line Tarbela – Burhan). Supervision of Construction of 220kV Transmission line including foundations, erection and stringing (220kV Transmission line Mangla - Burhan).

Membership in Professional Associations and Publications:

1. Member, Pakistan Engineering Council (PEC).
2. Member Institute of Electrical & Electronic Engineers of Pakistan (I.E.E.E.P).

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent
Punjabi	Excellent	Excellent	Excellent
Russian	Good	Fair	Fair

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Expert	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
To assist the team leader for data collection, review of data, coordination of field survey teams, survey and mapping experts, all activities including design and construction phases relating to 500kV Transmission Line Dasu-Mansehra-Islamabad. Coordination among local and international Experts, NTDC and other organizations where necessary. To look after the job of team leader during contingencies and monitoring of field and office activities.	Served in EHV for surveying of 500 kV & 220 kV Transmission Line in the capacity of SDO, XEN for about 7 (Seven) years. Management, Coordination among NTDC, Consultants, Contractors and other organizations in the capacity of Executive Engineer T/L Construction (EHV), Project Director GSC and Chief Engineer GSC as in the Employment Record.

Expert's Contact information:

Email: bhattiashiq@yahoo.com, enmasse@enmassepakistan.com

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Ashiq Hussain Bhatti		30-12-2014
Name of Expert	Signature	Date
Park, Jae Heun		30-12-2014
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6(B)
Assignment-B

CURRICULUM VITAE (CV)

Position Title and No.	Construction Supervision Expert (K-8)
Name of Expert	Ryu, Injin
Date of Birth	December 21, 1968
Country of Citizenship / Residence	Republic of Korea

Education:

1991, Sun-Cheon University, Korea, B.E in Electrical Engineering
Transmission line design education at the Wisconsin University in USA (2006)
Overseas business education Al the Akhawayn University in Morocco (2013)

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2007 to date	KEPCO	Senior Manager	Overseas Project Development Department
1993 to 2007	KEPCO	Transmission Engineer	Power System Construction Office

Membership in Professional Associations and Publications

Project Management Professional (PMI, USA)
Member of KIEE (Korea Institute of Electrical Engineers))

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native
French	Good	Excellent	Good

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task B2: Contracts Management Sub-Task B2.2: Quality Assurance Sub-Task B2.6: Review and Reporting Task B3: Construction Supervision Sub-Task B3.1: Mobilization of Contractors Sub-Task B3.2: Site Instructions and Co-ordination with the Contractor Sub-Task B3.3: Supervision of the Construction/Installation Work Sub-Task B3.4: Workshop Inspection and Acceptance Sub-Task B3.7: Know-How Transfer and Training	**Name of assignment or project**: West African Power Pool Interconnection Transmission line Route Survey & ESIA project **Year**: 2011 ~ Present **Location**: Benin, Cote D'ivior, Sierra Leone, Liberia, Guinea **Client**: West African Power Pool (WAPP), Benin **Main project features**: 1,411km(length), Transmission Line Route Detail Survey, ESIA Study **Positions held**: Transmission Line Expert **Activities performed**: Transmission line route investigate, Detail surveys, and ESIA Study
	Name of assignment or project: Consulting service for Feasibility Study of Myanmar 500kV Transmission System **Year**: 2011~2012 **Location**: Myanmar **Client:** MEPE (Myanmar Electric Power Enterprise) **Main project features**: Demand forecasting, a master plan, basic design, Cost Estimate, Feasibility study, and etc. **Positions held**: Transmission Line Expert r **Activities performed**: Feasibility study, data collection and review, system improvement plan, establish 500kV line and tower design criteria, basic design of the 500kV transmission system, review equipment specification and technical drawings
	Name of assignment or project: Feasibility Study for Han-Bobo Dioulasso-Sikasso-Bamako Interconnection Project in WAPP **Year**: 2007~2010 **Location**: Ghana, Burkina Faso, Mali **Client**: West African Power Pool (WAPP) **Main project features**: Feasibility Study for the 225kV and 330kV transmission system(742km, 5 substations) **Positions held**: Transmission Line Expert

	Activities performed: Feasibility Study for 225/330kV high voltage transmission line (about 750km) from Han in the northern part of Ghana to the capital of Mali, Bamako, through Bobo Dioulasso in Burkina Faso, and Sikasso in Mali with associated high voltage substation, Data collection and review, prepare design criteria and cost estimate, study economical, social and environmental feasibility
	Name of assignment or project: 345kV Sinchungju Intersection transmission line project **Year:** 2008~2009 **Location:** Korea **Client:** Ministry of Knowledge, Economy, Korea **Main project features:** 88 C-km (Line length), 63 (Towers), ACSR 480 ㎟ 4-cond bundle **Positions held:** Transmission Line Expert (Project Manager) **Activities performed:** Transmission line route surveys, basic line designs and drawings, equipment specifications, construction schedule, cost estimates, bid documents, environmental impact assessment, resettlement plan, procurement, construction supervision
	Name of assignment or project: 154kV Tong-young Transmission line Project **Year**: 2007~2009 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features**: The project is to construct the 154kV transmission line from 154kV Tong-Young Substation to 154kV Gosung Substation **Positions held**: Transmission Line Expert (Project Manager) **Activities Performed**: Preparation of preliminary engineering design, equipment specifications, cost estimates and carrying out basic design (conductor size, tower structure, tower, hardware, insulator) and Construction Supervision
	Name of assignment or project: Insulation coordination consulting for GECOL transmission system and EMTP training **Year:** 2004~2006 **Location**: Libya **Client:** GECOL

	Main project features: Establish insulation coordination study for 400/230/66/33kV transmission system in Libya, Insulation coordination and EMTP software training **Positions held**: Transmission Line Expert **Activities performed**: Establishment of design criteria, Insulation coordination study for 400/230/66/33kV transmission system
	Name of assignment or project: 345V Sin Onyang Transmission line Project **Year**: 2004~2007 **Client**: Ministry of Commerce, Industry and Energy **Location**: Korea **Main project features**: The project is to construct the 345kV transmission line from 345kV Sin Onyang Substation to 345kV Chung-yang Substation **Positions held**: Transmission Line Expert **Activities Performed**: Preparation of preliminary engineering design, equipment specifications, cost estimates and carrying out basic design (conductor size, tower structure, tower, hardware, insulator) and Construction Supervision
	Name of assignment or project: 345kV Yonghung Transmission Line Project **Year:** 1998-2004 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: The project is to construct 345kV over sea transmission line from Yonghung thermal power plant to 345kV Sinsihung substation after demand forecasting, master plan, feasibility study, design and bidding document etc. **Positions held**: Transmission Line Engineer(Project Manager for Lot2) **Activities performed**: - Preparation of tender document - Preparation of basic design (conductor size, tower structure, tower, hardware, insulator) - Procurement, Construction schedule and budget control - Checking & approval of design drawings/document - Issuing instructions to Contractors - Construction Supervision

	Name of assignment or project: 765kV Dangin - Sinseosan Transmission System Project **Year:** 1996-1999 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: The project is to construct of 765kV transmission line from Dangjin substation to Sinanseosan substation after demand forecasting, master plan, feasibility study, design and bidding document etc. **Positions held**: Transmission Line Engineer(Assistant Project Manager) **Activities performed**: - Preparation of tender document - Preparation of basic design (conductor size, tower structure, tower, hardware, insulator) - Procurement, Construction schedule and budget control - Checking & approval of design drawings/document - Issuing instructions to Contractors - Construction Supervision
	Name of assignment or project: West African Power Pool Interconnection Transmission line Route Survey & ESIA project **Year**: 2011 ~ Present **Location**: Benin, Cote D'ivior, Sierra Leone, Liberia, Guinea **Client**: West African Power Pool (WAPP), Benin **Main project features**: 1,411km(length), Transmission Line Route Detail Survey, ESIA Study **Positions held**: Transmission Line Engineer **Activities performed**: Transmission line route investigate, Detail surveys, and ESIA Study

	Name of assignment or project: Consulting service for Feasibility Study of Myanmar 500kV Transmission System **Year**: 2011~2012 **Location**: Myanmar **Client:** MEPE (Myanmar Electric Power Enterprise) **Main project features**: Demand forecasting, a master plan, basic design, Cost Estimate, Feasibility study, and etc. **Positions held**: Transmission Line Engineer **Activities performed**: Feasibility study, data collection and review, system improvement plan, establish 500kV line and tower design criteria, basic design of the 500kV transmission system, review equipment specification and technical drawings

Expert's Contact information:

Email: injin@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Name of Expert	Signature	Date
In Jin Ryu	Ryu	2014. 12. 26

Name of Authorized Representative of the Consultant	Signature	Date
Park, Jae Heun		2014.12.26

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FORM TECH-6 (A & B)

CURRICULUM VITAE (CV)

Position Title and No.	Local Design Expert [K-8(A) & K-12(B)]
Name of Expert	Munawar Ahmad Malik
Date of Birth	May 25, 1950
Country of Citizenship / Residence	Pakistan

Education:

B.Sc Engineering (Electrical) University of Engineering & Technology, Lahore

Employment Record:

Period	Employing organization and your title/ position. Contact information for references	Country	Summary of activities performed relevant to the Assignment
June 2007 to date (intermittent)	EnMasse Pvt. Ltd. / Principal Design Engineer	Pakistan	Designing of 220kV and 132kV Transmission Line Projects for ADB funded: Tranche-II and Tranche-III Projects in MEPCO and HESCO jurisdiction.
September 2007 to May 2011	National Transmission and Despatch Company (NTDC) / General Manager (GSC) Grid System Construction WAPDA House Lahore	Pakistan	• Construction of 500/ 220 kV Sub-Station. • Construction of 500 kV and 220 kV Transmission lines.
December 2005 to September 2007	NTDC / Chief Engineer (EHV) Construction Project NTDC	Pakistan	• Construction of 500/ 220 kV Sub-Station. • Construction of 500 kV and 220 kV Transmission lines.
November 2004 to December 2005	NTDC / Manager Design Transmission lines WAPDA House, Lahore.	Pakistan	• Design of Transmission lines.
April 2003 to November 2004	NTDC / Project Director (EHV)	Pakistan	• Construction of 500/ 220 kV Sub-Station. • Construction of 500 kV and 220 kV Transmission lines.

June 1999 to April 2003	T&G WAPDA / Director GSC	Pakistan	• Construction of 132/ 66 kV Sub-Station. • Construction of 132 kV and 66 kV Transmission lines.
July 1998 to June 1999	T&G WAPDA / Deputy Director Transmission lines WAPDA, Lahore.	Pakistan	• Design of 220 kV, 132 kV and 66 kV Transmission lines.
March 1995 to July 1998	T&G WAPDA / Executive Engineer Transmission Line Division, GSC, Lahore.	Pakistan	• Construction of 220/132/ 66 kV Sub-Station. • Construction of 220 kV, 132 kV and 66 kV Transmission lines.
August 1978 to March 1995	T&G WAPDA / Deputy Director Transmission lines WAPDA, Lahore.	Pakistan	• Design of 500 kV, 220 kV, 132 kV and 66 kV Transmission Lines.
July 1977 to August 1978	T&G WAPDA/ Sub Divisional Officer (EHV) Construction Sub Division, Gatti Faisalabad	Pakistan	• Construction of 500/ 220 kV Sub-Station Gatti.
March 1976 to July 1977	T&G WAPDA / Sub Divisional Officer (EHV) Construction Sub Division, Rawalpindi.	Pakistan	• Construction of 500/ 220 kV Sub-Station. • Construction of 500 kV and 220 kV Transmission lines.
March 1974 to March 1976	T&G WAPDA / Assistant Engineer (EHV) WAPDA House, Lahore	Pakistan	• Design and Construction of 500/ 220/132/66 kV Sub-Station and Transmission lines.

Membership in Professional Associations and Publications:

Member, Pakistan Engineering Council.

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent
Punjabi	Good	Good	Good

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
a) All aspects of engineering and design of Transmission Lines b) Construction of EHV Transmission Lines	• Rendered Consultancy Services as Design Engineer for the Design and Construction of 66kV, 132kV, 220kV and 500kV Transmission Line which included: • Evaluation of Bids for the supply of material and construction of 132 and 220kV Transmission Line. • Approval of Transmission Line Route and Profile Drawings. • Preparation of Bid Documents for the procurement of 132kV and 220kV Towers, Conductors, Hardwares, Insulators and Earthing Material etc. • Preparation of Bid Documents for the Construction of 132kV and 220kV Transmission Lines. • Review and monitoring of the progress of the work for all under construction transmission lines • Approval of Technical data and Drawings for 132kV and 220kV Towers Conductors etc. Submitted by Suppliers. • Issued instructions and material to field formation for Rehabilitation of transmission line collapse from time to time during occurrence of tornadoes. • Carried out inspection of prototype tower for 132kV, 220kV and 500kV Transmission Line supplied by local as well as foreign suppliers. • Design and prepare new design specification for 132 and 220kV conductor, steel tower, 132kV Steel poles and material for 220kV Steel Towers. • Design and prepare foundation drawings for 132kV and 220kV steel towers and 132kV Steel poles. • Carried out constructions supervision including spread foundation, pile foundation, erection of towers, stringing works, testing and commissioning of various Transmission Lines in the country which includes: ➢ 500kV Muzaffar-Gatti Transmission Lines ➢ 500kV Rawat-Lahore Transmission Lines ➢ 500kV interconnection at NKI KHI ➢ 500kV Interconnection at Ghakkar ➢ 220kV Mangla Burhan Transmission Lines ➢ 220kV Gatti-Ludewala Transmission Lines ➢ 220kV Barotha-Shahibagh Transmission Lines ➢ 220kV Muzaffargarh-Bahawalpur Transmission Lines ➢ 220kV Jamshoro-T.M. Khan Transmission Lines ➢ 220kV Daud Khel – Bannu Transmission Lines ➢ 220kV Chashma-Ludewala Transmission Lines

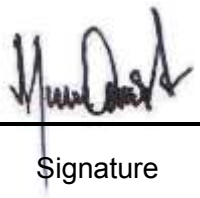
	• Carried out construction supervision of 220kV interconnection transms9sion lines for Atlas and Engro Power Transmisson Lines. • Carried out construction supervision of 132kV interconnection transmission line for Atoka, Sapphire, HUBCO, Helmore, Liberty, Nishat Power, Nishat Chunia, Sapphire, and Fauji Power Plants. • **VISIT ABROAD:** • Carried out inspection of 132kV and 220kV Transmission Line Hardware (France) • Carried out inspection of Stock Bridge Dampers (*Germany*) • Carried out inspection of Prototype Tower 220kV *(Yugoslavia)* • Carried out inspection of 132kV Towers *(China)* • Carried out inspection of SBD for 132 and 220kV and Spacer Dampers *(Italy)* • Carried out inspection of 220kV Towers *(China)* • Carried out inspection of Transmission Line Hardware and Stringing equipment *(Malaysia)* • Carried out inspection of 500kV Hardware *(Austria)* • Carried out inspection of 500kV Spacer Dampers *(Italy)* • Carried out inspection of CT, PT Circuit Breakers *(Germany)* • Carried out inspection of Disconnecting Switches *(Holland)* • Carried out inspection of Transmission Line Hardware *(Switzerland, Germany)* • Carried out inspection of Transmission Line Damper *(Switzerland)* • Carried out inspection of Transmission Line Hardware *(Czech Republic)* • Attended SARI Conference *Nepal* • Carried out inspection of 500kV Towers *(China)* • Carried out inspection of 500kV Hardware *(Austria)* • Carried out inspection of 500kV CT, PT, CCVT *(Canada)* • Carried out Prototype inspection of 220kV Steel Poles *(South Korea)*

Expert's Contact information:

Email: enmasse@enmassepakistan.com

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Munawar Ahmad Malik		30-12-2014
Name of Expert	Signature	Date

Park, Jae Heun		30-12-2014
Name of Authorized Representative of the Consultant	Signature	Date

FORM TECH-6 (A & B)

CURRICULUM VITAE (CV)

Position Title and No.	Environment/ Resettlement Expert [K-9(A) & K-13(B)]
Name of Expert	Mumtaz Hussain
Date of Birth	4-3-1947
Country of Citizenship / Residence	Pakistan

- **Education:**
  - **M.Sc. Environmental Engineering / Public Health Engineering** **1986**
    UET Lahore
  - **B.Sc. (Honors) War Studies** **1980**
    University of Balochistan, Pakistan
  - **B.Sc. (Civil) Engineering** **1979**
    Military College of Engineering, Risalpur, Pakistan
  - **M.Sc. (Honors) Soil Science** **1971**
    University of Agriculture, Faisalabad, Pakistan
  - **B.Sc. (Honors) Agriculture** **1967**
    University of Agriculture, Faisalabad, Pakistan

- **Employment Record:**

Period	Employing organization and title/ position	Country	Summary of activities performed relevant to the Assignment
2010 to date (Intermittent Period)	EnMasse Pvt. Ltd. 239 – P Model Town Extension Lahore. / Environment & Resettlement Expert	Pakistan	Supervised the preparation, reviewed and processed Initial Environmental Examination (IEE) / Environmental Impact Assessment (EIA) and Environmental Social Assessment (ESA) Reports for the following Projects: ➢ 500kV Shikarpur Grid Station and allied 500kV Transmission Lines ➢ 220kV Dera Murad Jamali Grid Station and allied 220kV Transmission Lines ➢ 220kV UCH-Sibbi Transmission lines • Carried out External Monitoring of Environmental/ Social Safeguard Matters (Third Party Validation) of 220kV Kassowal Grid Station and Transmission Lines financed by World Bank • Third Party Validation of Environmental Management & Monitoring Plan and Resettlement Plan of 36 Grid Stations and allied Transmission Lines of LESCO under World Bank financing.

2012 to 2013	SAMPAK International / Chief Environmental Engineer	Pakistan	• Environmental Impact Assessment (EIA) and Social Impact Assessment (SIA) studies for Lai Expressway and Katchehri Chowk Underpass at Rawalpindi • Environmental Impact Assessment (EIA) and Social Impact Assessment (SIA) studies for Multimodal Bus Terminal Project at Lahore.
2011 2012	Development & Management Consultants (DMC) / (i) Chief Environmental Engineer (ii) Chief Environmentalist	Pakistan	(i) Environmental Monitoring and Evaluation of Golen Gol Hydro Project in Chitral District (ii) Environmental Impact Assessment (EIA) and Social Impact Assessment (SIA) studies of **Dasu Hydropower Project** funded by World Bank.
2010 to 2011	Electra Consultants / National Environmental Specialist	Pakistan	Environmental Monitoring and Evaluation of Grid Stations and allied components including Transmission Lines, Residential Colonies and treatment plants in six Distribution Companies in Pakistan (Tranche-I) funded by Asian Development Bank.
2007 to 2008	Development & Management Consultants (DMC) / Principal Environmental & Resettlement Specialist	Pakistan	Environmental Impact Assessment of LBOD Project in four Districts of Sindh (Nawabshah, Mirpur Khas, Sanghar and Badin).
2007 to 2008	Freelance Consultant / Team Leader/ Chief Environmentalist	Pakistan	Prepared Environmental Assessment Reports of Industrial Units, Housing Schemes, Multistory Buildings/ Plazas, etc.
2006 to 2007	National Development Consultants (NDC) / Environmental Specialist	Pakistan	• Environmental Impact Asessment of Hingol Dam • Environmental Monitoring of Khan Khawar Hydropower Project. • EIA Study of Keyal Khawar Hydropower Project

2003 to 2006	BARQAAB Consulting Services (Pvt.) Ltd. / Environmental & Resettlement Specialist	Pakistan	Environmental and Social Impact Assessment of following projects: ➢ 6th STG Project, LESCO (funded by World Bank) for carrying out subject studies of six Grid Stations in Lahore Area. ➢ 500kV Grid Station at Rahimyar Khan and allied Transmission Lines, ➢ 220kV Grid Station at Gujrat ➢ 220kV Grid Station Vehari- Chishtian ➢ 220kV Grid Station Shalamar, Lahore. (funded by Japan Bank for International Corporation. ➢ 220kV Transmission Line from Chashma Nuclear Power Plant to Luddewala (Sargodha) ➢ Environmental Monitoring of 220kV Grid Station and allied Transmission Lines, Kassowal
2003 to 2004	NESPAK / Environmental Specialist	Pakistan	EIA Study of Mahmood Booti Solid Waste Dumping Site, Lahore.
2003 to 2003	Development & Management Consultants (DMC) / Environmental & Resettlement Specialist	Pakistan	• Environmental and Resettlement Studies of Chodwan, Daraban and Sheikh Haider Zam small Dams and High Head Hydro Power Projects. • Environmental and Resettlement Studies of Chashma Right Bank Irrigation Canal (CRBC), Lift-Cum-Gravity Project in KPK.
2002 to 2003	Indus Associated Consultants Lahore / Environmental Specialist	Pakistan	• Environmental Assessment Study of 21 Towns of Southern Punjab for Solid Waste, Water Supply, Roads, Slaughter Houses, Sewerage Disposal, etc. • Environmental Impact Assessment Study of Roads extending from Dara Tang, KPK to Punjab Province border, existing carriageway of Lahore-Sahiwal Road and Rural Roads in KPK.
2002 to 2002	BARQAAB Consulting Services (Pvt.) Ltd. / Environmental & Resettlement Specialist	Pakistan	Environmental Impact Assessment of Winder and Suklegi Dams in Balochistan.
2000 to 2002	Development & Management Consultants (DMC) / Environmental & Resettlement Specialist	Pakistan	• EIA Study of Balochistan effluent Disposal Study • EIA Study of Flood protection and drainage of Peshawar Valley Project in KPK • EIA Study of Khiner and Botogah Dam Projects.

1999 to 2000	Freelance Environment Consultant / Self Employed	Pakistan	Prepared Environmental Assessment Reports of Industrial Units, Housing Schemes, Multistory Buildings/ Plazas, etc.
1998 to 1998	Mot MacDonald Pakistan (MMP) / Principal Environmentalist	Pakistan	• EIA Study of Soil and Water pollution in fuel storages and outlets. • EIA Study of Solid Waste Management of Gujrat City.
1996 to 1998	Institution of Engineers, Pakistan (IEP) Members Co-operative Housing Society Lahore / Manager Planning	Pakistan	Planning of all types of Development Projects including land scaping and development aspects. Maintained close liaison with various organizations for settlement of different problems.
1996 to 1998	IEP / Convener Environment Cell / Director General Publications & Public Relations	Pakistan	• This Cell had been created on my perception and was appointed as the Convener. I arranged many Seminars and Lectures on multiple issues pertaining to environment. Ministers, Administrators, Environmentalists and Representatives of NGOs participated in the proceedings. Recommendations made during such meetings were forwarded to Federal and Provincial Governments. • Projection of performance of IEP
1993 to 1996	Military Engineer Services (PAF) / Commander	Pakistan	• In this capacity, I was Advisor to Air Officer Commanding (Southern Air Command) on the Construction as well as maintenance works in Samungli (Quetta), Jacobabad and Sukkur areas under different terrain and environmental conditions. I have the honour of completing projects of PAF BASE JACOBABAD and Sukkur Areas and **improving by alleviating salinity, sodicity and waterlogging problems.** • I was awarded Chief of the Air Staff commendation for the exemplary quality of Construction and the utmost honesty exhibited during the currency of the works.
1987 to 1993	Pakistan Army / Command and Staff Appointments	Pakistan	- -

1985 to 1987	Pakistan Navy / Garrison Engineer (Navy)	Pakistan	• Construction and maintenance works at Karachi, Pansi, Gwardar etc. for Pakistan Navy Establishment. • Advisor to Commander Pakistan Fleet on Military Engineer Services matter.
1983 to 1984	Saudi Arabia / Secondment	Saudi Arabia	Civil Engineering Works in Sandy Areas
1969 to 1982	Pakistan Army / Served in various unit and staff appointments.	Pakistan	Carried out engineering tasks in various terrains in Pakistan.

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Urdu	Excellent	Excellent	Excellent
Punjabi	Excellent	Excellent	Excellent
Arabic	Fair	Fair	Fair
Persian	Fair	Fair	Fair

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Expert	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
To carry out the site specific Environmental Impact Assessments (EIAs) as required under the Pakistani Laws. The project is to be implemented following Pakistani Environmental Regulations and the World Bank Guidelines and Operational Policies (Operational Policies 4.01) and guidelines of other international financial institutions. Preparation of documents to meet requirements for environmental assessment of 500kV Dasu – Mansehra – Islamabad Transmission Line Project as well as cumulative environmental impacts of the Project and development programs in the area.	**Year** : 2010-date (Intermittent) **Client:** NTDC (National Transmission and Despatch Company) **Main Project Features:** Supervised the preparation, reviewed and processed Initial Environmental Examination (IEE) / Environmental Impact Assessment (EIA) and Environmental Social Assessment (ESA) Reports for the following Projects: • 500kV Shikarpur Grid Station and allied 500kV Transmission Lines • 220kV Dera Murad Jamali Grid Station and allied 220kV Transmission Lines • 220kV UCH-Sibbi Transmission lines Carried out External Monitoring Of Environmental/ Social Safeguard Matters (Third Party Validation) of 220kV Kassowal Grid Station and Transmission Lines financed By World Bank and Third Party Validation of Environmental Management & Monitoring Plan and

	Resettlement Plan of 36 Grid Stations and allied Transmission Lines of LESCO under World Bank financing. **Year :** 2008- 2009(Intermittent) **Client:** General Manager (South) WAPDA **Main Project Features:** The LBOD passes through districts Nawabshah, Sanghar and Mirpur Khas of Sindh Province. The drainage system collects and evacuates drainage effluent from irrigated areas to the sea and reclaims waterlogged areas. Environmental impact assessment had been carried out for Re-Design of LBOD Stage-I in Badin area. A number of environmental and socio-economic issues had arisen due to heavy rains and sea intrusion/ Cyclones. These have been addressed in the Present Study. **Year :** 2007 –2008 **Client:** LESCO &NTDC **Main Project Features:** (i) 6th STG project consists of construction of 5 new grid stations of 132 KV each erection of associated transmission lines and extension & augmentation of grid stations. The project is finance by the World Bank. (ii) National Transmission and Dispatched Company undertakes Construction of 220 KV and 500 KV Grid Stations and Transmission Lines. The NTDC engaged consultants have carried out environmental and social impact assessment of grid stations and transmission lines at Chistian, ludewala and Rahim Yar Khan. **Year:** 2007- 2007 **Client:** WAPDA **Main Project Features:** Hingol Dam will be constructed on Hingol River in District Lasbela of Balochistan. It is the largest perennial river in Balochistan. It will provide irrigation water for 65, 000 acre of land, can store 1.206 MAF water and generate electricity to the tune of 1.37 MW. **Year:** 2006- 2007 **Client**: WAPDA **Main Project features:** I. Winder Dam is located near Goth Amaan, District Lesbella of Balochistan Province. Its storage capacity is 0.36 MAF. It will irrigate 20,000 acres of land

	II. Sukelgi Dam is located in Bolan District of Balochistan Province near Rind Town on National Highway. Its live storage capacity is 34,000 acre feet. It will irrigate 13,290 acres. **Year:** 2004 –2005 **Client:** Irrigation and Power Department NWFP and WAPDA **Main Project Features:** Feasibility Study **Year:** 2003 –2004 **Client**: Lahore City District Government **Main Project Features:** Carried our EIA and suggested mitigation measures including composting of solid waste **Year:** 2003 –2003 **Client:** WAPDA **Main Project Features:** Feasibility Study **Year:** 2002-2003 **Client:** Highway Department Punjab , NWFP **Main Project Features:** It is a rural road which connects Mianwali area with Punjab through Chashma Barrage. Its length is about 15 Km. **Year :** 2002 –2002 **Client**: WAPDA **Main Project Features:** Appraisal Studies **Year :** 2001 –2002 **Client:** WAPDA **Main Project Features:** Feasibility Studies **Year:** 2000 –2001 **Client:** Chief Engineering Advisor, Ministry of Water and Power **Main Project Features:** Environmental and Resettlement Aspects of Drainage Projects. **Year :** 1999 – 2000 **Client:** Self Employed **Main Project Features:** Environmental and Resettlement aspects of development projects.

Contact Email:

Email: enmasse@enmassepakistan.com

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Mumtaz Hussain		30-12-2014
Name of Expert	Signature	Date

Park, Jae Heun		30-12-2014
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6(B)
Assignment-B
CURRICULUM VITAE (CV)

Position Title and No.	Testing & Commissioning Expert (K-10)
Name of Expert	Yun, Ki-seob
Date of Birth	April 22, 1963
Country of Citizenship / Residence	Republic of Korea

Education:

2005, Korea University, Korea, Ph.D. in Electrical Engineering
2000, Korea University, Korea, M.A. in Electrical Engineering
1990, Busan Technical University, Korea, B.S degree in Electrical Engineering
Power Engineering Technology, Pennsylvania State University, USA (2003)
Substation Automation, Korea University, Switzerland & Sweden (2000)

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2014 to date	KEPCO	General Manager	Facility Diagnosis Center
2013 to 2014	KEPCO	General Manager	EPCM Renewable Energy Team
2010 to 2012	KEPCO.	General Manager	Substation Operation Team of District Division
1982 to 2006	KEPCO.	Senior Manager /Commissioning Engineer	Power System Protection Team of District Division

Membership in Professional Associations and Publications

Member of KEEA (Korea Electric Engineers Association)

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task B1: Design Review Sub-Task B1.1: Review and Approval of Contractor's Design Sub-Task B1.2: As Built Drawings Sub-Task B1.3: Punch List Task B2: Contracts Management Sub-Task B2.1: Monitoring of Project Progress Sub-Task B2.6: Review and Reporting Task B3: Construction Supervision Sub-Task B3.5: Test and Commissioning Sub-Task B3.7: Know-How Transfer and Training	**Name of assignment or project:** Commissioning Test for Power System of Domestic Power Facilities **Year**: 2014 **Location**: Korea **Client:** KEPCO **Main project features**: - 345kV SVC in Sin-paju Substation - 345kV Dongunsan Substation - 765kV Bukkyungnam Substation **Positions held**: General Manager **Activities performed**: Review and certification of technical design and drawings, commissioning test for power transformers, GIS and Protective Relays
	Name of assignment or project: Bid for Jordan Fujeij Wind Power Plant Construction **Year**: 2013 **Location**: Jordan **Client:** MEMR (Ministry of Energy & Mineral Resources) **Main project features**: Construction of 90MW Wind Power Plant and Operation for 20 years. **Positions held**: General Manager **Activities performed**: Prepare Technical Proposal for the Wind Farm Construction and Operation
	Name of assignment or project: Establishment of Substation Automation Specification **Year**: 2009-2012 **Location**: Korea **Client:** KEPCO **Main project features**: Establishment of Specification **Positions held**: General Manager **Activities performed**: Prepare detailed design and technical specification for the substation automation
	Name of assignment or project: Establishment of countermeasure against Blackout in Yeosu Industrial Complex **Year**: 2009 **Location**: Korea **Client:** KEPCO **Main project features**: Establishment of Countermeasure against Blackout

	Positions held: General Manager **Activities performed**: Regional Blackout cause and effect analysis and propose preventive measurement
	Name of assignment or project: 765kV Sin Ansung substation project **Year:** 2008 **Location:** Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features:** Construction of one 765kV Transformer and two Line Bays **Positions held :** Commissioning Engineer **Activities performed :** Review and certification of technical design and drawings, commissioning test for power transformer, GIS and protective system, energizing test
	Name of assignment or project: 765 kV Uljin Substation Commissioning Project **Year:** 2006-2007 (10 months) **Location**: Gangwon-do, Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: - 765/345/23 kV 2000 MVA Transformer 3 Banks - 800 kV GIS 8CBs, 362 kV GIS 16CBs, 25.8 kV GIS 4CBs **Positions held**: Commissioning Engineer **Activities performed**: Commissioning test for 800 kV GIS, 765 kV Power Transformers
	Name of assignment or project: 345 kV Sin Siheung Substation Commissioning Project **Year:** 2004 (10 months) **Location**: Gyeonggi-do, Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: - 345/154/23 kV 500 MVA Transformer 3 Banks - 362 kV GIS 8CBs, 154 kV GIS 12CBs, 25.8 kV GIS 18CBs **Positions held**: Commissioning Engineer **Activities performed**: Commissioning test for 362 kV GIS, 345 kV Power Transformers
	Name of assignment or project: 345 kV Sin-Onsan Substation Construction **Year:** 1998-2001 **Location**: Gyeong-gi do, Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: - 345 kV transformer(3) - 345 kV circuit breakers - 345 kV disconnecting switches, etc. **Positions held**: Substation Engineer (Project Manager)

	Activities performed: Project management, basic design, preparation of equipment specifications, procurement, construction schedule and budget control, site supervision
	Name of assignment or project: 345 kV Sin Namwon Substation Construction **Year:** 1993-1995 **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: - 345 kV transformer(2) - 345 kV circuit breakers - 345 kV disconnecting switches, etc. **Positions held**: Substation Engineer (Project Manager) **Activities performed**: Project management, basic design, preparation of equipment specifications, procurement, construction schedule and budget control, site supervision
	Name of assignment or project: 154 kV Seo Sihwa Substation Construction **Year:** 1990-1991 **Location**: Gyeong-gi do, Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: - 154 kV transformer(2) - 154 kV circuit breakers - 154 kV disconnecting switches, etc. **Positions held**: Substation Engineer (Assistant Project Manager) **Activities performed**: Project management, basic design, preparation of equipment specifications, procurement, construction schedule and budget control, site supervision
	Name of assignment or project: Commissioning and Operation and Maintenance **Year:** 1984-1988 **Location**: Seoul, Korea **Client:** Ministry of Commerce, Industry and Energy **Main project features**: 345kV and 154kV substation commissioning and operation in Seoul district office of transmission **Positions held**: Substation Engineer(Commissioning Team) **Activities performed**: Commissioning and Operation of substations in Seoul District office

Expert's Contact information:

Email: yunkso@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Yun, Kiseok	Yun	2014.12.26
Name of Expert	Signature	Date

Park, Jae Heun		2014.12.26
Name of Authorized Representative of the Consultant	Signature	Date

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Form TECH-6(B)
Assignment-A

CURRICULUM VITAE (CV)

Position Title and No.	Load Flow Expert (K-10)
Name of Expert	Oh, Du-Bong
Date of Birth	June 3, 1976
Country of Citizenship / Residence	Republic of Korea

Education:

1990, Chosun University, Korea, M.S in Electrical Engineering

Employment Record:

Period	Employing Organization	Title/Position	Summary of Activities
2013 to date	KEPCO	Senior Manager	Power Grid Planning Department
2011 to 2013	KEPCO	Power system Engineer	Gwangju Power Transmission District Office
2005 to 2011	KEPCO.	Power system Engineer	Power System Construction Office
2003 to 2005	KEPCO	Power system Engineer	Gwangju Power Transmission District Office.

Membership in Professional Associations and Publications

Member of KEEA (Korea Electric Engineers Association)
Member of KECA (Korea Electrical Contractors Association)

Language Skills:

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Korean	Native	Native	Native

Adequacy for the Assignment:

Detailed Tasks Assigned on Consultant's Team of Express	Reference to Prior Work/Assignments that Best illustrates Capability to Handle the Assigned Tasks
Task A1: Review of Feasibility Study & Use of Existing Information Task A3: Preparation of Tender Level Engineering Design Sub-Task A3.1: Collection of Data and Investigations Sub-Task A3.2: Design Criteria Task A4: Preparation of Tender Documents Task A7: Technical Training	**Name of assignment or project:** Renewable Generation Expansion and Interconnection Planning **Year**: 2013 ~ Present **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy, Korea **Main project features**: Renewable Generation Expansion and Interconnection Planning, Review and Approval **Positions held**: Project Manager(Power System Engineer/Leader) **Activities performed**: • Rewable generation database modeling • Established planning criteria and planning methodology • Established power system database (data collection) and renewable expansion plan
	Name of assignment or project: Development of master plan for power system (2013~2027) **Year**: 2012 ~ 2013(12 months) **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy, Korea **Main project features**: Revision of 5th national power system development master plan **Positions held**: Power System Engineer **Activities performed**: • Established planning criteria and planning methodology • Established the reactive power compensation plan • Established power system database (data collection) and transmission expansion plan • Reviewed national and regional demand forecast • Developed transmission system over 15 year expansion plans

	Name of assignment or project: : Development of KEPCO World wide Power System Solution (KW-PSS) **Year**: 2011 ~ 2013(12 months) **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy, Korea **Main project features**: Development of KEPCO World wide Power System Solution (KW-PSS) **Positions held**: Stability Study Analysis Engineer & Co-developer **Activities performed**: • Establish the Stability Analysis Module • Beta Tester of Program
	Name of assignment or project: Development of master plan for power system (2010~2024) **Year**: 2009 ~ 2010(6 months) **Location**: Korea **Client:** Ministry of Commerce, Industry and Energy, Korea **Main project features**: Revision of 4th national power system development master plan **Positions held**: Power System Engineer **Activities performed**: • Established planning criteria and planning methodology • Established the reactive power compensation plan • Established power system database (data collection) and transmission expansion plan • Reviewed national and regional demand forecast • Developed transmission system over 15 year
	Name of assignment or project: Consultancy Agreement on Planning Electrical Power System Studies for Libya **Year:** 2007-2008 **Location**: Tripoli, Libya **Client:** MEPE (Myanmar Electric Power Enterprise) **Main project features**: The consulting project consists of long term electricity load forecast study, generation expansion study and training.

	Positions held: Power System Engineer **Activities performed**: · Database modeling · Established planning criteria and planning methodology · Established the reactive power compensation plan · Established power system database (data collection) and transmission expansion plan · Carried out the trained on power system development for GECOL staffs
	Name of assignment or project: Power Development Master Plan and Institutional Strengthening in Cambodia **Year:** 2005-2006 **Location**: Cambodia **Client:** EDC **Main project features**: Establishment the Power development master plan and 230kV transmission system design **Positions held**: Power System Engineer **Activities performed**: • Provided data collection and establishment of simulation DB • Analyzed the power system (power flow, stability, fault level calculation, loss calculation, etc) • Reviewed and established a power master plan (period: 2006 ~ 2025) • Reviewed system design criteria (contingency, voltage control, frequency control, load shedding, stability, communication, protection, and control, etc) • Provided the planning methodology and training of power system analysis • Carried out feasibility study for two cross-border interconnected transmission lines

Expert's Contact information:

Email: bongnan@kepco.co.kr

Certification:

I, the understand, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Oh, Du-Bong		2014. 11. 18
Name of Expert	Signature	Date

Park, Jae Heun		2014. 11. 18
Name of Authorized Representative of the Consultant	Signature	Date

Form TECH-6A

TEAM COMPOSITION, ASSIGNMENT AND KEY EXPERTS' INPUTS

(Assignment-A) (Rev.1)

No.	Name	Expert's input (in person/month) per each Deliverable (List in TECH-5)												Total time-input (in Months)		
		Position		D-1	D-2	D-3	D-4	D-5	D-6	D-7	D-8	D-9	D-10	D-11	Home	Field	Total
KEY EXPERTS																	
K-1	Park, Jaeheun	Project Manager / Team Leader (KEPCO)	Home						1			1			2		12
			Field	1	1	0.5	1	1	1	0.5	0.5	1	2	0.5		10	
K-2	Park, Dugman	Transmission Line Design & Engineer Expert (KEPCO)	Home												0		6
			Field	1			1	1	1			1	1			6	
K-3	Choi, Byangchur	Transmission Line Survey & Mapping Expert (KEPCO)	Home												0		4
			Field				0.5	3		0.5						4	
K-4	Munir Ahmed Baig	Geological Expert (EnMasse)	Home					0.5		0.5					1		3
			Field				0.5	0.5		0.5	0.5					2	
K-5	Jung, Yongseok	Project Planning and Contract Expert (KEPCO)	Home									0.5		0.5	1		2
			Field									0.5	0.5			1	
K-6	Afzal Hayat Khan	Social/Resettlement Expert (EnMasse)	Home							0.5	0.5				1		3
			Field					2								2	
K-7	Ashiq Hussain Bhatti	Project Co-ordinator	Home	0.5	0.5	0.5		0.5		0.5	0.5				3		4
			Field					0.5			0.5					1	
K-8	Munnawar Ahmed Malik	Local Design Expert	Home	1	0.5		1	0.5	1			0.5		0.5	5		6
			Field					1								1	
K-9	Mumtaz Hussain	Envirenmantal Expert (EnMasse)	Home					1							1		3
			Field							1	1					2	
K-10	Oh, Du Bong	Principle Engineer Load flow study (KEPCO)	Home	0.5					0.5						1		2
			Field	0.5					0.5							1	
												Sub Total			**15**	**30**	**45**

Form TECH-6A

TEAM COMPOSITION, ASSIGNMENT AND KEY EXPERTS' INPUTS

(Assignment-A) (Rev.1)

No.	Name	Expert's input (in person/month) per each Deliverable (List in TECH-5)													Total time-input (in Months)		
		Position		D-1	D-2	D-3	D-4	D-5	D-6	D-7	D-8	D-9	D-10	D-11	Home	Field	Total
SUPPORT STAFF																	
N-1	T.B.N	Junior Engineer - 1	Home	0.5	1				0.5						2		6
			Field					2		1	1					4	
N-2	T.B.N	Junior Engineer - 2	Home	0.5			0.5					1			2		6
			Field					2	2							4	
N-3	T.B.N	Junior Engineer - 3	Home												0		3
			Field					1		1	1					3	
N-4	T.B.N	Junior Engineer Survey	Home												0		4
			Field					3		0.5	0.5					4	
N-5	T.B.N	Junior Engineer IEA/SA/RAP	Home												0		2
			Field							1	1					2	
N-6	T.B.N	Junior Engineer Mapping	Home				0.5	0.5		0.5	0.5				2		2
			Field													0	
N-7	T.B.N	Surveyor	Home												0		3
			Field					3								3	
N-8	T.B.N	Auto CAD Operator	Home				1	1.5	0.5	0.5	0.5				4		4
			Field													0	
												Sub Total			**10**	**20**	**30**
												Total			**25**	**50**	**75**

1. For the Foreign Consultants, Home means work in the office in the expert's country of residence. Field work means work carried out in the Client's country or any other country outside the expert‘s country of residence.

2. For the Local Sub-Consultants, Home means work in the office in the Client's country. Field work means work carried out at the project working site.

Form TECH-6A

TEAM COMPOSITION, ASSIGNMENT AND KEY EXPERTS' INPUTS

(Assignment-B) (Rev.1)

No.	Name	Expert's input (in person/month) per each Deliverable (List in TECH-5)											Total time-input (in Months)		
		Position		D-1	D-2	D-3	D-4	D-5	D-6	D-7	D-8	D-9	D-10	Home	Field	Total
KEY EXPERTS																
K-1	Park, Jaeheun	Project Manager / Team Leader (KEPCO)	Home		5							1		6		40
			Field	3	15	3	2	3	2	2	1	2	1		34	
K-2	Park, Dugman	Transmission Line Design & Engineering Expert (KEPCO)	Home											0		18
			Field	2	8	1	1	1	1	1	1	1	1		18	
K-3	Abdul Waheed Mir	Transmission Line Survey Expert (EnMasse)	Home											0		6
			Field		3	1				2					6	
K-4	Kim, Donghyun	Engineer Expert (Civil) Structure (KEPCO)	Home		3			0.5			0.5			4		8
			Field		3			1							4	
K-5	Munir Ahmed Baig	Geological Expert (EnMasse)	Home		0.5	0.5								1		4
			Field		1	2									3	
K-6	Jung, Yongseok	Project Planning and Contract Expert (KEPCO)	Home	0.5	0.5							0.5	0.5	2		4
			Field	1		1									2	
K-7	Riaz Hussain Chaudhary	Financial Management Expert (EnMasse)	Home	2	2	1						1		6		6
			Field												0	
K-8	Ryu, Injin	Construction Supervision Expert (KEPCO)	Home											0		24
			Field	1	15	2		1	1	2	1	1			24	
K-9	Afzal Hayat Khan	Social / Resettlement Expert (EnMasse)	Home	0.5		0.5								1		6
			Field			4				1					5	
K-10	Yun, Kiseob	Testing & Commissioning Expert (KEPCO)	Home							0.5		0.5		1		3
			Field							2					2	
K-11	Ashiq Hussain Bhatti	Project Co-ordinator	Home	3	3	3	3	1		1	1	1		16		20
			Field		1	2				1					4	
K-12	Munawar Ahmed Malik	Local Design Expert	Home	1	2		1	1	1	1	1	1	1	10		12
			Field		1	0.5			0.5						2	
K-13	Mumtaz Hussain	Envirenmantal Expert (EnMasse)	Home			1								1		6
			Field			5									5	
										Sub Total				**48**	**109**	**157**

Form TECH-6A

TEAM COMPOSITION, ASSIGNMENT AND KEY EXPERTS' INPUTS

(Assignment-B) (Rev.1)

No.	Name	Expert's input (in person/month) per each Deliverable (List in TECH-5)											Total time-input (in Months)		
		Position		D-1	D-2	D-3	D-4	D-5	D-6	D-7	D-8	D-9	D-10	Home	Field	Total
SUPPORT STAFF																
N-1	T.B.N	Sr. Engineer Construction	Home											0		30
			Field	1	24	1		1	1	1	1				30	
N-2	T.B.N	Jr. Engineer Civil	Home											0		24
			Field	3	15	3		1	1	1					24	
N-3	T.B.N	Jr. Engineer Electrical / Mechanical - 1	Home											0		18
			Field	1	14			1	1	1					18	
N-4	T.B.N	Jr. Engineer Electrical / Mechanical - 2	Home											0		18
			Field	1	14			1	1	1					18	
										Sub Total				**0**	**90**	**90**
										Total				**48**	**199**	**247**

1. For the Foreign Consultants, Home means work in the office in the expert's country of residence. Field work means work carried out in the Client's country or any other country outside the expert‘s country of residence.
2. For the Local Sub-Consultants, Home means work in the office in the Client's country. Field work means work carried out at the project working site.

Form FIN-3 BREAKDOWN OF REMUNERATION
Assignment - A (Rev.1)

When used for Lump-Sum contract assignment, information to be provided in this Form shall only be used to demonstrate the basis for the calculation of the Contract's ceiling amount; to calculate applicable taxes at contract negotiations; and, if needed, to establish payments to theConsultant for possible additional services requested by the Client. This Form shall not be used as a basis for payments under Lump-Sum contracts.

A. Remuneration

No.	Name	Position		US Dollars	Home	Field	Total (US $)	PK Rupees	Home	Field	Total (PK Rupees)
KEY EXPERTS											
K-1	Park, Jaeheun (KEPCO)	Project Manager / Team Leader	Home	24,200	2		48,400				
			Field	26,200		10	262,000				
K-2	Park, Dugman (KEPCO)	Design & Engineering Expert	Home	21,780	0		-				
			Field	23,580		6	141,480				
K-3	Choi, Byangchur (KEPCO)	Survey Mapping	Home	21,780	0		-				-
			Field	23,580		4	94,320				-
K-4	Munir Ahmed Baig (EnMasse)	Geological Expert	Home				-	435,600	1		435,600
			Field				-	479,600		2	959,200
K-5	Jung, Yongseok (KEPCO)	Project Planning and Contract Expert	Home	21,780	1		21,780				-
			Field	23,580		1	23,580				-
K-6	Afzal Hayat Khan (EnMasse)	Social / Resettlement Expert	Home					594,000	1		594,000
			Field					654,000		2	1,308,000
K-7	Ashiq Hussain Bhatti (EnMasse)	Project Coordinator	Home					594,000	3		1,782,000
			Field					654,000		1	654,000
K-8	Munawar Ahmed Malik (EnMasse)	Local Design Expert	Home					792,000	5		3,960,000
			Field					872,000		1	872,000
K-9	Mumtaz Hussain (EnMasse)	Envirenmantal Expert	Home					594,000	1		594,000
			Field					654,000		2	1,308,000
K-10	Oh, Du Bong (KEPCO)	Load Flow Study Expert	Home	21,780	1		21,780				-
			Field	23,580		1	23,580				-
Sub-Total (A)					**4**	**22**	**636,920**		**11**	**8**	**12,466,800**

A. Remuneration											
No.	**Name**	**Position**		**US Dollars**	**Home**	**Field**	**Total (US $)**	**PK Rupees**	**Home**	**Field**	**Total (PK Rupees)**
SUPPORT STAFF											
N-1	(EnMasse)	Jr. Engineer	Home					198,000	2		396,000
			Field					218,000		4	872,000
N-2	(EnMasse)	Jr. Engineer	Home					198,000	2		396,000
			Field					218,000		4	872,000
N-3	(EnMasse)	Jr. Engineer	Home					198,000			-
			Field					218,000		3	654,000
N-4	(EnMasse)	Jr. Engineer Survey	Home					178,200			-
			Field					196,200		4	784,800
N-5	(EnMasse)	Jr. Engineer IEA	Home					178,200			-
			Field					196,200		2	392,400
N-6	(EnMasse)	Jr. Engineer Mapping	Home					178,200	2		356,400
			Field					196,200			-
N-7	(EnMasse)	Surveyor	Home					128,700			-
			Field					141,700		3	425,100
N-8	(EnMasse)	AutoCad Operator	Home					148,500	4		594,000
			Field					163,500			-
Sub-Total (B)					**0**	**0**	**0**		**10**	**20**	**5,742,700**
Total (A+B)					**4**	**22**	**636,920**		**21**	**28**	**18,209,500**

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2. For the Local Sub-Consultants, Home means work in the office in the Client's country. Field work means work carried out at the project working site.

Form FIN-3 BREAKDOWN OF REMUNERATION
Assignment - B (Rev.1)

When used for Lump-Sum contract assignment, information to be provided in this Form shall only be used to demonstrate the basis for the calculation of the Contract's ceiling amount; to calculate applicable taxes at contract negotiations; and, if needed, to establish payments to theConsultant for possible additional services requested by the Client. This Form shall not be used as a basis for payments under Lump-Sum contracts.

A. Remuneration

No.	Name	Position		US Dollars	Home	Field	Total (US $)	PK Rupees	Home	Field	Total (PK Rupees)
KEY EXPERTS											
K-1	Park, Jaeheun (KEPCO)	Project Manager / Team Leader	Home	24,200	6		145,200				
			Field	26,200		34	890,800				
K-2	Park, Dugman (KEPCO)	Design & Engineering Expert (T/Line)	Home	21,780	0		-				
			Field	23,580		18	424,440				
K-3	Abdul Waheed Mir (EnMasse)	Transmission Line Survey Expert	Home					594,000			-
			Field					654,000		6	3,924,000
K-4	Kim, Donghyun (KEPCO)	Engineer Expert (Civil) Structure	Home	21,780	4		87,120				
			Field	23,580		4	94,320				
K-5	Munir Ahmed Baig (EnMasse)	Geological Expert	Home					435,600	1		435,600
			Field					479,600		3	1,438,800
K-6	Jung, Yongseok (KEPCO)	Project Planning and Contract Expert	Home	21,780	2		43,560				
			Field	23,580		2	47,160				
K-7	Riaz Ahmed (EnMasse)	Financial Management Expert	Home					594,000	6		3,564,000
			Field					654,000		0	-
K-8	Ryu, Injin (KEPCO)	Construction Supervision Expert	Home	21,780			-				
			Field	23,580		24	565,920				
K-9	Afzal Hayat Khan (EnMasse)	Social / Resettlement Expert	Home					594,000	1		594,000
			Field					654,000		5	3,270,000
K-10	Yun, Kiseob (KEPCO)	Testing & Commissioning Expert	Home	24,200	1		24,200				
			Field	26,200		2	52,400				

A. Remuneration											
No.	**Name**	**Position**		**US Dollars**	**Home**	**Field**	**Total (US $)**	**PK Rupees**	**Home**	**Field**	**Total (PK Rupees)**
K-11	Ashiq Hussain Bhatti (EnMasse)	Project Coordinator	Home					594,000	16		9,504,000
			Field					654,000		4	2,616,000
K-12	Munawar Ahmed Malik (EnMasse)	Local Design Expert	Home					792,000	10		7,920,000
			Field					872,000		2	1,744,000
K-13	Mumtaz Hussain (EnMasse)	Environmental Expert	Home					594,000	1		594,000
			Field					654,000		5	3,270,000
Sub-Total (A)					**13**	**84**	**2,375,120**		**35**	**25**	**38,874,400**
SUPPORT STAFF											
N-1	T.B.N (EnMasse)	Sr. Engineer Construction	Home					396,000			-
			Field					436,000		30	13,080,000
N-2	T.B.N (EnMasse)	Jr. Engineer Civil	Home					178,200			-
			Field					196,200		24	4,708,800
N-3	T.B.N (EnMasse)	Jr. Engineer Electrical / Mechanical	Home					198,000			-
			Field					218,000		18	3,924,000
N-4	T.B.N (EnMasse)	Jr. Engineer Electrical / Mechanical	Home					178,200			-
			Field					196,200		18	3,531,600
Sub-Total (B)					**0**	**0**	**0**		**0**	**90**	**25,244,400**
Total (A+B)					**13**	**84**	**2,375,120**		**35**	**115**	**64,118,800**

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2. For the Local Sub-Consultants, Home means work in the office in the Client's country. Field work means work carried out at the project working site.

Consultant's Representations Regarding Costs and Charges
(Model Form I)
Assignemnt-A (Rev.1)
(Expressed in {insert name of currency*)

No.	Personnel		1	2	3	4	5	6	7	8
	Name	Position	Basic Remuneration Rate per Working Month /Day/Year	Social Charges[1]	Overhead[1]	Subtotal	Profit[2]	Premium	Proposed Fixed Rate per Working Month/Day/Hour Home	Proposed Fixed Rate per Working Month/Day/Hour[1] Field
Kepco			USD	20%	100%		10%	20%		
K-1		Team Leader	10,000	2,000	10,000	22,000	2,200	2,000	24,200	26,200
K-2		Design and Engineering Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-3		Survey Mapping	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-5		Project Planning Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-10		Load Flow Study Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
EnMasse			PK Rupees	10%	70%		10%	20%		
K-4		Geo Expert	220,000	22,000	154,000	396,000	39,600	44,000	435,600	479,600
K-6		Env Social	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-7		Project Coordinator	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-8		Local Design Expert	400,000	40,000	280,000	720,000	72,000	80,000	792,000	872,000
K-9		Envirenmantal Expert	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
N-1	TBN	Junior Engineer - 1	100,000	10,000	70,000	180,000	18,000	20,000	198,000	218,000
N-2	TBN	Junior Engineer - 2	100,000	10,000	70,000	180,000	18,000	20,000	198,000	218,000
N-3	TBN	Junior Engineer - 3	100,000	10,000	70,000	180,000	18,000	20,000	198,000	218,000
N-4	TBN	Junior Engineer (Survey)	90,000	9,000	63,000	162,000	16,200	18,000	178,200	196,200
N-5	TBN	Junior Engineer (EP)	90,000	9,000	63,000	162,000	16,200	18,000	178,200	196,200
N-6	TBN	Junior Engineer Mapping	90,000	9,000	63,000	162,000	16,200	18,000	178,200	196,200
N-7	TBN	Surveyor	65,000	6,500	45,500	117,000	11,700	13,000	128,700	141,700
N-8	TBN	Cad Operator	75,000	7,500	52,500	135,000	13,500	15,000	148,500	163,500

Consultant's Representations Regarding Costs and Charges (Model Form I)
Assignemnt-B (Rev.1)
(Expressed in {insert name of currency*)

No.	Personnel		1	2	3	4	5	6	7	8
	Name	Position	Basic Remuneration Rate per Working Month/Day/Year	Social Charges[1]	Overhead[1]	Subtotal	Profit[2]	Premium	Proposed Fixed Rate per Working Month/Day/Hour Home	Proposed Fixed Rate per Working Month/Day/Hour[1] Field
Kepco			USD	20%	100%		10%	20%		
K-1		Team Leader	10,000	2,000	10,000	22,000	2,200	2,000	24,200	26,200
K-2		Design and Engineering Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-4		Engineer Expert (Civil) Structure	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-6		Project Planning and Contract Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-8		Construction Supervision Expert	9,000	1,800	9,000	19,800	1,980	1,800	21,780	23,580
K-10		Testing & Commissioning Expert	10,000	2,000	10,000	22,000	2,200	2,000	24,200	26,200
EnMasse			PK Rupees	10%	70%		10%	20%		
K-3		Transmission Line Survey Expert	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-5		Geo Expert	220,000	22,000	154,000	396,000	39,600	44,000	435,600	479,600
K-7		Financial Management Expert	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-9		Social / Resettlement Expert	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-11		Project Co-ordinator	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
K-12		Local Design Expert	400,000	40,000	280,000	720,000	72,000	80,000	792,000	872,000
K-13		Environmental Expert	300,000	30,000	210,000	540,000	54,000	60,000	594,000	654,000
N-1	TBN	Sr. Engineer Construction	200,000	20,000	140,000	360,000	36,000	40,000	396,000	436,000
N-2	TBN	Junior Engineer Civil	90,000	9,000	63,000	162,000	16,200	18,000	178,200	196,200
N-3	TBN	Jr.Engineer Electrical / Mechanical - 1	100,000	10,000	70,000	180,000	18,000	20,000	198,000	218,000
N-4	TBN	Jr.Engineer Electrical / Mechanical - 2	90,000	9,000	63,000	162,000	16,200	18,000	178,200	196,200

FORM FIN-4 BREAKDOWN OF REIMBURSABLE
ASSIGNMENT- A (Rev.1)

When used for Lump-Sum assignment, information to be provided in this Form shall only be used to demonstrate the basis for the calculation of the Contract's ceiling amount; to calculate applicable taxes at contract negotiation; and, if needed, to establish payments to the Consultant for possible additional services requested by the Client. This Form shall not be used as a basis for payments under Lump-Sum contracts

B. Reimbursable Expenses

N*	Type of Reimbursable Expenses	Unit	Unit Cost	Quantity	Currency No.1 as in FIN-2 ($)	Currency No.2 as in FIN-2 (Rupees)	{Local Currency as in FIN-2}
1	Per diem allowances	{Day}	270	660	178,200		
2	International flights*	{Ticket}	1,500	16	24,000		
3	In/Out Airport and Land transportation *	{Trip}	150,000	12 x 4		7,200,000	Rupees
4	Communication costs (between Lahore-Seoul)*	{Month}	50,000	12		600,000	Rupees
5	Reproduction of reports*	{Month}	100,000	12		1,200,000	Rupees
6	Office rent*	{Month}	500,000	12		6,000,000	Rupees
7	Other expenses (Foreign)*	{Lot}	-	1	30,000		
8	Other expenses (Local)*	{Lot}	-	1		14,000,000	Rupees
9	Training of the Client's personnel as per TOR*	{Lot}	-	1	60,000		
10	Cost of software (as required under TOR) along with validation upto three years	{Lot}	-	1	25,000		
11	Cost of 3 Laptops for installation of Software (as required under TOR) with latest market specifications along with genuine windows	{Nos}	1,500	3	4,500		
12	Cost of 2 handheld GPS of 1m accuracy to be supplied to client	{Nos}	3,000	2	6,000		
13	**Total Costs***				**327,700**	**29,000,000**	

Legend:

* Detail as per Annexure AA (attached)

FORM FIN-4 BREAKDOWN OF REIMBURSABLE ASSIGNMENT- B (Rev.1)

When used for Lump-Sum assignment, information to be provided in this Form shall only be used to demonstrate the basis for the calculation of the Contract's ceiling amount; to calculate applicable taxes at contract negotiation; and, if needed, to establish payments to the Consultant for possible additional services requested by the Client. This Form shall not be used as a basis for payments under Lump-Sum contracts

B. Reimbursable Expenses

N*	Type of Reimbursable Expenses	Unit	Unit Cost	Quantity	Currency No.1 as in FIN-2 (US$)	Currency No.2 as in FIN-2 (PK Rupees)	{Local Currency as in FIN-2}
1	Per diem allowances	{Day}	270	2,520	680,400		
2	International flights*	{Ticket}	1,500	52	78,000		
3	In/Out Airport and Land transportation *	{Month}	240,000	40		9,600,000	Rupees
4	Communication costs (between Lahore-Seoul)*	{Month}	50,000	48		2,400,000	Rupees
5	Reproduction of reports*	{Month}	100,000	48		4,800,000	Rupees
6	Office Rent*	{Month}	500,000	48		24,000,000	Rupees
7	Other expenses (Foreign)*	{Lot}	-	1	15,000		
8	Other expenses (Local)*	{Month}	200,000	48		9,600,000	Rupees
9	Training of the Client's personnel as per TOR*	{Lot}	-	1	60,000		
10	**Total Costs***				**833,400**	**50,400,000**	

Legend:

* Detail as per Annexure "BB" (attached)

Annexure "AA"

Sr. No.	Type of Reimbursable Expenses	Detail of Expenses		
1	Per diem allowances	Per diem allowance is paid for each night the expert is away from his/her place of residence.		
2	International flights	It will be actual; the number added in FIN-4 is tentative. It shall commensurate with the work assignment and subject to approval of client.		
3	In/Out Airport and Land transportation	It includes rental of 4 Nos. Vehicles for 12 Months fuel, lubricants, driver's pay and toll taxes etc.		
4	Communication costs (between Lahore-Seoul)	It includes Internet, Telephone and Courier Cost etc.		
5	Reproduction of reports	It includes all types of stationary and photocopy etc.		
6	Office rent	It includes rental of Office Building, all utility bills, 4 Security Staff and 1 Sweeper.		
7	Other expenses (Foreign)	It includes cost of Satellite Imagery as per TOR, purchase of camera, multimedia, softwares and Misc. expenses etc.		
8	Other expenses (Local)	Field Survey	Rs. 10,500,000	It includes rental of Surveying instruments supply and fixing of markers, pegs, tentage, labour, transportation and other petty items.
		Misc. Items	Rs. 3,500,000	It includes cost of non-technical Office Staff (1 Office Boy, 1 Office Secretary, 1 Computer Operator cum dispatch clerk, 1 accounts assistant) and purchase of generator, fuel, computers, printers, scanners, photocopiers, office furniture and such like other items for office.
9	Training of the Client's personnel as per TOR	Training of 6 Nos. Trainees nominated by NTDC for 2 (Two) weeks in Korea.		
10	Cost of software (as required under TOR) along with validation upto three years	As per Fin-4 (Assignment A).		
11	Cost of 3 Laptops for installation of Software (as required under TOR) with latest market specifications along with genuine windows and other requirements for proper running of the software	As per Fin-4 (Assignment A).		
12	Cost of 2 (two) handheld GPS of 1 (One) meter accuracy to be supplied to client for survey verification purposes	It includes cost of two handheld GPS units without base stations. Accuracy will be 0-5m standalone and ± 10mm when used with base stations.		
13	Total	Above details of expenses is tentative. Inter adjustment may have to be made within total amount of reimbursable expenses and month wise expenses will be adjusted accordingly.		

Annexure “BB”

Sr. No.	Type of Reimbursable Expenses	Detail of Expenses
1	Per diem allowances	Per diem allowance is paid for each night the expert is away from his/her place of residence.
2	International flights	It will be actual; the number added in FIN-4 is tentative. It shall commensurate with the work assignment and subject to approval of client.
3	In/Out Airport and Land transportation	It includes maintenance, fuel, lubricants of Vehicles, driver’s pay and toll taxes etc.
4	Communication costs (between Lahore-Seoul)	It includes Internet, Telephone and Courier Cost etc.
5	Reproduction of reports	It includes all types of stationary and photocopy etc.
6	Office rent	It includes rental of Office Building, all utility bills, 4 Security Staff and 1 Sweeper.
7	Other expenses (Foreign)	It includes purchase of camera, multimedia, software and Misc. expenses etc.
8	Other expenses (Local)	It includes cost of non-technical Office Staff (1 Office Boy,1 Office Secretary, 1 Computer Operator cum dispatch clerk, 1 accounts assistant) and purchase of generator, fuel, computers, printers, scanners, photocopiers, office furniture and such like other items for office.
9	Training of the Client's personnel as per TOR	Training of 6 Nos. Trainees nominated by NTDC for 2 (Two) weeks in Korea.
10	Total	Above details of expenses is tentative. Inter adjustment may have to be made within total amount of reimbursable expenses and month wise expenses will be adjusted accordingly.

FORM FIN-2 SUMMARY OF COSTS Assignement-A (Rev.1)		
Item	**Cost**	
	US$	PK Rupees
(A) Remuneration	636,920	18,209,500
(B) Reimbursable	292,200	29,000,000
(C) Deliverables	35,500	-
Total Cost excluding Tax:	**964,620**	**47,209,500**
All the Indirect Local Tax shall be paied by Client as per prevailing law of Pakistan as below;		
(i) General sales tax (GST) @ 16% on (A)	101,907.20	2,913,520.00
(ii) Income tax on non-resident experts @ 15% on (A)	130,381.27	-
Total Estimate for Indirect Local Tax:	232,288.47	2,913,520.00

FORM FIN-2 SUMMARY OF COSTS Assignement-B (Rev.1)		
Item	**Cost**	
	US$	PK Rupees
(A) Remuneration	2,375,120	64,118,800
(B) Reimbursable	833,400	50,400,000
Total Cost excluding Tax:	**3,208,520**	**114,518,800**
All the Indirect Local Tax shall be paied by Client as per prevailing law of Pakistan as below;		
(i) General sales tax (GST) @ 16% on (A)	380,019.20	10,259,008.00
(ii) Income tax on non-resident experts @ 15% on (A)	486,201.04	-
Total Estimate for Indirect Local Tax:	**866,220.24**	**10,259,008.00**

APPENDIX E - FORM OF ADVANCE PAYMENTS GUARANTEE
[See Clause GCC 41.2.1 and SCC 41.2.1]

{Guarantor letterhead or SWIFT identifier code}

Bank Guarantee for Advance Payment

Guarantor:_____ *[insert commercial Bank's Name, and Address of Issuing Branch or Office]*

Beneficiary: *National Transmission & Despatch Company (NTDC), WAPDA House, Lahore – Pakistan*

Date: _____*[insert date]*____

ADVANCE PAYMENT GUARANTEE No.: _____*[insert number]*_____

We have been informed that *Korea Electric Power Corporation (KEPCO)* (hereinafter called "the Consultant") has entered into Contract No. _____ *[reference number of the contract]* dated ___*[insert date]*_____ with the Beneficiary, for the provision of *Dasu Hydropower Transmission Line Project (DHTP)* (hereinafter called "the Contract").

Furthermore, we understand that, according to the conditions of the Contract, an advance payment in the sum of _____ *[insert amount in figures]* () *[amount in words]* is to be made against an advance payment guarantee.

At the request of the Consultant, we, as Guarantor, hereby irrevocably undertake to pay the Beneficiary any sum or sums not exceeding in total an amount of _____ *[amount in figures]* () *[amount in words]*[1] upon receipt by us of the Beneficiary's complying demand supported by the Beneficiary's written statement, whether in the demand itself or in a separate signed document accompanying or identifying the demand, stating that the Consultant is in breach of their obligation under the Contract because the Consultant:

(a) has failed to repay the advance payment in accordance with the Contract conditions, specifying the amount which the Consultant has filed to repay.

It is a condition for any claim and payment under this guarantee to be made that the advance payment referred to above must have been received by the Consultant on their account number _____ at _____ *[name and address of bank]*.

The maximum amount of this guarantee shall be progressively reduced by the amount of the advance payment repaid by the Consultant as indicated in certified statements or invoices

[1] The Guarantor shall insert an amount representing the amount of the advance payment and denominated either in the currency(ies) of the advance payment as specified in the Contract, or in a freely convertible currency acceptable to the Client.

marked as "paid" by the Client which shall be presented to us. This guarantee shall expire, at the latest, upon our receipt of the payment certificate or paid invoice indicating that the Consultant has made full repayment of the amount of the advance payment, or on the __ day of _*[month]*_____, *[year]*__,[2] whichever is earlier. Consequently, any demand for payment under this guarantee must be received by us at this office on or before that date.

This guarantee is subject to the Uniform Rules for Demand Guarantees (URDG) 2010 revision, ICC Publication No. 758.

_____

[signature(s)]

{Note: All italicized text is for indicative purposes only to assist in preparing this form and shall be deleted from the final product.}

[2] Insert the expected expiration date. In the event of an extension of the time for completion of the Contract, the Client would need to request an extension of this guarantee from the Guarantor. Such request must be in writing and must be made prior to the expiration date established in the guarantee. In preparing this guarantee, the Client might consider adding the following text to the form, at the end of the penultimate paragraph: "The Guarantor agrees to a one-time extension of this guarantee for a period not to exceed [six months][one year], in response to the Client's written request for such extension, such request to be presented to the Guarantor before the expiry of the guarantee."